SEVEN OPEN UNIT EQUILATERAL TRIANGLES DO NOT COVER A REGULAR UNIT HEXAGON

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ABSTRACT. We prove that the regular hexagon of side length one cannot be covered by seven open equilateral triangles of side length one. Seven distinguished points force one center role and six vertex roles. A finite classification then routes every hypothetical cover to one of four geometric mechanisms: trace deficit, monotone transfer, area loss, or a nine-point support obstruction. The reader-facing proof states the exact geometric interfaces, the resulting inequalities, and the exhaustive routing argument. Support-line derivations, polynomial reductions, monotonicity calculations, and exact certificates are isolated in technical appendices. The two final mixed cap overlaps are verified by a formally incorporated exact symbolic certificate using integer, rational, and $\mathbb{Q}(\sqrt{3})$ arithmetic.

CONTENTS

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1. Proof Architecture

This section gives the complete logical map before any technical verification. The body defines every object used in the proof, records the exact outputs of the calculations, explains the four geometric mechanisms, and performs the final case assembly. The TeX appendices and formally incorporated electronic supplements prove the stated formulas and inequalities. Thus a reader can follow the contradiction from the body alone and consult the verification modules only to check a calculation or a finite branch certificate.

1.1. **The theorem and the seven forced roles.**

Let

$$O = (0,0), \qquad V_i = \left(\cos\frac{i\pi}{3}, \sin\frac{i\pi}{3}\right), \qquad H = \operatorname{conv}\{V_0, \dots, V_5\},$$

where indices lie in $\mathbb{Z}/6\mathbb{Z}$. Write

$$e_{i,i+1} = [V_i, V_{i+1}], \qquad r_i = [O, V_i], \qquad M_i = \frac{1}{2}V_i,$$

and put

$$D = \bigcup_{i=0}^{5} r_i, \qquad S = \partial H \cup D.$$

Then $\mathcal{H}^1(\partial H) = \mathcal{H}^1(D) = 6$ and $\mathcal{H}^1(S) = 12$.

Theorem 1.1 (Main theorem)**.** *The hexagon H cannot be covered by seven open equilateral triangles of side length one.*

The compactness and scaling argument in Proposition C.1 gives the equivalent closed version.

Corollary 1.2 (Expanded closed formulation)**.** *For every $L > 1$, the regular hexagon $H_L = LH$ cannot be covered by seven closed unit equilateral triangles.*

The seven points $O, V_0, \dots, V_5$ are pairwise at distance at least one. Consequently a hypothetical open cover has seven distinct roles: an open triangle U_C containing O and open triangles U_i containing V_i. Their closures are denoted by T_C, T_i.

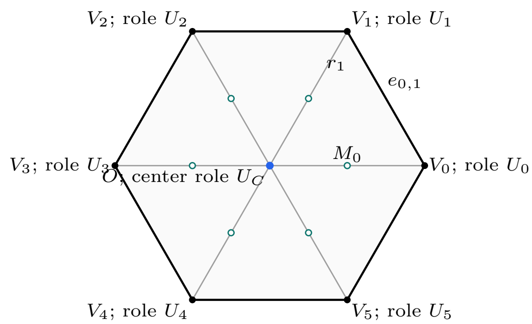


FIGURE 1. The center anchor, six vertex anchors, boundary edges, and radial arms. The role labels are forced; triangle orientations are not.

1.2. **The finite dictionary.** The center type is the number of boundary edges on which T_C has a positive-length trace. The only possibilities are $0, 1, 2$, denoted by $\mathrm{CE0}, \mathrm{CE1}, \mathrm{CE2}$.

For a vertex role, let o be the number of its triangle vertices outside H and let n be the number of adjacent radial arms met in positive length. The exhaustive types are

type	(o, n)
Vd0	$(1,0), (2,0), (3,0)$
Vd1	$(1,1)$
Vd2	$(1,2)$
T3-like	$(2,1)$.

For an original vertex role, let A_i, B_i, C_i be its actual maximal incoming boundary, outgoing boundary, and own-radial reaches. The row is *supercritical* when $A_i + B_i > 1$, and

$$N_+ = |\{i : A_i + B_i > 1\}| .$$

Lowercase (a_i, b_i, c_i) always denotes selected lower demands and never the actual maxima defining N_+.

On $e_{i,i+1}$, parametrized from V_i to V_{i+1}, the two incident open traces have the form $[0, B_i)$ and $(1 - A_{i+1}, 1]$. Their complement

$$[B_i, 1 - A_{i+1}]$$

is a boundary gap; equality gives a singleton gap, which is still uncovered. For CE1/CE2, let gr be the number of positive center traces containing such a gap.

Call a vertex role *short* if it is supercritical or has positive-length trace on an adjacent arm. Let m count the roles of type $\mathrm{Vd1}, \mathrm{Vd2}, \mathrm{T3\text{-}like}$, and let q count short roles. The structural calculus gives

$$q = N_+ + m.$$

1.3. **The exhaustive routing table.** Table 1 is read from top to bottom in the CE1/CE2 part. The first applicable row is the unique route for the hypothetical cover. The structural proof is summarized in Proposition 2.1 and given in full in Appendix C.

center and row count	exhaustive refinement	route
CE0, $N_+ = 0$	all vertex types	1
CE0, $N_+ = 1$	all six roles Vd0	4
CE0, $N_+ = 1$	a Vd1 or Vd2 role	1
CE0, $N_+ = 1$	no Vd1/Vd2 role and a T3-like role	3
CE0, $N_+ \geq 2$	all vertex types	3
CE1/CE2	$q \geq 3$	1
CE1/CE2, $N_+ = 0$	all six roles Vd0	2
CE1/CE2, $N_+ = 0$	a Vd1 or Vd2 role	1
CE1/CE2, $N_+ = 0$	no Vd1/Vd2 role and a T3-like role	2
CE1/CE2, $N_+ = 1$	all Vd0, zero gaps	4
CE1/CE2, $N_+ = 1$	all Vd0, a positive gap	2
CE1/CE2, $N_+ = 1$	exactly one T3-like, no Vd1/Vd2	2
CE1/CE2, $N_+ = 1$	exactly one Vd1/Vd2, no T3-like	1 / 1+2

TABLE 1. The complete terminal routing. Route j means Strategy j. In the last row the first entry is for CE1 and the second for CE2.

For the two all-Vd0 CE1/CE2 rows, the same six cells are organized by

	gr = 0	gr = 1	gr = 2
$N_+ = 0$	strict identity cycle	one-side endpoint chain	paired endpoint chain
$N_+ = 1$	nine-point support	five-row chain	paired endpoint chain.

The last column is CE2-only.

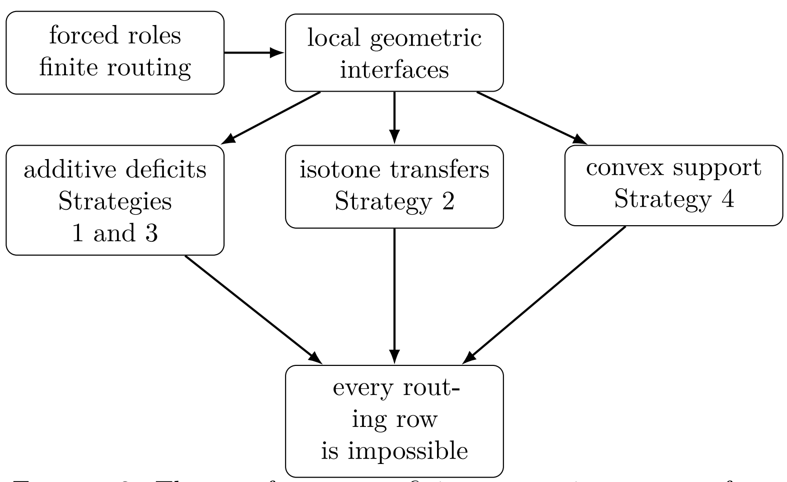


FIGURE 2. The proof separates finite geometric structure from three certificate classes. Strategy names are retained because they make the routing table readable.

1.4. **Four mechanisms.** Strategy 1 bounds one-dimensional traces on ∂H or S. Strategy 2 propagates lower reaches with monotone maps obtained from one local admissible set. Strategy 3 sums forced area losses. Strategy 4 forces nine points into the center role and applies an equilateral enclosure obstruction. Sections 3–6 contain these arguments; Section 7 performs the final audit.

2. Geometric Interfaces Used by the Proof

This section records the exact outputs of the structural and local calculations. Their derivations are postponed. The distinction is deliberate: the body uses only the statements below, while the appendices verify them on explicit real-variable domains.

2.1. Structural reduction.

Proposition 2.1 (Reader structural reduction)**.** *Every hypothetical cover has the seven distinct roles described in Section 1. The center has exactly one of the types* CE0, CE1, CE2*, every vertex role has exactly one of the types* Vd0, Vd1, Vd2, T3-like*, and the actual reaches determine exactly one row of Table 1. In the CE1/CE2 case the center contains exactly one radial midpoint and*

$$q = N_+ + m.$$

If the vertex roles alone cover the boundary, strict handoffs can be selected while preserving whether the actual supercritical pattern has none, exactly one, or at least two indices.

Proof. Distinctness is Lemma C.2; the center and vertex classifications are Propositions C.3 and C.5; the unique midpoint is Proposition C.11; the short-role identity is Lemma G.16; strict selection is Proposition C.9; and the disjoint routing assertion is Proposition C.13. Their geometric proofs are collected in Appendix C. □

2.2. The local admissible set.

Take unit vectors

$$u = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \qquad v = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right),$$

so u and v meet at 120° and $u + v$ is radial. For selected demands $0 \le a, b, c \le 1$, put

$$K(a, b, c) = \operatorname{conv}\{0, au, bv, c(u + v)\}$$

and define

$$\mathcal{A} = \{(a, b, c) \in [0, 1]^3 : K(a, b, c) \text{ lies in a closed unit equilateral triangle}\}.$$

Actual reaches dominate selected demands coordinatewise; they must not be substituted for them when N_+ is defined.

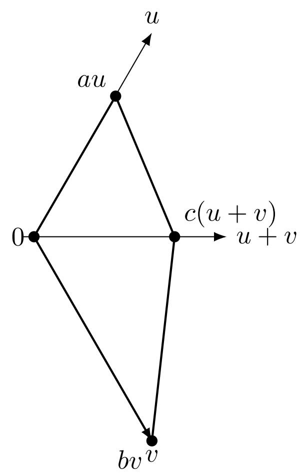


FIGURE 3. The local demand hull. Its least equilateral enclosure side is the single scalar controlling both Strategy 2 transfers and Strategy 4 witnesses.

For a compact set K, let R be rotation through $2\pi/3$ and define

$$\Lambda(K) = \frac{2}{\sqrt{3}} \min_{\|n\|=1} \sum_{j=0}^{2} h_K(\mathsf{R}^j n), \qquad h_K(n) = \max_{x \in K} \langle x, n \rangle.$$

Proposition D.1 says that $\Lambda(K)$ is the least side length of an enclosing closed equilateral triangle.

Proposition 2.2 (Explicit admissible-set description)**.** *Set*

$$s = a + b, \qquad D = \sqrt{a^2 + ab + b^2},$$

and, when $s > 0$ and $cs > ab$, set

$$R_a = \sqrt{a^2 - ac + c^2}, \qquad R_b = \sqrt{b^2 - bc + c^2}.$$

Define

$$L_{OA} = b + \max\{a, c\}, \qquad L_{BO} = a + \max\{b, c\},$$

$$L_{AC} = \begin{cases} (a^2 + bc)/R_a, & a \geq c, \\ c(a + b)/R_a, & a < c \leq a + b, \\ c^2/R_a, & c > a + b, \end{cases}$$

and let L_{CB} be obtained by exchanging a and b. Then the least side length of an equilateral triangle containing $K(a, b, c)$ is

$$L_{\min}(a, b, c) = \begin{cases} c, & a = b = 0, \\ D, & s > 0 \text{ and } cs \leq ab, \\ \min\{L_{OA}, L_{BO}, L_{AC}, L_{CB}\}, & s > 0 \text{ and } cs > ab. \end{cases}$$

Consequently

$$\boxed{\mathcal{A} = \{(a, b, c) \in [0, 1]^3 : L_{\min}(a, b, c) \leq 1\}.}$$

Proof. If $cs \le ab$, the radial point lies in $\operatorname{conv}\{0, au, bv\}$ and an explicit equilateral triangle of side D contains that hull. Otherwise the hull has four sides, and the finite-caliper principle reduces the minimum to the four edge-normal support values displayed above. The complete support calculation, the explicit triangular-hull enclosure, and the equivalent three polynomial cells are proved in Appendix E. □

2.3. **The transfer alphabet and its inequality interface.** For incoming defect $x = 1 - a$ and radial demand c, define

$$g_c(x) = \max\{y : (1-x, y, c) \in \mathcal{A}\}, \qquad \widehat{g}_c(x) = \min\{g_c(x), x\}.$$

For a map f, set

$$f^\vee(a) = 1 - f(1-a).$$

Thus g_c is the raw outgoing map, the hat imposes the nonsupercritical cap, and $\vee$ returns to incoming-reach coordinates. At zero radial demand,

$$g_0(x) > x \quad (0 < x < 1), \qquad \widehat{g}_0(x) = x.$$

The identity belongs to the hatted map.

If a center interval $J \subseteq [0,1]$ helps cover the handoff edge, let $\mathcal{R}_J(p)$ be the uncovered residual to the right of the component of $[0,p] \cup J$ containing 0, and define

$$g^\vee_{c,J}(a) = \mathcal{R}_J(g_c(1-a)), \qquad \widehat{g}^\vee_{c,J}(a) = \mathcal{R}_J(\widehat{g}_c(1-a)).$$

Finally, for $0 \le c < 1/2$, put

$$g^{\mathrm{sc}}_c = \sup_{\{x : g_c(x) > x\}} g_c(x) = \frac{c + \sqrt{c^2 - 8c + 4}}{2}.$$

Panels (a)–(e) of Figure 8 give a visual reading of this alphabet. In particular, x is a complementary coordinate, not necessarily a geometrically uncovered segment, and $g_c(x)$ is an extremal admissible output rather than the actual reach B.

Proposition 2.3 (Transfer interface)**.** *Suppose an actual row has incoming reach at least a, radial reach at least c, and outgoing reach B.*

(1) *Always,*

$$B \le g_c(1-a).$$

If the outgoing edge is center-free, then

$$A_{\mathrm{next}} \ge g^\vee_c(a).$$

If the row is nonsupercritical, then

$$B \le \widehat{g}_c(1-a),$$

and on a center-free outgoing edge

$$A_{\mathrm{next}} \ge \widehat{g}^\vee_c(a) \ge a.$$

(2) *With a center interval J on the handoff edge,*

$$A_{\mathrm{next}} \ge g^\vee_{c,J}(a),$$

and the hatted version holds for a nonsupercritical row.

(3) *If the row is strictly supercritical, then*

$$B < g_c^{\mathrm{sc}}.$$

Only when its outgoing edge is center-free may this be complemented to

$$A_{\mathrm{next}} > 1 - g_c^{\mathrm{sc}}.$$

(4) *If nondecreasing maps $\underline{\Phi}_j$ are pointwise lower bounds for the exact row transfers Φ_j, then replacing any slots in a composition by $\underline{\Phi}_j$ preserves a valid lower bound for the returned demand.*

Proof. Items 1 and 2 are Lemmas D.2 and D.4. Item 3 is (27), with the additional center-free hypothesis used only in (28). Item 4 is Lemma D.5. □

2.4. **One signed CE1/CE2 interface.** Normalize a positive center trace to $e_{0,1}$. The common signed parameters are

$$0 < R < 1, \qquad W = 1 - R,$$
$$E = \sqrt{1 - RW}, \qquad \eta = 1 - E, \qquad P = E(1 - E),$$

with center slacks $\alpha, \delta \geq 0$. Put

$$\Delta_R = P - \alpha - W\delta, \qquad \Delta_L = P - R\alpha - \delta.$$

Proposition 2.4 (Signed center interface)**.** *The normalization has $\Delta_R > 0$. The normalized trace and its possible companion have lengths*

$$\ell_R = \frac{\Delta_R}{R}, \qquad \ell_L = \frac{[\Delta_L]_+}{W},$$

and

$$T_C \text{ is CE1} \iff \Delta_L \leq 0, \qquad T_C \text{ is CE2} \iff \Delta_L > 0.$$

The six radial exits from O are

i	0	1	2	3	4	5
d_i^C	$E - \alpha - \delta$	δ/R	δ	$\min\{\alpha/R, \delta/W\}$	α	$\alpha/W.$

The full center boundary contribution is

$$L_{\partial H}(T_C) = \frac{\Delta_R}{R} + \frac{[\Delta_L]_+}{W}.$$

In particular,

$$L_{\partial H}(T_C) \leq P \leq \frac{\sqrt{3}}{2} - \frac{3}{4} \quad \text{in CE1},$$

whereas

$$L_{\partial H}(T_C) < \frac{1}{2} \quad \text{in CE2}.$$

Panels (f)–(j) of Figure 9 show the parameter dependencies and the companion-trace sign test. They also give two valid numerical center triangles, all six radial exits, and the scalar boundary budgets. The quantities η, P, Δ_R, and Δ_L are algebraic parameters; only ℓ_R, ℓ_L and the d_i^C are drawn as geometric lengths.

Proof. This is Proposition F.1 and Corollary F.3. Appendix F derives the side equations, traces, exits, and sign classification. □

3. Strategy 1: Trace Budgets

For a closed triangle T and a rectifiable target E, write

$$L_E(T) = \mathcal{H}^1(T \cap E).$$

If the seven open roles cover E, closure and subadditivity give

$$\mathcal{H}^1(E) \le L_E(T_C) + \sum_{i=0}^{5} L_E(T_i).$$

Strategy 1 contradicts this inequality on either ∂H or the full skeleton S.

Figure 4. The perimeter and the conditional full-skeleton target. The proof uses only the numerical trace caps stated below.

Proposition 3.1 (Trace-cap register)**.** *Under the hypotheses indicated in each row,*

role and target	*upper bound*
CE1/CE2 center on S	$L_S(T_C) < 3/2$
supercritical vertex role on S	$L_S(T_i) < 3/2$
vertex role with positive adjacent-arm support on S	$L_S(T_i) < 3/2$
nonsupercritical role with no adjacent-arm support on S	$L_S(T_i) \le 2$
CE1 center on ∂H	$L_{\partial H}(T_C) \le \sqrt{3}/2 - 3/4$
CE2 center on ∂H	$L_{\partial H}(T_C) < 1/2$
supercritical Vd0 role on ∂H	$A_i + B_i \le 2/\sqrt{3}$
Vd1 or Vd2 role on ∂H	$A_i + B_i < 1/2$
nonsupercritical Vd0 role on ∂H	$A_i + B_i \le 1$
T3-like role on ∂H	$A_i + B_i < 1$

Proof. The four skeleton bounds are Lemmas G.1, G.4, G.2, and G.3. The boundary bounds are the signed center formula and the vertex trace atlas in Appendix G. $\square$

Two abstract sums explain every Strategy 1 route. First, if there are $q \ge 3$ short roles, Proposition 3.1 gives

$$L_S(T_C) + \sum_i L_S(T_i) < \frac{3}{2} + q\frac{3}{2} + (6-q)2 = \frac{27-q}{2} \le 12,$$

contradicting $\mathcal{H}^1(S) = 12$.

Second, put

$$\theta = \frac{2}{\sqrt{3}} - 1.$$

Suppose a perimeter branch has:

- center contribution L_C;

- n supercritical Vd0 rows, each with contribution at most $2/\sqrt{3}$;
- $r \geq 1$ distinguished nonsupercritical rows with strict caps $q_1, \ldots, q_r < 1$;
- every remaining row with contribution at most 1.

If

$$L_C + n\theta \leq \sum_{j=1}^{r} (1 - q_j),$$

then the total perimeter contribution is strictly below six. This is the master perimeter deficit. The fourth hypothesis is part of the statement; it is supplied in every application by the vertex classification and the actual supercritical count.

Proposition 3.2 (Branches closed by Strategy 1)**.** *Strategy 1 closes:*

(1) CE0*,* $N_+ = 0$*;*
(2) CE0*,* $N_+ = 1$*, with a* Vd1/Vd2 *role;*
(3) *CE1/CE2 with* $q = N_+ + m \geq 3$*;*
(4) *CE1/CE2,* $N_+ = 0$*, with a* Vd1/Vd2 *role;*
(5) *CE1,* $N_+ = 1$*, with exactly one* Vd1/Vd2 *role and no* T3-like *role;*
(6) *the Vd2 neighboring-midpoint subcase of the CE2 hybrid row.*

Proof. The first row is the cyclic boundary sum. The third row is the skeleton sum above. The remaining rows substitute the center and distinguished vertex caps into the master perimeter deficit, with all unlisted rows bounded by one. The exact substitutions and strictness are Proposition G.11 and Corollary G.17. □

4. Strategy 2: Monotone Transfer Certificates

Strategy 2 turns boundary coverage into lower bounds propagated around the six rows. The geometric input is Proposition 2.3; the remaining work is to retain a small number of exact endpoint maps, replace other center-free nonsupercritical rows by lower relaxations, and prove one terminal inequality.

For maps listed in geometric row order, write

$$[\Phi_1 \mid \cdots \mid \Phi_r](x) = (\Phi_r \circ \cdots \circ \Phi_1)(x).$$

The leftmost slot acts first, and I denotes the identity map.

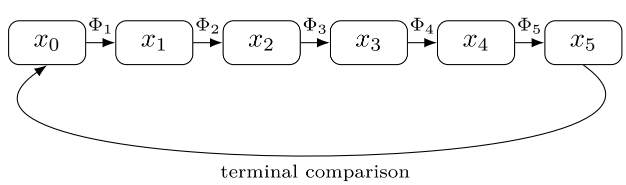


Figure 5. An exact row chain may be weakened slot by slot. The internal edges used by an identity relaxation are center-free; endpoint center intervals are first absorbed into residual demands.

4.1. **The all-Vd0 kernel.**

The two row counts and the active-gap rank form the single kernel

	gr = 0	gr = 1	gr = 2
$N_+ = 0$	I^6 cycle	two endpoints	paired endpoints
$N_+ = 1$	nine points	five-row chain	paired endpoints.

The last column is CE2-only. For $N_+ = 0$ and $\mathrm{gr} = 0$, the center has no trace on any missed handoff and the six incident vertex traces alone cover every edge. Thus

$$A_{i+1} > 1 - B_i \geq \widehat{g}^{\vee}_{C_i}(A_i) \geq A_i,$$

which gives the impossible strict cycle

$$A_0 < A_1 < \cdots < A_5 < A_0.$$

4.2. **Exact terminal certificates.** Put $k = \eta + \alpha + \delta$ for the signed center model.

Proposition 4.1 (All-Vd0 endpoint certificates)**.** *The following inequalities hold on the exact signed domains.*

(1) *Suppose $N_+ = 0$, all six roles are Vd0, and exactly the normalized right center trace contains a V-gap. Put*

$$s = \frac{k}{R}, \qquad q = R - \delta, \qquad c_1 = 1 - \frac{\delta}{R}, \qquad c_5 = 1 - \frac{\alpha}{W}.$$

Then the two exact hatted outgoing caps satisfy

$$\widehat{g}_{c_5}(1-s) + \widehat{g}_{c_1}(1-q) < 1.$$

The companion edge is center-free in CE1 and gap-free in CE2; the three middle rows lie on a center-free ordinary path and may be replaced by I^3.

(2) *Suppose the two CE2 traces both contain V-gaps and rows $T_1, \ldots, T_5$ are nonsupercritical Vd0. Put*

$$p = W - \alpha, \qquad q = R - \delta.$$

Then

$$\widehat{g}_{p/W}(1-p) + \widehat{g}_{q/R}(1-q) < 1.$$

Again the three middle rows form a center-free ordinary path.

Singleton gaps are included.

Proof. These are the one-side and paired-endpoint theorems in Appendix J, applied through the corrected boundary-path lemma, Lemma D.6. The endpoint residuals include every center contribution before the ordinary path is used. □

Proposition 4.2 (Five-row one-gap certificate)**.** *Assume all six roles are Vd0, $N_+ = 1$, and the normalized right center trace contains a V-gap; in CE2 assume the companion trace is gap-free. Put*

$$X = R - \delta, \qquad H = \frac{k}{2R}, \qquad m = \min\left\{\frac{\alpha}{R}, \frac{\delta}{W}\right\},$$

$$c_1 = 1 - \delta/R, \quad c_2 = 1 - \delta, \qquad c_3 = 1 - m,$$
$$c_4 = 1 - \alpha, \qquad c_5 = 1 - \alpha/W.$$

The actual rows satisfy

$$A_0 \geq Z, \qquad A_0 < 1 - H,$$

where

$$Z = [\widehat{g}^{\vee}_{c_1} \mid \widehat{g}^{\vee}_{c_2} \mid \widehat{g}^{\vee}_{c_3} \mid \widehat{g}^{\vee}_{c_4} \mid \widehat{g}^{\vee}_{c_5}](X).$$

Exact capped-map duality reduces the required contradiction $Z > 1 - H$ to

$$[\widehat{g}^{\vee}_{1-\alpha} \mid \widehat{g}^{\vee}_{1-m} \mid \widehat{g}^{\vee}_{1-\delta}](H) > 1 - X.$$

In CE1 the first two retained slots admit the affine lower bounds with coefficients $1-4\alpha$ *and* $1-5m$*; their output lies above* $e(\delta)$*, and the last threshold slot gives the strict target. In CE2,*

$$X > e(\alpha), \qquad \min\{e(\alpha), e(\delta)\} < H.$$

Therefore at least one of the α*- or* δ*-threshold slots may be chosen; every other slot is replaced by* I*. Both thresholds may be available, and no uniqueness is asserted. Reflection exchanges* (R, α) *with* (W, δ) *and reverses the row order.*

Proof. The actual-row induction and terminal diameter bound are Proposition F.5. The CE1 affine calculation is Proposition I.2; the CE2 alternatives are Proposition F.7. Every formal iterate is a lower bound for the corresponding actual incoming reach. □

Definition 4.3 (Support-isolated T3-like endpoint state)**.** This is a CE1/CE2 state with $N_+ = 0$, no Vd1/Vd2 role, and a normalized T3-like row T_0 supporting r_1. Remove from the two terminal boundary edges the connected components already covered by T_0 and the center. Let a_1^*, a_5^* be the remaining far-side incoming demands, and let c_1^*, c_5^* be the radial demands forced on the two endpoint rows by the corresponding radial components. Their exact formulas are those in Lemma L.4; the complete four-label branch proof is the incorporated supplement of Appendix M.

Proposition 4.4 (T3-like endpoint certificate)**.** *Every support-isolated T3-like endpoint state satisfies*

$$\widehat{g}_{c_5^*}(1-a_5^*) + \widehat{g}_{c_1^*}(1-a_1^*) < 1.$$

The exact endpoint function is audited on the four labels Full, $L, T_-, T_+^{\mathrm{hi}}$*; no unsupported common envelope replaces that audit. The three internal nonsupercritical rows lie on a center-free ordinary path and are replaced by* I^3*.*

Proof. This is Theorem M.1. □

Proposition 4.5 (Adjacent rescuer and Vd terminal certificates)**.** *The following exact geometric terminals hold.*

(1) *Let a T3-like or adjacent Vd1 rescuer have endpoint reach* a *and supported radial interval* $[c, u]$*, measured from the adjacent vertex toward* O*. Put*

$$\varepsilon = 1-u, \qquad \vartheta = \frac{a}{a+\varepsilon}.$$

Its local profile gives

$$a \le 1 - g_c^{\mathrm{sc}}, \qquad \vartheta \le 1 - g_c^{\mathrm{sc}}.$$

If radial isolation makes the center cover the O*-side interval of length* ε*, the opposite boundary demand is at least* g_c^{sc}*. The unique supercritical row has outgoing reach strictly below* g_c^{sc}*. The intervening four-row boundary path is center-free on its internal edges, so Lemma D.6 gives a contradiction.*

(2) *In the adjacent CE2 one-Vd1/Vd2 placement let* (a_0, b_0) *be the unique supercritical row, let*

$$I_L = [k/W, R+\alpha], \qquad I_R = [k/R, W+\delta],$$

and put

$$A = \mathcal{R}_{I_R}(b_0), \qquad H = \mathcal{R}_{I_L}(a_0).$$

Then

$$A + H < \frac{1}{2}, \qquad \delta < \frac{H}{4}.$$

The Vd row's possible endpoint on the target arm is below $1 - H$, while the next Vd0 row, which has boundary demands exceeding $1/2 + A$ and H, has own radial reach at most $1 - H/4$. Both are strictly below the center interval, which begins at $1 - \delta$. Hence that radial arm is uncovered.

(3) *In a nonadjacent placement set $T = \alpha + \delta$ and define the two center-residual boundary demands A, H as in Lemma H.7. Then*

$$A > T, \qquad H > T, \qquad d_\tau^C < \min\{A, H\}, \qquad c_\tau < 1 - \min\{A, H\}.$$

Thus the Vd radial trace and the center trace are disjoint.

(4) *In the remaining adjacent Vd1–supercritical placement, the pair can be replaced by two open nonsupercritical Vd0 roles while preserving every boundary and radial demand used by the all-Vd0 theorem.*

Proof. Items 1–3 are Corollary H.5 and Lemmas H.6 and H.7. Item 4 is Proposition M.2. □

branch	retained transfer data	terminal contradiction
$N_+ = 0$, all Vd0, gr $= 0$	I^6	strict cyclic ascent
$N_+ = 0$, all Vd0, gr $= 1$	two exact endpoints, I^3	endpoint sum < 1
CE2, gr $= 2$	paired exact endpoints, I^3	paired sum < 1
$N_+ = 0$, a T3-like role	exact four-label endpoints, I^3	endpoint sum < 1
$N_+ = 1$, all Vd0, one gap	exact five-row hatted chain	$Z > 1 - H$
one T3-like or adjacent Vd1	$1 - g_c^{\mathrm{sc}}$ and ordinary path	outgoing cap g_c^{sc}
adjacent Vd1/Vd2	two center residuals and quarter envelope	radial separation
nonadjacent Vd1/Vd2	two residuals and Vd corner margin	radial separation
Vd1–supercritical pair	explicit axis replacements	all-Vd0 contradiction

Table 2. The reader-facing Strategy 2 certificate register. Every identity slot lies on an internal edge for which center and nonincident traces have already been excluded.

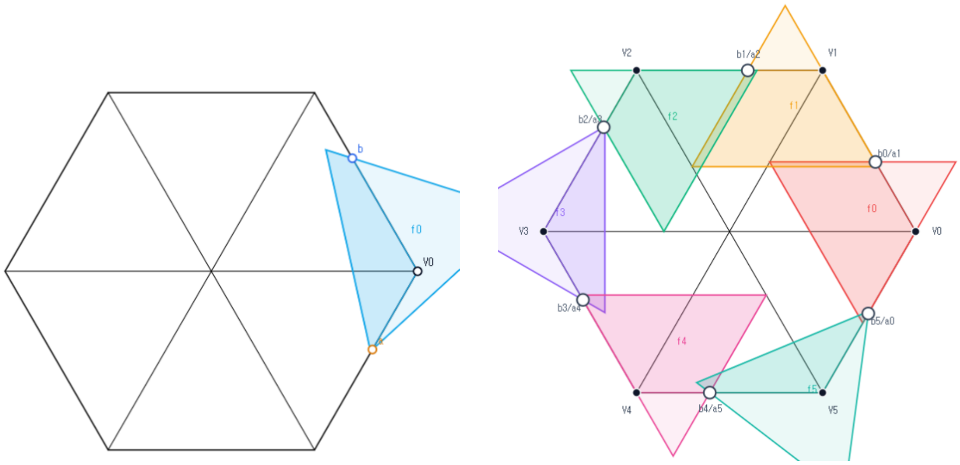
Proposition 4.6 (Branches closed by Strategy 2)**.** *Strategy 2 closes:*

(1) *the $N_+ = 0$ all-Vd0 and T3-like-without-Vd1/Vd2 branches for CE1 and CE2;*
(2) *every positive-gap $N_+ = 1$ all-Vd0 state;*
(3) *the exactly-one-T3-like $N_+ = 1$ state with no Vd1/Vd2 role;*
(4) *the complementary transfer placements in the CE2, $N_+ = 1$, exactly-one-Vd1/Vd2 branch.*

Proof. Use Propositions 4.1, 4.2, 4.4, and 4.5. The complete CE2 one-Vd placement assembly is Proposition N.1. All singleton-gap states are included. □

5. Strategy 3: Area Loss

Normalize the area of a unit equilateral triangle to one. Then H has area six. For a vertex role with selected adjacent boundary reaches a, b, let $\mathcal{L}(a, b)$ be the least normalized area forced outside the local hexagonal wedge.



(a) local loss (b) cyclic sum

FIGURE 6. Strategy 3. The picture supplies intuition only; the proof uses the three inequalities in Proposition 5.1.

Proposition 5.1 (Area-loss register)**.** *For every feasible selected pair,*

$$\mathcal{L}(a,b) \geq \min\{a,b\}^2.$$

If $a+b>1$*, then*

$$\mathcal{L}(a,b) \geq \max\{a,b\}^2.$$

For a T3-like *role,* $a+b\leq 1$*, and for* $0\leq m\leq 1/2$*,*

$$a,b\geq m \quad\Longrightarrow\quad \mathcal{L}(a,b)\geq 2m-4m^2.$$

Proof. These are Theorems O.2 and O.3. Appendix O reduces every orientation to two affine normal forms and proves the inequalities there. The body uses only the three displayed losses. □

Suppose strict boundary handoffs are selected and encoded by

$$(a_i,b_i)=(1-x_{i-1},x_i), \qquad 0<x_i<1.$$

Put

$$\mu=\min_i x_i, \qquad M=\max_i x_i.$$

Reflection and reversal replace x_i by $1-x_{-i-1}$, preserve every row loss, and interchange μ with $1-M$. Hence assume $\mu\leq 1-M$; then every coordinate x_i and $1-x_i$ is at least μ.

If at least two rows are supercritical, one ascent out of a minimum plateau contributes at least $(1-\mu)^2$, a second ascent contributes strictly more than $1/4$, and the remaining four rows contribute at least μ^2 each. The total loss is strictly larger than

$$4\mu^2+(1-\mu)^2+\frac{1}{4}=5\left(\mu-\frac{1}{5}\right)^2+\frac{21}{20}>1.$$

If there is exactly one supercritical row, its unique ascent runs from the minimum to the maximum. A distinct T3-like row then contributes at least $2\mu-4\mu^2$, the supercritical row contributes at least $(1-\mu)^2$, and the other four contribute at least μ^2 each. The sum is

$$(1-\mu)^2 + (2\mu - 4\mu^2) + 4\mu^2 = 1 + \mu^2 > 1.$$

Thus the six vertex triangles have total inside area strictly below five; the center triangle adds at most one.

Proposition 5.2 (Branches closed by Strategy 3)**.** *Strategy 3 closes:*

(1) *T_C is* CE0 *and $N_+ \geq 2$, with arbitrary vertex-role types;*
(2) *T_C is* CE0*, $N_+ = 1$, no role is* Vd1 *or* Vd2*, and at least one role is* T3-like*.*

Proof. In CE0 the six vertex roles cover the boundary, so Proposition 2.1 supplies strict selected handoffs preserving the actual supercritical pattern. The two cyclic sums above apply. This is Proposition O.5, whose local orientation calculations remain in Appendix O. ☐

6. Strategy 4: Direct Nine-Point Forcing

This section gives the geometric proof of the direct nine-point obstruction. The exact frontier calculations, moving-circle signs, Newton reduction, and adjacent cap overlaps are proved in Appendix P. The two mixed overlaps are completed by the formally incorporated exact certificate in Appendix Q. The body uses those inputs only through the forcing lemmas and the four-overlap proposition below.

Assume throughout that all six vertex roles are Vd0, that they cover ∂H, and that exactly one actual boundary row is supercritical. Rotate its index to 4. Lemma P.3 gives strict handoffs

$$x_3 < x_2 < x_1 < x_0 < x_5 < x_4, \tag{1}$$

and parameters

$$a = 1 - x_3, \qquad b = x_4, \qquad p = 1 - b, \qquad q = 1 - a \tag{2}$$

with

$$0 < a, b < 1, \qquad a + b > 1, \qquad a^2 + ab + b^2 < 1. \tag{3}$$

Every selected row has incoming demand at least p and outgoing demand at least q.

6.1. Direct forcing of the disk and three frontier points.

For $(x, y) \in [0, 1]^2$, write

$$c_{\max}(x, y) = \max\{c : (x, y, c) \in \mathcal{A}\}.$$

Let

$$c_* = c_{\max}(p, q), \qquad h = \frac{\sqrt{3}}{2}, \qquad \eta = h(1 - c_*).$$

The six radial points

$$D_i = (1 - c_*)V_i, \qquad 0 \leq i \leq 5,$$

are excluded from their matching roles by maximality, from the two adjacent roles by Vd0 locality, and from the other three roles by distance from their distinguished vertices. Lemma P.4 therefore gives

$$\mathcal{D}_\eta := \{X : \|X\| \leq \eta\} \subset U_C. \tag{4}$$

At the supercritical corner V_4, the strict AB-frontier consists of two circle pieces and two line pieces. The two line pieces meet at Q_0; their first intersections with the two actual adjacent-handoff unit circles are Q_- and Q_+. The exact moving-circle signs in Lemma P.5 exclude all three points from the two adjacent roles. Direct vertex-distance inequalities exclude the remaining three nondistinguished roles, and exact frontier membership excludes the supercritical role. Lemma P.6 therefore gives

$$Q_-, Q_0, Q_+ \in U_C. \tag{5}$$

Consequently the center role contains the compact witness set

$$K_{\text{wit}} = \mathcal{D}_\eta \cup \{Q_-, Q_0, Q_+\}. \tag{6}$$

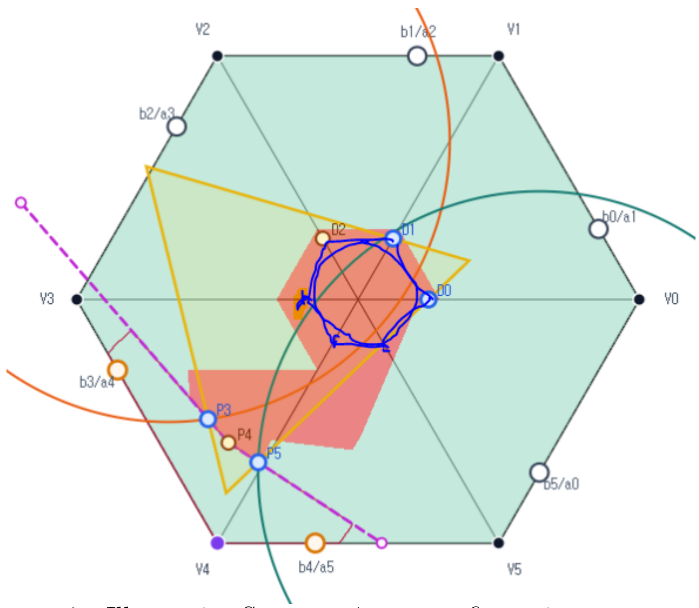


FIGURE 7. Illustrative Strategy 4 core configuration, not to scale. It is an example only; no exclusion, forcing, or enclosure inequality is inferred from the picture.

6.2. **Newton inner points.** The exact outer points Q_- and Q_+ contain two independent quadratic radicals. The construction in Lemma P.7 takes one Newton step inward on each frontier line. It gives points

$$A \in (Q_0, Q_-), \qquad B = Q_0, \qquad C \in (Q_0, Q_+)$$

whose coordinates are rational in a, b and the single radical

$$D = \sqrt{4(a^2 + ab + b^2) - 3}.$$

The same lemma gives

$$\widehat{K} := \operatorname{conv}(\mathcal{D}_\eta \cup \{A, B, C\}) \subseteq \operatorname{conv}(K_{\text{wit}}), \qquad \Lambda(K_{\text{wit}}) \geq \Lambda(\widehat{K}). \tag{7}$$

6.3. **The cap-chain lemma.** Use the enclosure gauge Λ from Proposition D.1, and let R denote rotation through $2\pi/3$. For a disk $\mathbb{D}(X, r)$ define its level-h cap

$$I(X, r) = \{n \in S^1 : \langle X, n\rangle + r \geq h\}.$$

Lemma 6.1 (Cyclic cap chain)**.** *Let K be compact and put*

$$S = K + \mathsf{R}K + \mathsf{R}^2K.$$

Suppose S contains the full R-orbits of

$$\mathbb{D}(A, 2\eta), \quad \mathbb{D}(B, 2\eta), \quad \mathbb{D}(C, 2\eta), \quad \mathbb{D}(W, \eta), \qquad W = C + \mathsf{R}A.$$

Assume that the rays

$$A,\ B,\ C,\ W,\ \mathsf{R}A$$

occur in this cyclic order, each corresponding cap is a connected arc of half-width less than $\pi/2$, and the four consecutive pairs

$$I(A, 2\eta) \leftrightarrow I(B, 2\eta), \qquad I(B, 2\eta) \leftrightarrow I(C, 2\eta),$$
$$I(C, 2\eta) \leftrightarrow I(W, \eta), \qquad I(W, \eta) \leftrightarrow I(\mathsf{R}A, 2\eta)$$

overlap across their intervening short sectors. Then $\Lambda(K) \geq 1$.

Proof. The four overlaps and the cyclic ray order make the five caps cover the sector from the ray of A to the ray of $\mathsf{R}A$. Their R-orbits cover the unit circle. For every unit normal n, one of the displayed disks therefore has support at least h in direction n. Since all those disks lie in S,

$$h_S(n) \geq h.$$

On the other hand,

$$h_S(n) = \sum_{j=0}^{2} h_K(\mathsf{R}^j n).$$

Proposition D.1 now gives $\Lambda(K) \geq 1$. □

6.4. **The four verified overlaps.** For $K = \widehat{K}$, the Minkowski sum in the cap-chain lemma contains the four required disk orbits: one witness point and two copies of the centered disk give the three radius-2η orbits, while $C + \mathsf{R}A$ and one disk copy give the radius-η orbit.

Proposition 6.2 (Four-overlap certificate)**.** *Assume* (3) *and $c_* > 2/3$. The Newton points above satisfy*

$$\|A\|, \|C\| > \eta,$$

the rays $A, B, C, W, \mathsf{R}A$ occur in that cyclic order, and all four consecutive cap overlaps in Lemma 6.1 hold.

Proof. The ray order, radius bounds, and the two equal-radius adjacent overlaps are proved analytically in Proposition P.10. The remaining two mixed overlaps are Theorem Q.3; its proof derives the exact Gram residuals, verifies the eight reduced integer polynomials by global positive-basis certificates, and then converts the residual signs back into common tangent support inequalities. These two parts supply precisely the four consecutive overlaps stated here. □

6.5. **Enclosure and the center-independent contradiction.**

Theorem 6.3 (Reader-facing witness enclosure)**.** *For every pair (a, b) satisfying (3),*

$$\Lambda(K_{\mathrm{wit}}(a, b)) \geq 1.$$

Proof. The disk alone gives

$$\Lambda(K_{\mathrm{wit}}) \geq \frac{3\eta}{h} = 3(1 - c_*),$$

so the result is immediate when $c_* \leq 2/3$. Assume $c_* > 2/3$. The Newton reduction (7), Proposition 6.2, and Lemma 6.1 give

$$\Lambda(K_{\mathrm{wit}}) \geq \Lambda(\widehat{K}) \geq 1.$$

□

Theorem 6.4 (Center-independent direct obstruction)**.** *There is no cover of H by open unit equilateral triangles $U_C, U_0, \ldots, U_5$ such that every U_i is a* Vd0 *role, the six vertex roles cover ∂H, and exactly one actual boundary row is supercritical.*

Proof. Direct forcing gives the compact containment $K_{\mathrm{wit}} \subset U_C$, whereas Theorem 6.3 gives $\Lambda(K_{\mathrm{wit}}) \geq 1$. A compact subset of an open unit equilateral triangle has positive clearance from all three sides and is therefore contained in a closed equilateral triangle of side strictly below one. This would give $\Lambda(K_{\mathrm{wit}}) < 1$, a contradiction. □

Proposition 6.5 (Branches closed by Strategy 4)**.** *Theorem 6.4 closes*

(1) *the* CE0*, $N_+ = 1$, all-*Vd0 *branch;*
(2) *the zero-gap* CE1 *and* CE2*, $N_+ = 1$, all-*Vd0 *branches.*

Proof. In the CE0 branch, a boundary point missed by the six vertex roles would belong to the open center role and create a positive center trace, contrary to CE0. Thus the vertex roles cover ∂H. In the CE1 and CE2 branches, zero gaps mean exactly that the adjacent vertex traces cover every boundary edge. The exact-one hypothesis supplies the unique supercritical row in all three cases, so Theorem 6.4 applies. □

7. Exhaustive Assembly

Proof of Theorem 1.1. Assume that seven open unit equilateral triangles cover H. Proposition 2.1 assigns the cover to exactly one row of Table 1, with N_+ computed from actual maximal reaches and, in the CE1/CE2 branch, $q = N_+ + m$.

The Strategy 1 rows contradict Proposition 3.2. The Strategy 3 rows contradict Proposition 5.2. The all-Vd0, T3-like, and ordinary transfer rows contradict Proposition 4.6; their exact retained maps and terminal inequalities are recorded in Table 2. The complete CE2 exactly-one-Vd1/Vd2 row, including the neighboring-midpoint length subcase and every complementary placement, contradicts Proposition N.1. The zero-gap all-Vd0 rows contradict Proposition 6.5; its two mixed cap overlaps are Theorem Q.3.

No routing row is omitted. Singleton gaps remain gaps, every center-assisted handoff is treated by a residual map, every identity path has center-free internal edges, and no actual criticality classification is inferred from a selected demand pair. Hence every row of Table 1 is impossible, a contradiction. □

Corollary 1.2 follows from Proposition C.1.

Appendix A. Technical Verification Roadmap

The body gives the complete geometric routing and identifies the exact terminal statement used in every branch. The following appendices establish those interfaces. The proof is therefore the body together with these verification modules and the formally incorporated electronic supplements; the body alone is not asserted to prove the calculation lemmas.

module	principal variables	output used in the body
structural reduction	finite incidence and trace data	role types, strict handoffs, exhaustive routing
local enclosure	$a, b, c \in [0, 1]$	explicit $\mathcal{A}$, polynomial cells, radial envelope
signed center	R, α, δ	CE sign, traces, exits, residuals
trace budgets	one- and two-variable caps	perimeter and skeleton deficits
transfer calculus	monotone maps and residual intervals	actual-row transfers, center-free path budgets, relaxed compositions
endpoint certificates	compact signed domains	one-side, paired, five-row, and four-label terminal inequalities
special placements	Vd/T3 local parameters	rescuer obstruction, radial separation, explicit Vd0 replacement
area loss	$a, b, t \in [0, 1]$	local square losses and cyclic sums
nine-point geometry	a, b on the strict frontier	forced witnesses, Newton placement, cap chain, adjacent overlaps
mixed-overlap certificate	U, z, D and exact sparse polynomials	two mixed cap intersections by exact symbolic verification

Table 3. Verification modules supporting the reader-facing proof.

Except for the finite geometric classification and placement reductions, the appendices pass from geometry to inequalities on explicit real-variable domains. Polynomial expansions, root selection, rational parametrizations, monotonicity checks, Gram reductions, and Bernstein identities occur only below or in the exact electronic supplements identified by repository path, Git blob, and authenticated transcript. A machine verifier is used as part of the proof only where its exact input and algorithm are formally incorporated; a bare script invocation or claimed output is not treated as a proof.

Appendix B. Illustrated Notation Atlas

This appendix supplies examples for the two notation-heavy interfaces in Sections 2. Each panel is a separate TikZ source so that it can be retained or removed independently. The diagrams explain the definitions and hypotheses; they are not substitutes for the inequalities in Propositions 2.3 and 2.4.

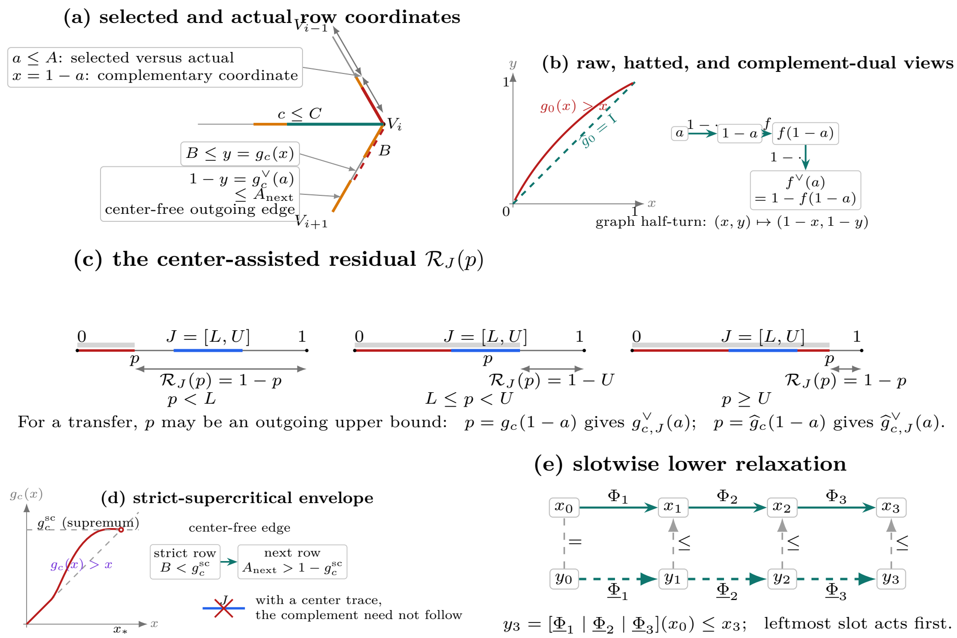


FIGURE 8. Reading the transfer alphabet. Panel (a) distinguishes selected demands from actual reaches: $x = 1 - a$ is a coordinate defect, and $g_c(x)$ is an extremal admissible output rather than the actual B. Panel (b) is the exact zero-radial graph and shows that the identity is $\widehat{g}_0$, not raw g_0; the right-hand diagram explains $\vee$ by complementing both coordinates. Panel (c) shows that only the component attached to 0 enters $\mathcal{R}_J$. Panel (d) is schematic: the strict-supercritical supremum is not attained, and its complementary next-reach bound needs a center-free edge. Panel (e) shows where nondecreasing lower relaxations enter a composition.

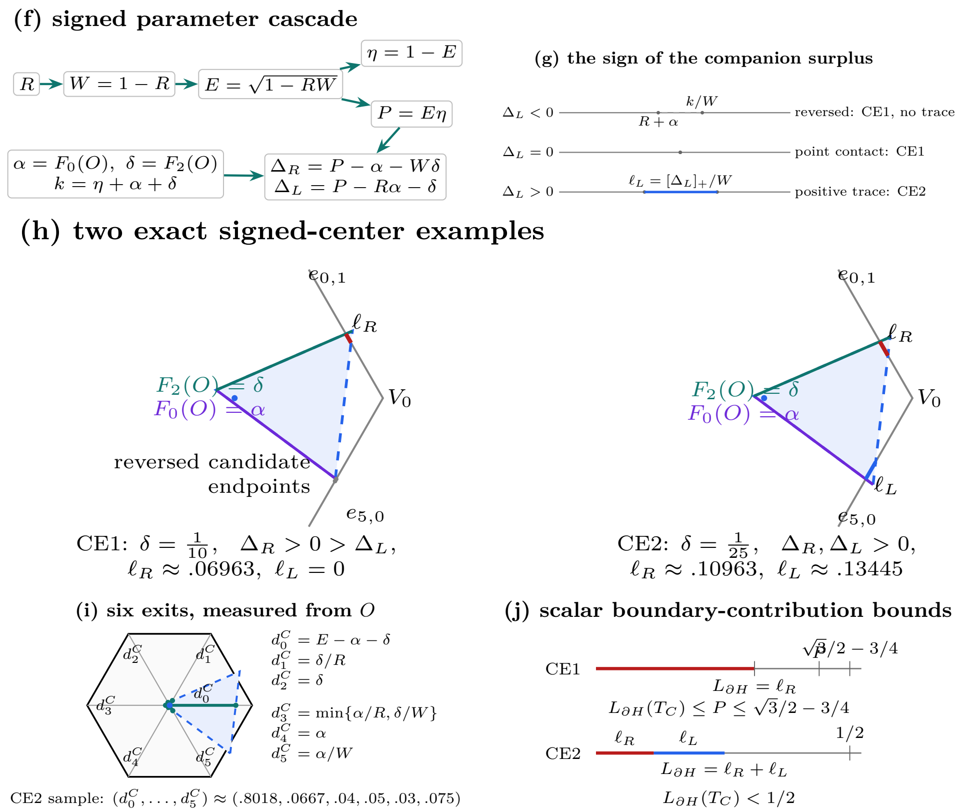


FIGURE 9. The signed CE1/CE2 interface. Panel (f) records which parameters are derived and identifies α, δ as affine side-slack values at O. Panel (g) explains the positive part $[\Delta_L]_+$ and keeps the $\Delta_L = 0$ point-contact state in CE1. Panel (h) uses the valid samples $R = 3/5$, $W = 2/5$, $\alpha = 3/100$, with $\delta = 1/10$ for CE1 and $\delta = 1/25$ for CE2; rounded coordinates are used only to draw the exact unit triangles. Panel (i) maps each exit d_i^C to its ray and measures it from O. Panel (j) consists of scalar comparison bars: they are not locations on ∂H, and $L_{\partial H}(T_C)$ is not the perimeter of T_C.

APPENDIX C. STRUCTURAL REDUCTION TO FINITE CASES

Throughout, indices are taken in $\mathbb{Z}/6\mathbb{Z}$. Put

$$O = (0,0), \qquad V_i = \left(\cos\frac{i\pi}{3}, \sin\frac{i\pi}{3}\right), \qquad H = \mathrm{conv}\{V_0, \ldots, V_5\}.$$

Thus H is the regular hexagon of side length one. We use the notation

$$e_{i,i+1} = [V_i, V_{i+1}], \qquad r_i = [O, V_i], \qquad M_i = \frac{1}{2}V_i,$$

and, when needed,

$$D = \bigcup_{i=0}^{5} r_i, \qquad S = \partial H \cup D, \qquad S_{1/2} = \partial H \cup \{O, M_0, \dots, M_5\}.$$

The arms r_i meet only at O, up to sets of one-dimensional measure zero, and consequently

$$\mathcal{H}^1(\partial H) = 6, \qquad \mathcal{H}^1(D) = 6, \qquad \mathcal{H}^1(S) = 12. \tag{8}$$

For $L > 0$, $H_L := LH$ denotes the concentric regular hexagon of side length L. All symmetries used below belong to the dihedral group D_6; applying a symmetry always transports the indices, orientations, and incoming and outgoing labels together.

C.1. **Open, closed, and scaled formulations.**

Proposition C.1 (Compactness and scaling equivalence). *The following assertions are equivalent.*

1. *The set H is covered by seven open equilateral triangles of side length* 1*.*
2. *For some $\varepsilon > 0$, the set H is covered by seven closed equilateral triangles of side length $1 - \varepsilon$.*
3. *For some $L > 1$, the set H_L is covered by seven closed equilateral triangles of side length* 1*.*

Proof. Suppose that open unit triangles $U_1, \dots, U_7$ cover H. Define

$$m_j(p) = \operatorname{dist}(p, \mathbb{R}^2 \setminus U_j), \qquad m(p) = \max_{1 \le j \le 7} m_j(p).$$

The continuous function m is positive on the compact set H, so $\eta := \min_{p \in H} m(p) > 0$. Choose

$$0 < \rho < \min\left\{\eta, \frac{\sqrt{3}}{6}\right\}.$$

Moving each side of U_j inward through distance ρ gives a closed equilateral triangle K_j of side $1 - 2\sqrt{3}\rho$. Every $p \in H$ has $m_j(p) \ge \eta > \rho$ for some j, and hence belongs to K_j. This proves (1)⇒(2), with $\varepsilon = 2\sqrt{3}\rho$.

Conversely, enlarging a closed triangle of side $1 - \varepsilon$ about its incenter to an open triangle of side 1 preserves containment, proving (2)⇒(1). Finally, dilation by $(1 - \varepsilon)^{-1}$ identifies (2) with (3), with

$$L = (1 - \varepsilon)^{-1} > 1.$$

□

We conduct the geometric argument in the open side-one model. Denote the original open triangles by U_C and U_i for $0 \le i \le 5$, and put

$$T_C = \overline{U_C}, \qquad T_i = \overline{U_i}.$$

The distinction between U_i and T_i will be maintained: closure may add endpoints or tangencies, although it does not change area or one-dimensional trace measure.

Lemma C.2 (Distinct roles)**.** *The seven triangles can be labelled so that*

$$O \in U_C, \qquad V_i \in U_i \quad (0 \le i \le 5),$$

and these seven roles are distinct. Consequently

$$O \in \operatorname{int}(T_C), \qquad V_i \in \operatorname{int}(T_i).$$

Proof. The seven distinguished points $O, V_0, \dots, V_5$ are pairwise at distance at least 1. Any two points of an open unit equilateral triangle are at distance strictly less than its diameter 1. Thus one open triangle contains at most one distinguished point. Selecting a covering triangle for each of the seven points gives an injection from seven points to seven triangles, hence a bijection and the asserted labelling. □

C.2. **The center classification.** For a closed center role define

$$N_E(T_C) = |\{i : T_C \cap e_{i,i+1} \text{ contains a positive-length interval}\}|\,.$$

Point contacts, endpoint contacts, and tangencies do not contribute. We say that T_C has type CE0, CE1, or CE2 according as $N_E(T_C)$ is 0, 1, or 2.

Proposition C.3 (Exhaustive center types)**.** *Every center role has exactly one of the types CE0, CE1, and CE2. Equivalently,*

$$N_E(T_C) \le 2.$$

Proof. Among any three edges of the six-cycle ∂H, two have no common endpoint. It is therefore enough to show that relative-interior points of two nonadjacent edges are more than unit distance apart. Up to symmetry there are two cases. For $0 < t, s < 1$, put

$$x = (1-t)V_0 + tV_1.$$

If $y = (1-s)V_2 + sV_3$, direct calculation gives

$$\|x-y\|^2 - 1 = (1-t)(2-t) + s(s+t) > 0.$$

If instead $y = (1-s)V_3 + sV_4$ and $u = t+s$, then

$$\|x-y\|^2 - 1 = (u-1)^2 + 2 > 0.$$

A unit equilateral triangle has diameter 1. It therefore cannot contain relative-interior points on two nonadjacent hexagon edges. Three positive-length edge traces would supply such a pair, a contradiction. □

C.3. **Vertex types and the T3-like normalization.** For an original vertex role T_i, let

$$o(T_i) = |\{\text{vertices of } T_i \text{ outside } H\}|$$

and let $n(T_i)$ be the number of adjacent arms r_{i-1}, r_{i+1} having positive-length intersection with T_i. We use the names

type	(o, n)
Vd0	$(1,0), (2,0), (3,0)$
Vd1	$(1,1)$
Vd2	$(1,2)$
T3-like	$(2,1)$.

Lemma C.4 (Local wedge)**.** *Every original vertex role has $1 \le o \le 3$. If it has positive-length intersection with one adjacent arm, at least one of its vertices belongs to H; if it has positive-length intersection with both adjacent arms, at least two of its vertices belong to H.*

Proof. Normalize to $i = 0$ and write

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0).$$

Then $V_0 = (0, 0)$,

$$\|(x, y)\|^2 = x^2 + y^2 - xy,$$

and, within the closed unit ball about V_0, membership in H is equivalent to membership in the wedge

$$W = \{(x, y) : x \geq 0, y \geq 0\}.$$

Indeed,

$$x^2 + y^2 - xy = \left(y - \frac{x}{2}\right)^2 + \frac{3x^2}{4} = (x - y)^2 + xy$$

shows that $x, y \geq 0$ and norm at most one imply the remaining hexagon inequalities. The adjacent arms are

$$r_1 = \{(1, y) : 0 \leq y \leq 1\}, \qquad r_5 = \{(x, 1) : 0 \leq x \leq 1\}.$$

Suppose $T_0 \cap r_1$ has positive length, and let m be the largest x-coordinate of a vertex of T_0. If $m > 1$, a maximizing vertex (m, y) lies in the unit ball and hence satisfies $0 < y < 1$; it belongs to H. If $m = 1$, positive-length contact with the supporting line $x = 1$ forces a side of T_0 to lie there, and both endpoints satisfy $1 + y^2 - y \leq 1$, hence belong to H. The same argument applies to r_5. If both arms are met but only one triangle vertex belonged to H, that vertex would have to have both $x > 1$ and $y > 1$. This is impossible because $(x - y)^2 + xy > 1$.

Finally, if all three triangle vertices belonged to the convex set H, then $T_0 \subset H$. This is incompatible with the extreme point V_0 lying in the interior of T_0. Hence $o \geq 1$. □

Proposition C.5 (Exhaustive vertex types)**.** *Every original vertex role has exactly one of the four types Vd0, Vd1, Vd2, and T3-like.*

Proof. We have $n \in \{0, 1, 2\}$. Lemma C.4 gives $o \leq 2$ when $n \geq 1$, and $o \leq 1$ when $n = 2$. Together with $o \geq 1$, the possibilities are

n	possible o
0	$1, 2, 3$
1	$1, 2$
2	$1,$

which are precisely the four named classes. □

The next normalization is useful for trace estimates, but its final warning is essential.

Proposition C.6 (T3-like closed-trace domination)**.** *Let T be the closure of an original T3-like V_i-role. There is a translate $\widehat{T}$ of T, with the same orientation and side length, such that*

$$V_i \in \operatorname{relint}(\textit{a side of } \widehat{T}), \qquad T \cap H \subsetneq \widehat{T} \cap H,$$

and $\widehat{T}$ remains of type $(o, n) = (2, 1)$. Every old open interior point in $H \setminus \{V_i\}$ remains an interior point of $\widehat{T}$. The point V_i itself becomes a boundary point, so $\widehat{T}$ is a closed-trace majorant and is not a replacement open role.

Proof. Use the wedge coordinates of Lemma C.4. Set

$$D_t = 1 - t + t^2, \qquad z = \sqrt{D_t}, \qquad 0 \le t \le 1.$$

After relabelling triangle vertices, every orientation belongs to one of the following two families:

	first side vector	second side vector
I	$z^{-1}(1,t)$	$z^{-1}(1-t,1)$
II	$z^{-1}(t,-(1-t))$	$z^{-1}(1,t)$.

For Type I write

$$U = \alpha + y - tx, \qquad V = \beta + x - (1-t)y, \qquad T = \{U \ge 0, V \ge 0, U + V \le z\}.$$

Here $\alpha, \beta > 0$ and $\alpha + \beta < z$. Direct inversion shows that if exactly two triangle vertices are outside W, the sole wedge vertex has both coordinates strictly below 1: in the two possible cases its coordinates are bounded respectively by $(1-t)/z$ and t/z, with the endpoint strictness supplied by $\alpha > 0$ or $\beta > 0$. The support argument in Lemma C.4 then excludes positive-length contact with either $x = 1$ or $y = 1$. Thus Type I cannot be T3-like.

For Type II write

$$U = \alpha + (1-t)x + ty, \qquad V = \beta + tx - y, \qquad T = \{U \ge 0, V \ge 0, U + V \le z\}.$$

Reflecting in $x = y$ if necessary, assume the supported adjacent arm is r_1. The two outside vertices have respectively a negative x- and a negative y-coordinate, while the unique wedge vertex is

$$Q = \left(\frac{z - \alpha - t\beta}{D_t}, \frac{t(z-\alpha) + (1-t)\beta}{D_t} \right).$$

The endpoint cases $t = 0, 1$ give no positive adjacent interval, so $0 < t < 1$. The exact axis reaches are

$$A = \beta, \qquad B = z - \alpha - \beta,$$

and T3-like support is equivalent to

$$Q_x > 1, \qquad Q_y \le 1.$$

Define $\widehat{T}$ by replacing α with 0 and keeping t, β fixed. Since the linear map $(x, y) \mapsto (U, V)$ is nonsingular, this changes only the translation. At the origin, $\widehat{U} = 0$ and $0 < \widehat{V} = \beta < z$, so V_0 is in the relative interior of a side. For $(x, y) \in W$,

$$\widehat{U} = (1-t)x + ty \ge 0, \qquad \widehat{V} = V, \qquad \widehat{U} + \widehat{V} = U + V - \alpha.$$

Thus $T \cap W \subset \widehat{T} \cap W$. The old x-axis reach is B and the new one is $B + \alpha$; because $B < 1$ and $\alpha > 0$, the inclusion is strict inside H.

The two outside vertices remain outside, whereas V_0 and the new wedge vertex lie in H, so $o = 2$. Moreover $\widehat{Q}_x > Q_x > 1$. From $Q_x > 1$ one gets $t\beta < z - \alpha - D_t$, and hence

$$tD_t(1 - \widehat{Q}_y) > D_t(1 - z) + (1-t)\alpha > 0.$$

Thus $\widehat{Q}_y < 1$ and $n = 1$. Finally, for every nonzero point of W, $\widehat{U} > 0$; the displayed slack relations show that every old interior point other than the origin remains interior. $\square$

C.4. **Reach coordinates, supercritical rows, and gaps.** For the original vertex role U_i, define its actual maximal incoming, outgoing, and radial reaches by

$$A_i = \sup\{t \in [0,1] : V_i + t(V_{i-1} - V_i) \in U_i\},$$
$$B_i = \sup\{t \in [0,1] : V_i + t(V_{i+1} - V_i) \in U_i\},$$
$$C_i = \sup\{t \in [0,1] : V_i + t(O - V_i) \in U_i\}.$$

Thus the closure T_i contains the corresponding endpoints. We reserve lowercase (a_i, b_i, c_i) for selected or prescribed lower-bound demands whenever actual and selected values occur together. A realizing role then satisfies

$$A_i \geq a_i, \qquad B_i \geq b_i, \qquad C_i \geq c_i.$$

An actual row is supercritical if $A_i + B_i > 1$, and

$$N_+ = |\{i : A_i + B_i > 1\}|\,. \tag{9}$$

This definition never counts a smaller demand pair in place of the actual row.

Parametrize $e_{i,i+1}$ by

$$X_i(x) = V_i + x(V_{i+1} - V_i), \qquad 0 \leq x \leq 1.$$

The two incident open traces are

$$U_i \cap e_{i,i+1} = [0, B_i), \qquad U_{i+1} \cap e_{i,i+1} = (1 - A_{i+1}, 1]$$

in this coordinate. A *boundary gap* on this edge is the nonempty closed complement between these two traces,

$$[B_i, 1 - A_{i+1}] \qquad \text{when } B_i \leq 1 - A_{i+1}.$$

In particular, equality gives a singleton gap. This convention records the actual open traces; a point missing from both open roles is not erased merely because both closures contain it.

Lemma C.7 (Exhaustiveness of the gap split)**.** *On each edge the complement of the two incident vertex-role traces is either empty or one closed interval; a singleton interval counts as a gap. Every edge having a gap has positive-length intersection with the open center role. Consequently a* CE0 *center permits no gaps, a* CE1 *center permits at most one, and a* CE2 *center permits at most two.*

Proof. The two incident traces are intervals issuing from the two endpoints, so their complement is exactly the interval displayed above when nonempty. A nonincident vertex role cannot contain a relative-interior point of the edge: that point is more than unit distance from the role's distinguished vertex. Thus every point of a gap lies in U_C. Because U_C is open, its intersection with that edge contains a relative neighborhood and hence has positive length. Different gap edges therefore give different edges counted by $N_E(T_C)$. Proposition C.3 gives the three stated bounds, including singleton gaps. □

Lemma C.8 (T3-like rows are nonsupercritical)**.** *Every original T3-like row satisfies*

$$A_i + B_i \leq 1. \tag{10}$$

In particular, it is not counted by N_+.

Proof. The proof of Proposition C.6 showed that a T3-like role has Type II orientation with

$$A_i = \beta, \qquad B_i = z - \alpha - \beta, \qquad z = \sqrt{1 - t + t^2} \le 1.$$

Thus $A_i + B_i = z - \alpha \le 1$. The same conclusion holds after reflection. □

We record the precise transfer from actual rows to strict handoff demands. Its hypothesis that the six vertex roles themselves cover the boundary is part of the statement.

Proposition C.9 (Strict boundary handoffs)**.** *Assume that $U_0, \dots, U_5$ cover ∂H. On each edge choose*

$$x_i \in (1 - A_{i+1}, B_i)$$

and define

$$a_i = 1 - x_{i-1}, \qquad b_i = x_i.$$

Then $0 < a_i < A_i$, $0 < b_i < B_i$, the two selected anchors lie in U_i, and

$$a_i^2 + a_i b_i + b_i^2 < 1.$$

Moreover:

(1) *if exactly one actual row, indexed by p, is supercritical, every strict selection has exactly one selected supercritical row, also indexed by p;*
(2) *if at least two actual rows are supercritical, some simultaneous strict selection has at least two selected supercritical rows.*

Proof. Because V_i is interior to a diameter-one closed triangle,

$$0 < A_i < 1, \qquad 0 < B_i < 1.$$

No nonincident vertex role contains a relative-interior point of $e_{i,i+1}$, since such a point is more than unit distance from its distinguished vertex. The two displayed incident traces therefore cover the edge and, being relatively open and nonempty, overlap. Hence each interval $(1 - A_{i+1}, B_i)$ is nonempty. The anchor and distance assertions follow from the choice and from the fact that two points of U_i are at distance strictly less than one.

Put $\ell_i = 1 - A_{i+1}$ and $u_i = B_i$. Then

$$A_i + B_i > 1 \iff \ell_{i-1} < u_i, \qquad a_i + b_i > 1 \iff x_{i-1} < x_i. \tag{11}$$

If actual row i is nonsupercritical, then

$$x_i < u_i \le \ell_{i-1} < x_{i-1},$$

so every selected version of it is strictly subcritical. Also

$$\sum_{i=0}^{5} (a_i + b_i) = \sum_{i=0}^{5} (1 - x_{i-1} + x_i) = 6. \tag{12}$$

If p is the sole actual supercritical index, the other five selected sums are strictly below one; equation (12) forces the pth sum above one. This proves (1).

For (2), first take two nonadjacent supercritical indices p, q. For each $r \in \{p, q\}$ choose

$$\ell_{r-1} < z_r < u_r,$$

then choose

$$x_{r-1} \in (\ell_{r-1}, \min\{u_{r-1}, z_r\}), \quad x_r \in (\max\{\ell_r, z_r\}, u_r).$$

The two pairs of variables are disjoint and give ascents at p, q. Any set of at least three indices on a six-cycle contains two nonadjacent indices. It remains to consider exactly two adjacent supercritical indices $p, p+1$. For the other four nonsupercritical rows, successive inequalities $\ell_i < u_i \le \ell_{i-1}$ give

$$\ell_{p-1} < \ell_{p+1}.$$

Together with supercriticality at $p, p+1$, this implies

$$\max\{\ell_{p-1}, \ell_p\} < \min\{u_p, u_{p+1}\}.$$

Choose x_p between these bounds and then choose $x_{p-1} < x_p < x_{p+1}$ in their respective overlap intervals. By (11), both adjacent rows are selected supercritical. $\square$

C.5. **Midpoint forcing.** We first need a local fact for vertex roles. Take unit vectors

$$u = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \qquad v = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right).$$

Lemma C.10 (Self-midpoint obstruction)**.** *Let a closed unit triangle contain*

$$0, \qquad au, \qquad bv, \qquad \frac{u+v}{2}.$$

Then $a + b \le 1$. Consequently, for every vertex role,

$$M_i \in T_i \quad \Longrightarrow \quad A_i + B_i \le 1.$$

If the two demanded boundary points and the midpoint are contained with a positive margin, the conclusion is strict.

Proof. Suppose $a + b > 1$ and set

$$K = \operatorname{conv}\left\{0, au, \frac{u+v}{2}, bv\right\}.$$

For outward unit normals n_0, n_1, n_2 separated by $120°$, the least equilateral triangle with those normals and containing K has side length

$$\frac{2}{\sqrt{3}} \sum_{j=0}^{2} h_K(n_j).$$

The finite-caliper theorem, Theorem P.1, reduces its minimum to an outward normal of one of the four edges of K.

For the edge from 0 to au, direct projection gives

$$S_{0A} = \max\left\{\frac{\sqrt{3}}{4}, \frac{\sqrt{3}a}{2}\right\} + \frac{\sqrt{3}b}{2} > \frac{\sqrt{3}}{2};$$

if $a \ge 1/2$ this is $(\sqrt{3}/2)(a+b)$, and if $a < 1/2$ then $b > 1/2$. The edge through bv is symmetric. For the edge from au to $(u+v)/2$, put $D_a = \sqrt{4a^2 - 2a + 1}$. Projection gives

$$S_{AM} = \begin{cases} \dfrac{\sqrt{3}(a+b)}{2D_a}, & a \le \frac{1}{2}, \\ \dfrac{\sqrt{3}(2a^2+b)}{2D_a}, & a > \frac{1}{2}. \end{cases}$$

The first value is strictly above $\sqrt{3}/2$ because $D_a \le 1$. In the second case, $b > 1-a$ and

$$(2a^2 - a + 1)^2 - D_a^2 = a^2(2a-1)^2 \ge 0,$$

so again $S_{AM} > \sqrt{3}/2$. The fourth edge is symmetric. Every enclosing equilateral triangle therefore has side greater than one, a contradiction.

If equality $a+b=1$ occurred while all relevant points had positive margin, both boundary demands could be increased slightly; the closed conclusion would then be violated. □

The corresponding center fact is exact and uses the interior condition at O.

Proposition C.11 (Unique center midpoint)**.** *Let T_C be a closed unit equilateral triangle with $O \in \operatorname{int}(T_C)$. If T_C has positive-length intersection with a boundary edge of H, then it contains exactly one of $M_0, \dots, M_5$.*

Proof. Normalize the positive trace by a dihedral symmetry. Proposition F.1 gives a single signed side-slack model for both CE1 and CE2 and proves directly that its midpoint set is $\{M_0\}$. Undoing the normalization proves the assertion. □

Remark C.12. The hypothesis $O \in \operatorname{int}(T_C)$ cannot be weakened to $O \in T_C$: the triangle $\operatorname{conv}\{O, V_0, V_1\}$ has a boundary-edge overlap and contains both M_0 and M_1.

Proposition C.13 (Exhaustive structural reduction)**.** *Every hypothetical cover by seven open unit equilateral triangles admits the distinct center and vertex roles used in Table 1, and its actual reaches determine exactly one row of that table.*

Proof. Lemma C.2 gives the seven roles. Propositions C.3 and C.5 give the exhaustive and mutually exclusive center and vertex types. The actual reaches determine exactly one of $N_+ = 0$, $N_+ = 1$, and $N_+ \ge 2$. For CE0 these data and the vertex classification give the five displayed rows.

Now suppose the center is CE1 or CE2. It contains exactly one radial midpoint by Proposition C.11. Lemma G.16 gives $q = N_+ + m$, and Lemma G.5, together with the disjoint count, shows that $N_+ \ge 2$ forces a third short role. Hence every such branch either enters the first CE1/CE2 row or has $q \le 2$ and $N_+ \le 1$. If $N_+ = 0$, the vertex classification gives, in order, all Vd0, at least one Vd1/Vd2, or no such role and at least one T3-like. If $N_+ = 1$, then $q \le 2$ implies $m \le 1$: the roles are all Vd0, exactly one T3-like, or exactly one Vd1/Vd2. In the all-Vd0 case, Lemma C.7 supplies the exhaustive zero-gap/positive-gap split. These alternatives are disjoint, proving the claim. □

Appendix D. Universal Enclosure and Transfer Calculus

This section collects the geometry shared by the transfer and direct-witness arguments. The same equilateral enclosure functional governs the local admissible set of Strategy 2 and the forced witness set of Strategy 4. The transfer statements below distinguish three situations that must not be conflated: a center-free edge, an edge with a specified center interval, and a free strict-supercritical outgoing envelope.

D.1. **The equilateral enclosure gauge.** Let R denote counterclockwise rotation through $2\pi/3$. For a nonempty compact set $K \subset \mathbb{R}^2$, put

$$h_K(n) = \max_{x\in K}\langle x, n\rangle$$

and

$$\Lambda(K) = \frac{2}{\sqrt{3}} \min_{\|n\|=1} \sum_{j=0}^{2} h_K(\mathsf{R}^j n). \tag{13}$$

Proposition D.1 (Equilateral enclosure gauge)**.** *The number $\Lambda(K)$ is the least side length of a closed equilateral triangle containing K. It is monotone under inclusion, translation invariant, and positively homogeneous. If K is a finite polygonal hull, a minimizing normal may be chosen perpendicular to one of its hull edges.*

Proof. For a fixed unit normal n, the three supporting half-planes with outward normals $n, \mathsf{R}n, \mathsf{R}^2 n$ form an equilateral triangle. Its side is $2/\sqrt{3}$ times the sum of the three support numbers, proving (13). Monotonicity and homogeneity are immediate, while translation invariance follows from

$$n + \mathsf{R}n + \mathsf{R}^2 n = 0.$$

On an open normal-direction cell with three fixed support points, the support sum has the form $A\cos\theta + B\sin\theta$ and has second derivative equal to its negative. In the nondegenerate case an interior critical point is a maximum. Hence a minimum occurs at a cell boundary, where one support line contains a hull edge. □

For the local demand hull

$$K(a, b, c) = \operatorname{conv}\{0, au, bv, c(u + v)\},$$

where u and v are the two unit boundary directions meeting at 120°, the admissible set is

$$\mathcal{A} = \{(a, b, c) \in [0, 1]^3 : \Lambda(K(a, b, c)) \le 1\}. \tag{14}$$

The complete support calculation is given in Appendix E.

D.2. **One equilateral radical.** For $0 \le x \le 1$, define

$$\omega(x) = \sqrt{1 - x + x^2}, \qquad \sigma(x) = 1 - \omega(x). \tag{15}$$

Then

$$\omega(1 - x) = \omega(x), \qquad \sigma(x) = \frac{x(1 - x)}{1 + \omega(x)}, \tag{16}$$

and direct differentiation gives

$$\sigma''(x) = -\frac{3}{4\omega(x)^3} < 0. \tag{17}$$

Thus σ is strictly concave. In the signed center calculus,

$$E = \omega(R), \qquad \eta = \sigma(R).$$

For $0 < x < 1$, put $z = \sigma(x)/x$. Eliminating the square root gives

$$x = \frac{1 - 2z}{1 - z^2}, \qquad \sigma(x) = \frac{z(1 - 2z)}{1 - z^2}, \qquad 0 < z < \frac{1}{2}. \tag{18}$$

D.3. **One g-family.** For $0 \le x, c \le 1$, define

$$g_c(x) = \max\{y \in [0,1] : (1-x, y, c) \in \mathcal{A}\}, \qquad \widehat{g}_c(x) = \min\{g_c(x), x\}. \tag{19}$$

For any map $f : [0,1] \to [0,1]$, put

$$f^{\vee}(a) = 1 - f(1-a). \tag{20}$$

Hence

$$\widehat{g}_c^{\vee}(a) = 1 - \widehat{g}_c(1-a) \ge a. \tag{21}$$

The exact appendix aliases are

$$B_c(a) = g_c(1-a), \qquad F_c(a) = \widehat{g}_c(1-a), \qquad G_c(a) = \widehat{g}_c^{\vee}(a). \tag{22}$$

Lemma D.2 (Raw and capped actual-row transfer)**.** *Suppose an actual row has incoming reach at least a, radial reach at least c, and outgoing reach B. Then*

$$B \le g_c(1-a).$$

If no center interval intervenes on the next edge, then

$$A_{\text{next}} \ge g_c^{\vee}(a).$$

If the row is nonsupercritical, then

$$B \le \widehat{g}_c(1-a),$$

and, on a center-free next edge,

$$A_{\text{next}} \ge \widehat{g}_c^{\vee}(a) \ge a.$$

Proof. The same triangle realizes (a, B, c), so the definition of g_c gives the outgoing bound. On a center-free edge, coverage gives $A_{\text{next}} \ge 1 - B$, and complementation proves the raw transfer. Under nonsupercriticality, $B \le 1 - A \le 1 - a$, so $B \le \min\{g_c(1-a), 1-a\} = \widehat{g}_c(1-a)$. The final assertion follows in the same way. □

Proposition D.3 (Zero-radial distinction)**.** *For $0 < x < 1$,*

$$g_0(x) > x, \qquad \widehat{g}_0(x) = x,$$

and therefore

$$g_0^{\vee}(a) < a, \qquad \widehat{g}_0^{\vee}(a) = a \quad (0 < a < 1).$$

Proof. The exact diameter formula gives

$$g_0(x) = \frac{x - 1 + \sqrt{1 + 6x - 3x^2}}{2}.$$

Since

$$1 + 6x - 3x^2 - (1+x)^2 = 4x(1-x) > 0,$$

one has $g_0(x) > x$. The remaining assertions follow from the definitions. □

D.4. **Center-assisted transfers and the free envelope.** Let $J \subseteq [0,1]$ be empty or a closed interval. For $0 \le p \le 1$, put

$$e_J(p) = \sup\{x \in [0,1] : [0,x] \subseteq [0,p] \cup J\}, \qquad \mathcal{R}_J(p) = 1 - e_J(p).$$

If $J = [L,U]$, then

$$\mathcal{R}_{[L,U]}(p) = \begin{cases} 1-p, & p < L, \\ 1 - \max\{p,U\}, & p \ge L, \end{cases} \qquad \mathcal{R}_\varnothing(p) = 1 - p. \tag{23}$$

The map is nonincreasing in p and under enlargement of J.

Lemma D.4 (Generalized boundary handoff)**.** *Suppose a unit boundary edge is covered by a left trace contained in $[0,p]$, a center trace J, and a right trace contained in $[1-q,1]$. Then*

$$q \ge \mathcal{R}_J(p).$$

For an actual row define

$$g^\vee_{c,J}(a) = \mathcal{R}_J(g_c(1-a)), \qquad \widehat{g}^\vee_{c,J}(a) = \mathcal{R}_J(\widehat{g}_c(1-a)).$$

Then

$$A_{\text{next}} \ge g^\vee_{c,J}(a), \tag{24}$$

and the hatted version holds when the row is nonsupercritical:

$$A_{\text{next}} \ge \widehat{g}^\vee_{c,J}(a). \tag{25}$$

Proof. The component of $[0,p] \cup J$ containing 0 is $[0, e_J(p)]$. If $1 - q > e_J(p)$, the intervening open interval is uncovered. The transfer statements follow from Lemma D.2 and the fact that $\mathcal{R}_J$ is nonincreasing. □

For $0 \le c < 1/2$, define

$$g^{\text{sc}}_c = \sup_{\{x : g_c(x) > x\}} g_c(x) = \frac{c + \sqrt{c^2 - 8c + 4}}{2}. \tag{26}$$

The supremum is not attained. Hence an actual strict-supercritical row with radial demand at least c satisfies

$$B < g^{\text{sc}}_c. \tag{27}$$

If, in addition, its outgoing edge is center-free, then coverage gives

$$A_{\text{next}} > 1 - g^{\text{sc}}_c. \tag{28}$$

The former notation was $B_{\text{sc}}(c) = g^{\text{sc}}_c$ and $A_{\text{sc}}(c) = 1 - g^{\text{sc}}_c$.

D.5. **Relaxed composition and path budgets.** For maps listed in geometric row order, write

$$[\Phi_1 \mid \cdots \mid \Phi_r](x) = (\Phi_r \circ \cdots \circ \Phi_1)(x). \tag{29}$$

The leftmost slot acts first. Write $\mathrm{I}(x) = x$.

Lemma D.5 (Relaxed composition)**.** *Suppose actual demands satisfy*

$$x_j \ge \Phi_j(x_{j-1}) \qquad (1 \le j \le r),$$

where every Φ_j is nondecreasing. If $\underline{\Phi}_j \le \Phi_j$, then

$$x_r \ge [\underline{\Phi}_1 \mid \cdots \mid \underline{\Phi}_r](x_0).$$

Proof. Set $y_0 = x_0$ and $y_j = \underline{\Phi}_j(y_{j-1})$. Monotonicity gives $x_j \geq \Phi_j(x_{j-1}) \geq \Phi_j(y_{j-1}) \geq y_j$ inductively. □

Lemma D.6 (Boundary-path budget)**.** *Let $E_0, \dots, E_k$ be consecutive boundary edges and let $T_1, \dots, T_k$ be the roles at their intermediate vertices. Suppose:*

(1) *on E_0, the component covered from the endpoint opposite T_1 by all roles other than T_1 has extent at most u;*
(2) *on E_k, the analogous component opposite T_k has extent at most v;*
(3) *on every middle edge, no role other than the two incident path roles has positive-length trace;*
(4) *the chain is covered.*

Then

$$\sum_{j=1}^{k}(A_j + B_j) \geq k + 1 - u - v. \tag{30}$$

Consequently, if all k internal rows are nonsupercritical, the chain is impossible whenever $u + v < 1$.

Proof. The first and last component hypotheses give $A_1 \geq 1 - u$ and $B_k \geq 1 - v$. On every middle edge the center and all nonincident roles have been excluded, so $B_j + A_{j+1} \geq 1$. Adding proves the formula. □

D.6. **Low roots, selected T_+ curves, and thresholds.** For the high-radial maps define

$$e(d) = \frac{1-d}{2}\left(1 - \sqrt{4(1-d)^2 - 3}\right) \qquad \left(0 < d < 1 - \frac{\sqrt{3}}{2}\right), \tag{31}$$

and, for the low-radial order transition,

$$h(c) = \frac{2c}{1 + \sqrt{16c^2 - 3}} \qquad \left(\frac{1}{2} < c \leq \frac{\sqrt{3}}{2}\right). \tag{32}$$

Proposition D.7 (Universal selected-T_+ curve)**.** *Fix $0 < d < 1/2$, put $c = 1 - d$, and suppose an input p belongs to a selected T_+ arc of $\widehat{g}_c^\vee$. Write*

$$q = \widehat{g}_c^\vee(p), \qquad \nu = q - p, \qquad x = \frac{q-d}{1-d}.$$

Then

$$\nu = \sigma(x) = 1 - \sqrt{1 - x + x^2}, \tag{33}$$

and

$$q = d + (1-d)x, \qquad p = d + (1-d)x - \sigma(x). \tag{34}$$

Consequently the selected arc is increasing and strictly concave.

Proof. The selected quadratic is

$$(1-q)(q-d) = (1-d)^2\nu(2-\nu).$$

Substitution gives $x(1-x) = \nu(2-\nu)$, and the selected component uses the smaller root $\nu = \sigma(x)$. The input function is increasing and strictly convex because σ is strictly concave; inversion gives the assertion. □

The corresponding chord bounds are

$$\widehat{g}^{\vee}_{1-d}(p) \geq p + \frac{d}{e(d)-d}(p-d), \tag{35}$$

and, with $t = h(1-d)$,

$$\widehat{g}^{\vee}_{1-d}(p) \geq p + \frac{1-2t}{t-d}(p-d). \tag{36}$$

For every coefficient λ proved valid on the relevant selected arc, write

$$\widehat{g}^{\vee,\lambda}_{1-d}(p) := p + \lambda(p-d) \leq \widehat{g}^{\vee}_{1-d}(p). \tag{37}$$

The rational form (18) gives, in the historical notation $x = 1-\beta$,

$$\beta = \frac{z(2-z)}{1-z^2}, \qquad m_\beta = \frac{1-z+z^2}{1-z^2}. \tag{38}$$

For $0 < d < 1-\sqrt{3}/2$, the exact high-demand threshold is

$$z > e(d) \quad \Longrightarrow \quad \widehat{g}^{\vee}_{1-d}(z) \geq 1-e(d). \tag{39}$$

Define

$$\widehat{g}^{\vee,\mathrm{th}}_{1-d}(z) = \begin{cases} z, & z \leq e(d), \\ 1-e(d), & z > e(d), \end{cases} \qquad \widehat{g}^{\vee,\mathrm{th}}_{1-d} \leq \widehat{g}^{\vee}_{1-d}. \tag{40}$$

Lemma D.8 (Threshold routing)**.** *Let $z_j = \Phi_j(z_{j-1})$, where every Φ_j is nondecreasing and extensive. If $\Phi_k = \widehat{g}^{\vee}_{1-d}$ and $z_{k-1} > e(d)$, then $z_j \geq 1-e(d)$ for all $j \geq k$. If a chain contains later occurrences of $\widehat{g}^{\vee}_{1-d_1}$ and $\widehat{g}^{\vee}_{1-d_2}$ and*

$$z_0 > e(d_1), \qquad \min\{e(d_1), e(d_2)\} < Q,$$

then at least one of those two rows forces every subsequent output above $1-Q$.

Proof. The first assertion is (39) followed by extensivity. If $e(d_1) < Q$, use the first threshold. Otherwise $z_0 > e(d_1) \geq Q > e(d_2)$, so the second threshold fires. The conclusion is "at least one", not uniqueness; both thresholds may be available. $\square$

Appendix E. Derivation of the Local Admissible Set

This section proves the explicit description stated in Proposition 2.2. It also derives the three polynomial cells used by the exact-demand appendix and records the component selectors introduced when a positive support equation is squared.

Let

$$u = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \qquad v = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right),$$

and put

$$O = 0, \qquad A = au, \qquad B = bv, \qquad C = c(u+v), \qquad K = \operatorname{conv}\{O, A, B, C\}.$$

Set

$$s = a+b, \qquad d = a^2 + ab + b^2.$$

E.1. **The triangular hull.** Assume first that $s > 0$ and $cs \le ab$. If $a, b > 0$, then

$$\frac{c}{a} + \frac{c}{b} \le 1,$$

so

$$C = \frac{c}{a}A + \frac{c}{b}B + \Big(1 - \frac{c}{a} - \frac{c}{b}\Big) O \in \operatorname{conv}\{O, A, B\}.$$

The zero-coordinate cases follow by continuity. The distance AB equals $\sqrt{d}$ and is the diameter of $\triangle OAB$. Put

$$T = -bu - av.$$

The triangle $\operatorname{conv}\{A, B, T\}$ is equilateral of side $\sqrt{d}$, and

$$O = \frac{b^2}{d}A + \frac{a^2}{d}B + \frac{ab}{d}T.$$

Thus it contains O and therefore also C. Every enclosing set has diameter at least AB, so

$$L_{\min}(a, b, c) = \sqrt{d}.$$

When $a = b = 0$, the hull is the segment $[0, c(u + v)]$ and its least enclosing equilateral side is c.

E.2. **The quadrilateral hull and the four calipers.** Assume now that $s > 0$ and $cs > ab$. The four hull vertices occur in the cyclic order

$$O,\ B,\ C,\ A.$$

By Proposition D.1, a minimizing orientation has an outward normal perpendicular to one of the four hull edges.

For the edge OA, the three supports give

$$L_{OA} = b + \max\{a, c\};$$

reflection gives

$$L_{BO} = a + \max\{b, c\}.$$

For the edge AC, put

$$R_a = \sqrt{a^2 - ac + c^2}.$$

An outward unit normal is

$$n_0 = \frac{1}{2R_a}(\sqrt{3}a,\ 2c - a).$$

If $x_+ = \max\{x, 0\}$, the support values for $n_0, \mathsf{R}n_0, \mathsf{R}^2 n_0$ are

$$\frac{\sqrt{3}ac}{2R_a}, \qquad \frac{\sqrt{3}a(a - c)_+}{2R_a}, \qquad \frac{\sqrt{3}c \max\{b, (c - a)_+\}}{2R_a}.$$

Multiplying their sum by $2/\sqrt{3}$ gives

$$L_{AC} = \begin{cases} (a^2 + bc)/R_a, & a \ge c, \\ c(a + b)/R_a, & a < c \le a + b, \\ c^2/R_a, & c > a + b. \end{cases}$$

Reflection exchanges a and b and gives L_{CB}. These four hull edges are exhaustive, so

$$L_{\min}(a, b, c) = \min\{L_{OA}, L_{BO}, L_{AC}, L_{CB}\}.$$

This proves Proposition 2.2, including all equality and degenerate cases by continuity.

E.3. **Equivalent polynomial cells.** For the exact transfer calculations put

$$m = \min\{a,b\}, \qquad M = \max\{a,b\}, \qquad q = s^4 - s^2 + ab,$$

and define

$$\begin{aligned} F_L(a,b,c) &= c^4 - c^2 + mc - m^2,\\ F_T(a,b,c) &= (s^2-1)c^2 + Mc - M^2,\\ F_S(a,b,c) &= (m^2-1)c^2 + (2mM^2+M)c + M^4 - M^2. \end{aligned}$$

We prove

$$\mathcal{A} = \mathcal{A}_L \cup \mathcal{A}_T \cup \mathcal{A}_S, \tag{41}$$

where

$$\begin{aligned} \mathcal{A}_L &= \{d \le 1,\ s \le 1,\ q \le 0,\ F_L \le 0\},\\ \mathcal{A}_T &= \{d \le 1,\ s \le 1,\ q \ge 0,\ F_T \le 0,\ c \le 2M\},\\ \mathcal{A}_S &= \{d \le 1,\ s > 1,\ F_S \le 0,\ c \le 1/2\}, \end{aligned}$$

with all variables in $[0,1]$.

Assume by symmetry that $a = m \le b = M$. Reading the active support points in the four-caliper formulas gives the following contact patterns:

range	active supports	unit-frontier equation
$s<1,\ q<0$	$\{O\},\{C\},\{A,C\}$	$F_L=0$
$s<1,\ q>0$	$\{O\},\{B,C\},\{A\}$	$F_T=0$
$s=1$	preceding pattern with an O,B tie	$c=M$
$s>1$	$\{B\},\{B,C\},\{A\}$	$F_S=0.$

The inactive-support inequalities reduce respectively to $q \le 0$, $q \ge 0$, and $s > 1$. The side-length equations have positive sides; squaring them gives exactly $F_L = 0, F_T = 0, F_S = 0$. It remains to identify the geometric connected component after squaring.

For Cell T, $0 < s < 1$, the roots of $F_T = 0$ are

$$c_- = \frac{2M}{1+\sqrt{4s^2-3}}, \qquad c_+ = \frac{2M}{1-\sqrt{4s^2-3}}.$$

The inequality $F_T \le 0$ contains the interval $0 \le c \le c_-$ and the formal high component $c \ge c_+$. The geometric component is the one connected to $c = 0$. One has

$$c_- \le 2M \le c_+,$$

with equality of all three quantities only when the two roots coalesce. Hence

$$c \le 2M$$

selects exactly the component containing $c = 0$. Equivalently one may use $M + 2(s^2-1)c \ge 0$.

For Cell S,

$$F_S(a,b,0) = M^2(M^2-1) < 0, \qquad F_S(a,b,M) = M^2(s^2-1) > 0.$$

Furthermore $4F_S(a,b,1/2)$ is increasing in m, and at $m = 1-M$ it equals

$$M^2(2M-1)^2.$$

Because $s > 1$ gives $m > 1 - M$, one has $F_S(a, b, 1/2) > 0$. The quadratic opens downward, so $1/2$ lies strictly between its two roots. Therefore $c \le 1/2$ selects the smaller-root component containing $c = 0$.

For Cell L, the fiber from $c = 0$ reaches one selected root in $[\sqrt{3}/2, 1]$; monotonicity there follows from $4c^3 - 2c + m > 0$. At $q = 0$ the L and T formulas meet and both give $c = s$. This proves (41) and, in particular, proves that $\mathcal{A}$ is coordinatewise down-closed.

E.4. **The radial envelope.** Solving the selected frontier equations gives

$$c_{\max}(a,b) = \begin{cases} 1, & a = b = 0, \\ C_L(m), & s \le 1,\ q \le 0,\ (a,b) \ne (0,0), \\ \dfrac{2M}{1 + \sqrt{4s^2 - 3}}, & s \le 1,\ q > 0, \\ \dfrac{M(2mM + 1) - M\sqrt{4d - 3}}{2(1 - m^2)}, & s > 1, \end{cases}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

At $q = 0$ the two subcritical formulas agree and equal s. These formulas are the radial-envelope inputs used later in Strategies 2 and 4.

Appendix F. Signed CE1/CE2 Reductions

This section gives one signed center model and the common one-gap row interface. The transfer notation has already been fixed in Section D: the exact nonsupercritical incoming-reach map is $\widehat{g}_c^{\vee}$, while g_c itself remains the historical boundary-defect map.

F.1. **One signed center normal form.**

Proposition F.1 (Signed CE1/CE2 center normal form)**.** *Let T_C be a closed unit equilateral triangle with $O \in \mathrm{int}(T_C)$ and a positive-length boundary trace. Normalize such a trace to $e_{0,1}$ and use affine coordinates*

$$X = V_0 + b(V_1 - V_0) + a(V_5 - V_0).$$

There are parameters

$$0 < R < 1, \qquad W = 1 - R,$$
$$E = \sqrt{1 - RW}, \qquad \eta = 1 - E, \qquad P = E(1 - E),$$

and nonnegative center slacks α, δ such that, with

$$k = \eta + \alpha + \delta,$$

the center triangle is

$$\boxed{\begin{aligned} F_0 &= R + \alpha - a + Wb, \\ F_1 &= Rb + Wa - k, \\ F_2 &= W + \delta - b + Ra, \end{aligned}} \qquad T_C = \{F_0, F_1, F_2 \ge 0\}. \tag{42}$$

Define

$$\Delta_R = P - \alpha - W\delta, \qquad \Delta_L = P - R\alpha - \delta. \tag{43}$$

Then

$$T_C \cap e_{0,1} = \left[\frac{k}{R}, W+\delta\right], \qquad \mathcal{H}^1(T_C \cap e_{0,1}) = \frac{\Delta_R}{R}, \tag{44}$$

and the possible companion trace is

$$T_C \cap e_{5,0} = \left[\frac{k}{W}, R+\alpha\right] \quad \text{when } \Delta_L > 0, \qquad \mathcal{H}^1(T_C \cap e_{5,0}) = \frac{[\Delta_L]_+}{W}. \tag{45}$$

The normalization requires $\Delta_R > 0$*, and*

$$T_C \text{ is CE1} \iff \Delta_L \le 0, \qquad T_C \text{ is CE2} \iff \Delta_L > 0. \tag{46}$$

The six center exits, measured from O*, are*

$$\boxed{\begin{aligned} &d_0^C = E - \alpha - \delta, && d_1^C = \frac{\delta}{R}, && d_2^C = \delta, \\ &d_3^C = \min\left\{\frac{\alpha}{R}, \frac{\delta}{W}\right\}, && d_4^C = \alpha, && d_5^C = \frac{\alpha}{W}. \end{aligned}} \tag{47}$$

For closures of original open center roles one has $\alpha, \delta > 0$.

Proof. Write

$$T_C \cap e_{0,1} = [s,t], \qquad 0 \le s < t \le 1.$$

The two sides through these endpoints have a positive slope ratio. Reflect across the perpendicular bisector of $e_{0,1}$ if necessary and call the ratio $R \le 1$; put $W = 1 - R$. Direct equations for the three side lines give

$$F_1 = Rb + Wa - Rs, \qquad F_2 = -b + Ra + t, \qquad F_0 = Wb - a + E + Rs - t,$$

where $E = \sqrt{1 - RW}$ is forced by the unit side-length relation. If $R = 1$, then $F_0(O) = s - t < 0$, contrary to $O \in \operatorname{int}(T_C)$; hence $0 < R < 1$.

Set

$$\alpha = F_0(O), \qquad \delta = F_2(O).$$

Then

$$t = W + \delta, \qquad Rs = \eta + \alpha + \delta = k,$$

and substitution gives (42). The gradients are the three unit-equilateral side-normal directions and

$$F_0 + F_1 + F_2 = E,$$

so the three inequalities define the same unit equilateral triangle.

On $e_{0,1}$ one has $a = 0$. The active inequalities are

$$Rb \ge k, \qquad b \le W + \delta,$$

which gives (44); its length is

$$W + \delta - \frac{k}{R} = \frac{P - \alpha - W\delta}{R}.$$

On $e_{5,0}$ one has $b = 0$, and the active inequalities are

$$Wa \ge k, \qquad a \le R + \alpha.$$

Their signed interval length is

$$R + \alpha - \frac{k}{W} = \frac{P - R\alpha - \delta}{W},$$

which proves (45).

The positive right trace gives

$$\delta < \frac{P}{W} = \frac{ER}{1+E} < \frac{R}{2}.$$

On $e_{1,2}$, parameterized by $b = 1+q$, $a = q$, the slack F_2 is

$$\delta - R - Wq < 0.$$

Thus a second positive trace cannot occur on $e_{1,2}$. The CE classification says that a center triangle has at most two positive boundary traces and that two such traces are adjacent. Hence $e_{5,0}$ is the only possible companion edge, and its signed length proves (46). Equality $\Delta_L = 0$ is only a point contact and is therefore CE1.

For the radial exits, substitute the six ray parametrizations in (42). On r_1 the decreasing slack is $\delta - Rq$; on r_2 it is $\delta - q$; on r_3 the two decreasing slacks are $\alpha - Rq$ and $\delta - Wq$. Reflection gives r_4, r_5, and on r_0 the decreasing slack is $E - \alpha - \delta - q$. This proves (47).

Finally,

$$W(\alpha + \delta) \le \alpha + W\delta < P,$$

so

$$E - \alpha - \delta > E - \frac{P}{W} = \frac{E(E-R)}{W} > \frac{1}{2},$$

where the last inequality is equivalent to $2E(E-R) - W = (E-R)^2 > 0$. Together with

$$\alpha < P < \min\left\{\frac{R}{2}, \frac{W}{2}\right\}, \qquad \delta < \frac{R}{2},$$

the midpoint tests give the exact midpoint set $\{M_0\}$. For the closure of an original open center role, all three center slacks are positive; hence $\alpha, \delta > 0$. □

Remark F.2 (Legacy variables). For CE1, take

$$\lambda = R, \qquad s = \frac{k}{R}, \qquad t = W + \delta, \qquad C_0 = \alpha, \qquad C_2 = \delta.$$

For CE2, take

$$x = \frac{k}{W}, \qquad \nu = R + \alpha, \qquad y = \frac{k}{R}, \qquad v = W + \delta.$$

If $S = x + y$ and $D = \sqrt{x^2 + xy + y^2}$, then

$$S = \frac{k}{RW}, \qquad D = \frac{Ek}{RW},$$

and the old coupling equation is

$$(\nu + v)S - xy = D.$$

Thus the CE1 and CE2 formula lists are two coordinate readings of Proposition F.1.

F.2. **Boundary contribution and diameter transfer.**

Corollary F.3 (Common center-boundary formula)**.** *The full center-boundary contribution is*

$$L_{\partial H}(T_C) = \frac{\Delta_R}{R} + \frac{[\Delta_L]_+}{W}. \tag{48}$$

In CE1,

$$L_{\partial H}(T_C) \le P \le \frac{\sqrt{3}}{2} - \frac{3}{4}.$$

In CE2,

$$L_{\partial H}(T_C) = \frac{E}{1+E} - \frac{E^2(\alpha+\delta)}{RW} < \frac{1}{2}.$$

Proof. Equation (48) is the sum of (44) and (45). If $\Delta_L \le 0$, then $R\alpha + \delta \ge P$. Multiplying by W and using $\alpha \ge RW\alpha$ gives

$$\alpha + W\delta \ge WP.$$

Thus $\Delta_R \le RP$, so the contribution is at most P. Since $E \ge \sqrt{3}/2$ and $E(1-E)$ decreases on this interval, the CE1 cap follows.

If $\Delta_L > 0$, direct addition gives

$$\begin{aligned} L_{\partial H}(T_C) &= \frac{P-\alpha-W\delta}{R} + \frac{P-R\alpha-\delta}{W} \\ &= \frac{P-E^2(\alpha+\delta)}{RW}. \end{aligned}$$

Here $W + R^2 = W^2 + R = E^2$ and $P/(RW) = E/(1+E)$, so the last expression is strictly below $1/2$. □

Lemma F.4 (Adjacent-edge diameter transfer)**.** *For $0 \le q \le 1$, put*

$$\beta(q) = \frac{-q + \sqrt{4-3q^2}}{2}.$$

If a unit equilateral triangle reaches parameters q, b on two adjacent boundary edges, then

$$b \le \beta(q).$$

For $q > 0$,

$$\beta(q) < 1 - \frac{q}{2}, \qquad 1 - \beta(q) > \frac{q}{2}.$$

Proof. The squared distance between the two boundary points is $q^2 + qb + b^2$, so it is at most one. Solving the quadratic in b gives the first assertion. For $q > 0$, $\sqrt{4-3q^2} < 2$, which gives both strict linear comparisons. □

F.3. **The local propagation interface.** The exact nonsupercritical reach transfer is $\widehat{g}_c^{\vee}$. It is nondecreasing and extensive:

$$\widehat{g}_c^{\vee}(z) \ge z. \tag{49}$$

The exact reflection duality is

$$\widehat{g}_c^{\vee}(a) \le z \iff \widehat{g}_c^{\vee}(1-z) \le 1-a. \tag{50}$$

Suppose an actual nonsupercritical row has maximal reaches (A_i, B_i, C_i) and

$$A_i \ge z, \qquad C_i \ge c_i.$$

Then its outgoing reach satisfies

$$B_i \le \widehat{g}_{c_i}(1-z).$$

If the next boundary handoff gives $A_{i+1} \ge 1 - B_i$, then

$$\begin{aligned} A_{i+1} &\ge 1 - B_i \\ &\ge 1 - \widehat{g}_{c_i}(1-z) \\ &= \widehat{g}^{\vee}_{c_i}(z). \end{aligned} \tag{51}$$

Every composition below is shorthand for repeated applications of this inequality to specified actual rows.

F.4. **The common one-gap interface.**

Proposition F.5 (Common five-row one-gap reduction)**.** *Assume every vertex role is Vd0, $N_+ = 1$, and the normalized right center trace contains a V-gap, possibly a singleton. In the CE2 case assume the companion trace contains no V-gap. Put*

$$X = R - \delta, \qquad H = \frac{k}{2R}, \qquad m = \min\left\{\frac{\alpha}{R}, \frac{\delta}{W}\right\},$$

and

$$\begin{aligned} &c_1 = 1 - \frac{\delta}{R}, \quad c_2 = 1 - \delta, \qquad c_3 = 1 - m, \\ &c_4 = 1 - \alpha, \quad\; c_5 = 1 - \frac{\alpha}{W}, \quad c_0 = k. \end{aligned}$$

Let

$$z_0 = X, \qquad z_i = \widehat{g}^{\vee}_{c_i}(z_{i-1}) \quad (1 \le i \le 5),$$

and $Z = z_5$. Then the actual reaches satisfy

$$A_0 \ge Z, \qquad A_0 \le g_k\left(1 - \frac{k}{R}\right) < 1 - H. \tag{52}$$

Consequently it is enough to prove

$$[\widehat{g}^{\vee}_{1-\alpha} \mid \widehat{g}^{\vee}_{1-m} \mid \widehat{g}^{\vee}_{1-\delta}](H) > W + \delta. \tag{53}$$

Proof. The exact-midpoint argument makes rows $1, \ldots, 5$ nonsupercritical and row 0 uniquely supercritical. Gap containment gives

$$B_0 \ge \frac{k}{R}, \qquad A_1 \ge X.$$

Suppose inductively that $A_i \ge z_{i-1}$. Radial coverage gives actual radial reach at least c_i, and the hatted outgoing cap gives

$$B_i \le \widehat{g}_{c_i}(1 - z_{i-1}).$$

The next boundary handoff therefore gives

$$A_{i+1} \ge 1 - B_i \ge \widehat{g}^{\vee}_{c_i}(z_{i-1}) = z_i.$$

For the final handoff, the CE1 companion trace is empty and the CE2 companion trace is gap-free, so $A_0 \ge z_5 = Z$.

The same row has boundary reach at least k/R and radial reach at least k. Reflection and down-closedness of $\mathcal{A}$ give

$$\left(\frac{k}{R}, A_0, k\right) \in \mathcal{A},$$

so

$$A_0 \le g_k \left(1 - \frac{k}{R}\right).$$

Lemma F.4 gives

$$g_k \left(1 - \frac{k}{R}\right) \le \beta\left(\frac{k}{R}\right) < 1 - \frac{k}{2R} = 1 - H,$$

which proves (52).

The first and fifth maps are extensive and the middle maps are nondecreasing, so it is enough to prove

$$[\widehat{g}^{\vee}_{c_2} \mid \widehat{g}^{\vee}_{c_3} \mid \widehat{g}^{\vee}_{c_4}](X) > 1 - H.$$

Three applications of (50) show that its negation is equivalent to the negation of (53). This proves the reduction. □

The CE1 sign $\Delta_L \le 0$ is completed by the selected-T_+ relaxation in Proposition I.2. We now give the shorter CE2 scalar clause.

Lemma F.6 (CE2 total slack and two thresholds)**.** *Assume* $\Delta_R > 0$ *and* $\Delta_L > 0$, *and put*

$$T = \alpha + \delta, \qquad Q = \frac{\eta + T}{2R}.$$

Then

$$T < \frac{\eta}{E}, \tag{54}$$

and

$$\min\{e(\alpha), e(\delta)\} < Q. \tag{55}$$

Proof. The CE2 inequalities are

$$\alpha + W\delta < P, \qquad R\alpha + \delta < P.$$

Multiply the first by W, the second by R, and add. Since $W + R^2 = W^2 + R = E^2$, this gives

$$E^2 T < P = E\eta,$$

which is (54).

Let $d = \min\{\alpha, \delta\}$. Since $d \le T/2$ and the rational low-root bound is increasing,

$$\min\{e(\alpha), e(\delta)\} < \frac{2d}{1 - 2d} \le \frac{T}{1 - T}.$$

It remains to prove

$$\frac{T}{1 - T} < \frac{\eta + T}{2R}.$$

Define

$$\Phi(t) = (\eta + t)(1 - t) - 2Rt.$$

This function is concave, so its minimum on $[0, \eta/E]$ is at an endpoint. Clearly $\Phi(0) = \eta > 0$. Direct simplification using $E^2 = 1 - R + R^2$ gives

$$\Phi\left(\frac{\eta}{E}\right) = \frac{R(E - R)^2}{E^2} > 0.$$

Thus $\Phi(T) > 0$, proving (55). □

Proposition F.7 (Short CE2 one-gap completion). *Under the CE2 hypotheses of Proposition F.5, the one-gap state is impossible in either orientation.*

Proof. The right-gap initial input is $z_0 = R - \delta$. The strict CE2 domain gives

$$Wz_0 = RW - W\delta = \eta + P - W\delta > \eta + \alpha.$$

For

$$\pi(q) = (\eta + q)(1 - 2q) - 2qW,$$

one has

$$\pi(0) = \eta > 0, \qquad \pi(P) = \eta(\eta + 2ER^2) > 0.$$

Concavity therefore gives $\pi(\alpha) > 0$, and the low-root bound yields

$$z_0 > \frac{\eta + \alpha}{W} > \frac{2\alpha}{1 - 2\alpha} > e(\alpha). \tag{56}$$

If $e(\alpha) < Q$, extensivity carries (56) to the row-4 map $\widehat{g}^{\vee}_{1-\alpha}$, which gives an output at least $1 - e(\alpha) > 1 - Q$. All later maps preserve it. If $e(\alpha) \geq Q$, Lemma F.6 gives $e(\delta) < Q$, and

$$z_0 > e(\alpha) \geq Q > e(\delta).$$

The row-2 map $\widehat{g}^{\vee}_{1-\delta}$ then gives an output above $1 - Q$, again preserved by all later maps. Thus $Z > 1 - Q = 1 - H$, contradicting (52).

Reflection exchanges

$$R \leftrightarrow W, \qquad \alpha \leftrightarrow \delta$$

and reverses the row order. The signed CE2 domain and the proof above are invariant under this exchange, so the left-gap orientation is impossible as well. □

Remark F.8 (Why the CE1 scalar clause remains separate). The geometric one-gap interface is common, but (55) is not valid on the full CE1 sign domain. Thus the merger stops at Proposition F.5; the CE1 selected-T_+ clause and the CE2 threshold clause are genuinely different scalar proofs of the same target (53).

Appendix G. Trace Budgets and Length Contradictions

This section develops the boundary and conditional-skeleton trace bounds used by the length mechanism.

For a closed triangle T and a rectifiable target E, write

$$L_E(T) = \mathcal{H}^1(T \cap E).$$

The passage from an original open role U to its closure $T = \overline{U}$ can only enlarge the trace. Hence a cover of E by the open roles implies

$$\mathcal{H}^1(E) \leq L_E(T_C) + \sum_{i=0}^{5} L_E(T_i). \tag{57}$$

All estimates in this section concern full traces. They therefore remain valid for any assigned measurable subsets of those traces.

The strategy uses only the perimeter ∂H and the full skeleton S, whose measures were recorded in (8). Every skeleton estimate below is conditional on its stated center and row hypotheses; no unconditional noncoverage assertion for S is made.

FIGURE 10. The two one-dimensional targets. The full skeleton is used only under the hypotheses of the corresponding trace budget.

G.1. **Conditional skeleton trace caps.** We now develop the skeleton estimate used only when the center is CE1 or CE2. The first lemma is the center contribution.

Lemma G.1 (Center skeleton cap). *If T_C is a CE1 or CE2 center role and contains exactly one radial midpoint, then*

$$L_S(T_C) < \frac{3}{2}.$$

Proof. Normalize the unique midpoint as M_0 and the positive boundary trace to $e_{0,1}$. Use the signed variables of Proposition F.1; in particular

$$W = 1 - R, \qquad E = \sqrt{1 - RW}, \qquad \eta = 1 - E, \qquad P = E(1 - E).$$

The positive trace condition is

$$\alpha + W\delta < P. \tag{58}$$

The two possible boundary contributions are

$$\ell_{01} = W + \delta - \frac{\eta + \alpha + \delta}{R}, \qquad \ell_{50} = \frac{[P - (R\alpha + \delta)]_+}{W}.$$

The six radial contributions are bounded by

i	0	1	2	3	4	5
$\mathcal{H}^1(T_C \cap r_i)$	$E - \alpha - \delta$	δ/R	δ	$\min\{\alpha/R, \delta/W\}$	α	$\alpha/W.$

Adding gives

$$L_S(T_C) \le K_R + \mathcal{E}(\alpha, \delta),$$

where

$$K_R = W + E - \frac{\eta}{R}$$

and

$$\mathcal{E}(\alpha, \delta) = -\frac{\alpha}{R} + \frac{\alpha}{W} + \delta + \min\left\{\frac{\alpha}{R}, \frac{\delta}{W}\right\} + \frac{[P - (R\alpha + \delta)]_+}{W}.$$

Splitting only according to the minimum and the positive part gives

$$\mathcal{E}(\alpha, \delta) \le \frac{P}{W}. \tag{59}$$

For example, if $\alpha/R \le \delta/W$, the expression without the positive part is $\alpha/W + \delta$; when the positive part is active, its excess over P/W is $\alpha - R\delta/W \le 0$. The reflected order gives excess $\delta - W\alpha/R \le 0$. If the positive part vanishes, (58) gives the same bound strictly.

Put $A = 1 + R - R^2$. Then

$$\frac{3}{2} - K_R - \frac{P}{W} = \frac{1 + A - 2EA}{2RW}.$$

Both sides of $1 + A > 2EA$ are positive, and

$$(1 + A)^2 - 4A^2E^2 = R^2W^2(5 + 4R - 4R^2) > 0.$$

Thus $K_R + P/W < 3/2$. □

Lemma G.2 (Positive-support skeleton cap)**.** *Let T_i be the closure of an original open vertex role, so $V_i \in \operatorname{int}(T_i)$. If T_i has positive-length intersection with an adjacent radial arm, then*

$$L_S(T_i) < \frac{3}{2}.$$

Proof. Normalize to $i = 0$ and translate the hexagon by V_0:

$$\widetilde{H} = V_0 + H, \qquad \widetilde{O} = V_0, \qquad W_j = V_0 + V_j.$$

Then $W_3 = O$, $W_2 = V_1$, and $W_4 = V_5$. The five original skeleton components on which a unit triangle containing V_0 can have positive-length trace are

$$[V_0, V_1],\ [V_0, V_5],\ [O, V_1],\ [O, V_0],\ [O, V_5].$$

All five belong to the skeleton $\widetilde{S}$ of $\widetilde{H}$. Since $V_0 = \widetilde{O}$ is interior to T_0, the triangle is a legitimate center role for $\widetilde{H}$. Its positive adjacent-arm trace is a positive boundary-edge trace of $\widetilde{H}$, so it is CE1 or CE2 by Proposition C.3. Proposition C.11 and Lemma G.1 therefore give

$$L_{\widetilde{S}}(T_0) < \frac{3}{2}.$$

A nonlocal component of S is more than unit distance from the interior point V_0, apart from zero-measure endpoints. Hence $L_S(T_0) \le L_{\widetilde{S}}(T_0)$. □

Lemma G.3 (No-support skeleton cap)**.** *If a vertex role has $A_i + B_i \le 1$ and no positive-length trace on either adjacent radial arm, then*

$$L_S(T_i) \le 2.$$

Proof. Its boundary contribution is $A_i + B_i \le 1$ by (62). For $P = sV_j \in r_j \setminus \{O\}$,

$$\|V_i - P\|^2 = 1 + s^2 - 2s\cos\frac{(j-i)\pi}{3} > 1$$

when $j \notin \{i-1, i, i+1\}$. The two adjacent arms have zero-length trace by hypothesis, so only the own unit arm can contribute positive length. This adds at most one. □

For the remaining cap we use one specialized part of the exact local equilateral enclosure criterion. In local directions u, v meeting at $120°$, suppose a closed unit triangle contains

$$0, \quad au, \quad bv, \quad c(u + v),$$

with $0 \le a \le b \le 1$ and $a + b > 1$. The support-line calculation gives the selected supercritical cell

$$\begin{aligned} &a^2 + ab + b^2 \le 1, \qquad c \le \frac{1}{2}, \\ &(a^2 - 1)c^2 + (2ab^2 + b)c + (b^4 - b^2) \le 0, \end{aligned} \tag{60}$$

where c lies on the connected component containing $c = 0$. To see the selector directly, the quadratic is negative at $c = 0$, positive at $c = 1/2$, and opens downward. For the middle assertion, four times its value at $1/2$ is increasing in a and, at $a = 1 - b$, equals $b^2(2b - 1)^2$; it is therefore positive when $a + b > 1$. Hence $c \le 1/2$

selects its smaller-root component. Formula (60) follows by fixing in the support sum the three active contacts $\{B\}$, $\{B, c(u+v)\}$, and $\{A\}$, and squaring the positive unit-side equation.

Lemma G.4 (Supercritical skeleton cap)**.** *If $A_i + B_i > 1$, then*

$$L_S(T_i) < \frac{3}{2}.$$

Proof. By Proposition C.5, a vertex role is Vd0, Vd1, Vd2, or T3-like. Proposition G.6 makes every Vd1/Vd2 row nonsupercritical, and Lemma C.8 does the same for T3-like rows. Hence a supercritical role is Vd0 and, by definition, has no positive-length trace on either adjacent arm. Thus, after normalizing and writing $a = A_i$, $b = B_i$, the skeleton trace has length $a + b + c$, where c is the own-arm reach. Reflect so that $a \le b$, and put

$$s = a + b, \qquad \delta = b - a, \qquad c_0 = \frac{3}{2} - s.$$

The diameter constraint gives

$$1 < s \le \frac{2}{\sqrt{3}}, \qquad 0 \le \delta \le \sqrt{4 - 3s^2}.$$

Let $P(c)$ denote the quadratic in (60). Substitution and multiplication by 16 give

$$\begin{aligned} 16P(c_0) = Q(s, \delta) = {} & \delta^4 + 8\delta^3 s - 6\delta^3 + 14\delta^2 s^2 - 18\delta^2 s + 5\delta^2 \\ & - 8\delta s^3 + 30\delta s^2 - 34\delta s + 12\delta \\ & + s^4 - 6s^3 - 19s^2 + 60s - 36. \end{aligned}$$

Its derivative factors as

$$\frac{\partial Q}{\partial \delta} = 2(2\delta + 4s - 3)\big(\delta^2 + \delta(4s - 3) + (s - 1)(2 - s)\big) \ge 0.$$

Therefore

$$Q(s, \delta) \ge Q(s, 0) = (s - 1)(s^3 - 5s^2 - 24s + 36) > 0.$$

Indeed, the cubic is decreasing on $[1, 2/\sqrt{3}]$ and at its right endpoint equals $88/3 - 136\sqrt{3}/9 > 0$. Hence $P(c_0) > 0$. Since (60) selects the smaller-root component, $P(c) \le 0$ forces $c < c_0$. Thus $a + b + c < 3/2$. □

Lemma G.5 (A positive-support rescuer is forced)**.** *Suppose T_C is CE1 or CE2 and, after symmetry, contains exactly M_0. If at least two vertex rows are supercritical, then a vertex role distinct from a chosen supercritical role has positive-length adjacent-arm support.*

Proof. Choose a supercritical index $s \ne 0$. Lemma C.10 gives $M_s \notin T_s$, and $M_s \notin T_C$ by the unique-midpoint property. In the original open cover, M_s is therefore contained by some other open vertex role. Diameter one restricts this role to U_{s-1} or U_{s+1}. Because it contains an interior point of the adjacent arm, its intersection with that arm has positive length. □

G.2. **Vd1/Vd2 corner estimates.** We next isolate the much stronger trace loss caused by Vd1 and Vd2 geometry. For the following two results use the temporary coordinates

$$X = V_0 + x(V_5 - V_0) + y(V_1 - V_0).$$

Thus $V_0 = (0, 0)$, $V_5 = (1, 0)$, $V_1 = (0, 1)$, $O = (1, 1)$, and $\|(x, y)\|^2 = x^2 + y^2 - xy$.

Proposition G.6 (Vd1/Vd2 corner normal form)**.** *Let T be the closure of an original Vd1 or Vd2 role at V_0, and let a, b be its exact reaches toward V_5, V_1. There is a unique $t > 0$ such that, with $d = \sqrt{t^2 + t + 1}$,*

$$T = \left\{ (x, y) : \begin{array}{r} x - (t+1)y \le a, \\ ty - (t+1)x \le tb, \\ tx + y \le d - a - tb \end{array} \right\}.$$

The raw traces, in coordinates measured from the indicated outer vertex, are

$$\begin{array}{c|c} r_0 & 0 \le s \le \dfrac{d - a - tb}{t+1} \\ r_1 & \dfrac{t(1-b)}{t+1} \le s \le \dfrac{d - a - tb - 1}{t} \\ r_5 & \dfrac{1-a}{t+1} \le s \le d - a - tb - t. \end{array}$$

Intersecting with $[0, 1]$ gives the actual traces. Moreover,

$$a + b < \frac{1}{2}. \tag{61}$$

Proof. A Vd1/Vd2 triangle has one vertex W outside H and two vertices U, Z in H. Since V_0 is an interior convex combination of them, both coordinates of W are negative. The two axis endpoints $(a, 0), (0, b)$ lie on distinct sides incident with W. Put $p = U - W$, $q = Z - W$. Both vectors have positive coordinates and satisfy

$$\|p\| = \|q\| = \|p - q\| = 1.$$

The order in which the two sides cross the positive axes selects the $60°$ rotation

$$q = (p_x - p_y, p_x), \qquad p_x > p_y > 0.$$

Writing $t = (p_x - p_y)/p_y$ gives uniquely

$$p = \frac{(t+1, 1)}{d}, \qquad q = \frac{(t, t+1)}{d}.$$

The two side lines through the axis endpoints and the parallel third side are exactly those in (G.6). Substitution of $(s, s), (s, 1), (1, s)$ gives (G.6).

If the r_1 trace has positive length, its lower endpoint is smaller than its upper endpoint. Clearing denominators gives

$$(t+1)a + tb < (t+1)(d-1) - t^2.$$

Since the left side is at least $t(a + b)$,

$$a + b < \frac{(t+1)(d-1) - t^2}{t} < \frac{1}{2}.$$

For the last inequality, both sides are positive and

$$\left(t^2 + \frac{3t}{2} + 1\right)^2 - (t+1)^2(t^2 + t + 1) = \frac{t^2}{4} > 0.$$

Reflection proves the same conclusion when the supported arm is r_5. □

Lemma G.7 (Vd2 neighboring-midpoint cap). *If an original Vd2 role contains either neighboring midpoint, then its exact boundary reaches satisfy*

$$a + b < \frac{1}{3}.$$

Proof. In Proposition G.6, reflection sends (t, a, b, M_1, M_5) to $(1/t, b, a, M_5, M_1)$, so assume $t \geq 1$ and put $U = a + tb$.

If $M_5 = (1, 1/2)$ is contained, the third half-plane in (G.6) gives

$$a + b \leq U \leq d - t - \frac{1}{2} \leq \sqrt{3} - \frac{3}{2} < \frac{1}{3};$$

the middle expression decreases for $t \geq 1$.

If $M_1 = (1/2, 1)$ is contained, the second and third half-planes give

$$b \geq \frac{t-1}{2t}, \qquad a + b \leq S(t) := d - t - \frac{1}{2t}.$$

Positive-length contact with the other adjacent arm r_5 gives, by (G.6),

$$a + b < R(t) := \frac{2(t+1)d - 3t^2 - t - 2}{2t}.$$

Direct differentiation shows that S is increasing and R decreasing on $[1, \infty)$. For example, the sign calculation for S' reduces to

$$(2t+1)^2 t^4 - (t^2 + t + 1)(2t^2 - 1)^2 = (t+1)(t^3 + 3t^2 - 1) > 0,$$

while $R' < 0$ follows from $d > t + 1/2$. Splitting at $t = 4/3$ yields

$$\begin{aligned} 1 \leq t \leq \frac{4}{3}: &\quad a + b \leq S(4/3) = \frac{8\sqrt{37} - 41}{24} < \frac{1}{3}, \\ t \geq \frac{4}{3}: &\quad a + b < R(4/3) = \frac{7\sqrt{37} - 39}{12} < \frac{1}{3}. \end{aligned}$$

The final inequalities follow respectively from $8\sqrt{37} < 49$ and $7\sqrt{37} < 43$. □

G.3. **Boundary trace caps.** The signed center normal form gives the CE1 and CE2 center contributions from one formula; no separate maximal-interval calculation is needed here.

Theorem G.8 (Boundary trace table). *For closures of original open roles, the following bounds hold:*

role	*hypothesis*	$L_{\partial H}$
T_C	CE0	0
T_C	CE1	$\leq \frac{\sqrt{3}}{2} - \frac{3}{4}$
T_C	CE2	$< \frac{1}{2}$
T_i	Vd0, $A_i + B_i \leq 1$	≤ 1
T_i	Vd0, $A_i + B_i > 1$	$\leq \frac{2}{\sqrt{3}}$
T_i	Vd1/Vd2	$< \frac{1}{2}$
T_i	*T3-like*	$< 1.$

Proof. For every vertex role,

$$L_{\partial H}(T_i) = A_i + B_i. \tag{62}$$

Indeed, the two incident traces are initial intervals of these lengths. A relative-interior point of any nonincident edge is more than unit distance from V_i, whereas V_i is interior to the diameter-one triangle T_i. This excludes any other positive-length boundary trace.

The CE0 row is the definition. Both remaining center rows are the two class specializations of the common signed formula in Corollary F.3.

For a nonsupercritical Vd0 row, (62) is enough. For a supercritical row, its incident endpoints are at squared distance $A_i^2 + A_iB_i + B_i^2 \le 1$, and hence

$$\frac{3}{4}(A_i + B_i)^2 \le 1.$$

This proves the $2/\sqrt{3}$ cap. The Vd1/Vd2 row is (61). Finally, the Type II calculation in Proposition C.6 gives for an original T3-like role

$$A_i + B_i = z - \alpha < z < 1$$

because $0 < t < 1$ and $\alpha > 0$. □

G.4. **Terminal length contradictions.**

Proposition G.9 (Boundary-length branches)**.** *The following branches are impossible:*

(1) *CE0 with* $N_+ = 0$*;*
(2) *CE0 with* $N_+ = 1$ *and at least one Vd1/Vd2 role;*
(3) *CE1 or CE2 with* $N_+ = 0$ *and at least one Vd1/Vd2 role;*
(4) *CE1 with* $N_+ = 1$ *and at least one Vd1/Vd2 role;*
(5) *CE2 with* $N_+ = 1$ *and at least two Vd1/Vd2 roles.*

Proof. In CE0, the center open role contains no relative-interior boundary point. No nonincident vertex role can contain such a point either. Hence on every edge the two incident open vertex traces form two nonempty relatively open sets covering a connected interval; they overlap in positive length. Thus

$$1 < B_i + A_{i+1}$$

on every edge. Summing gives $6 < \sum_i (A_i + B_i)$, contradicting $N_+ = 0$. This proves (1).

Items (2)–(5) are respectively items (1)–(4) of the common budget Corollary G.17. That corollary applies the one master perimeter deficit to the center cap and the indicated strict Vd1/Vd2 caps. □

Proposition G.10 (Vd2 neighboring-midpoint branch)**.** *Assume a CE2,* $N_+ = 1$ *configuration has exactly one Vd1/Vd2 role, that role is Vd2 and contains a neighboring midpoint, and all other vertex roles are Vd0. Then the boundary cannot be covered.*

Proof. This is item (5) of Corollary G.17, using Lemma G.7. □

Proposition G.11 (Branches closed by Strategy 1)**.** *Direct length-sum obstructions close the following routing rows and hybrid subcase:*

(1) *CE0,* $N_+ = 0$*;*

(2) *CE0, $N_+ = 1$, with a Vd1/Vd2 role;*
(3) *CE1/CE2 with $q = N_+ + m \geq 3$;*
(4) *CE1/CE2, $N_+ = 0$, with a Vd1/Vd2 role;*
(5) *CE1, $N_+ = 1$, with exactly one Vd1/Vd2 role and no T3-like role;*
(6) *the Vd2 neighboring-midpoint subcase of the CE2 hybrid row.*

Proof. Items (1), (2), (4), and (5) are Proposition G.9. Item (3) is the short-role count and the three-short-role theorem in Corollary G.17. Item (6) is Proposition G.10. This is only the indicated subcase of the exceptional CE2 row; its remaining placements are handled by the exact-demand strategy. □

G.5. **Common perimeter and skeleton budgets.** The signed center formula also removes repeated length arithmetic. We first state a generic perimeter deficit, then count all short roles before entering the CE1/CE2 case tree.

Lemma G.12 (Master perimeter deficit)**.** *Put*

$$\theta = \frac{2}{\sqrt{3}} - 1.$$

Suppose a branch has a center contribution L_C, n supercritical Vd0 rows, $m \geq 1$ distinguished nonsupercritical rows with strict boundary caps $q_1, \dots, q_m < 1$, and $6 - n - m$ remaining rows whose boundary contributions are at most 1. If

$$L_C + n\theta \leq \sum_{j=1}^{m} (1 - q_j), \tag{63}$$

then the seven roles do not cover ∂H.

Proof. The supercritical Vd0 cap is $2/\sqrt{3}$. Hence the total boundary contribution is strictly less than

$$\begin{aligned} L_C + n\frac{2}{\sqrt{3}} + \sum_{j=1}^{m} q_j + (6 - n - m) &= 6 + L_C + n\theta - \sum_{j=1}^{m} (1 - q_j) \\ &\leq 6. \end{aligned}$$

The strictness follows from $m \geq 1$ and the strict distinguished-row caps. A cover of the perimeter would give the opposite weak inequality by subadditivity. □

Lemma G.13 (Small center slacks in the one-Vd1/Vd2 branch)**.** *Assume there is exactly one supercritical Vd0 row, exactly one Vd1/Vd2 row, and four nonsupercritical Vd0 rows. Then every candidate surviving the perimeter budget is CE2 and satisfies*

$$\boxed{\alpha + \delta < \frac{1}{24}, \qquad \alpha + \delta < \frac{\min\{R, W\}}{6}}. \tag{64}$$

Proof. The Vd1/Vd2 boundary cap is strict and equals 1/2. If the perimeter were covered, Lemma G.12 would force

$$L_C + \theta > \frac{1}{2}.$$

Put

$$\kappa = \frac{1}{2} - \theta = \frac{3}{2} - \frac{2}{\sqrt{3}}.$$

Thus $L_C > \kappa$. The CE1 cap in Corollary F.3 gives

$$L_C + \theta \le \left(\frac{\sqrt{3}}{2} - \frac{3}{4}\right) + \left(\frac{2}{\sqrt{3}} - 1\right) < \frac{1}{2},$$

so every survivor is CE2.

Set $T = \alpha + \delta$. The CE2 formula gives

$$T < M(E) := \frac{RW}{E^2}\left(\frac{E}{1+E} - \kappa\right).$$

Since $RW = 1 - E^2$,

$$M(E) = \frac{1-E}{E^2}\big(E - \kappa(1+E)\big),$$

and

$$M'(E) = -\frac{E - 2\kappa}{E^3} < 0 \qquad \left(\frac{\sqrt{3}}{2} \le E < 1\right).$$

Here $2\kappa < \sqrt{3}/2$, because it is equivalent to $6\sqrt{3} < 11$, whose square is $108 < 121$. Therefore

$$T < M\left(\frac{\sqrt{3}}{2}\right) = \frac{8\sqrt{3}}{9} - \frac{3}{2} < \frac{1}{24}.$$

The last inequality is equivalent to $64\sqrt{3} < 111$, and follows after squaring from $12288 < 12321$.

For the second estimate, use $E/(1+E) < 1/2$:

$$T < \frac{RW}{E^2}\left(\frac{1}{2} - \kappa\right) = \frac{RW}{E^2}\theta.$$

Since

$$E^2 = W + R^2 = R + W^2,$$

one has $RW/E^2 \le R$ and $RW/E^2 \le W$. Also $\theta < 1/6$, equivalently $12 < 7\sqrt{3}$, whose square is $144 < 147$. This proves (64). $\square$

Lemma G.14 (CE2 initial endpoints dominate the total slack)**.** *Assume* $\Delta_R > 0$ *and* $\Delta_L > 0$*. Put*

$$T = \alpha + \delta, \qquad x = \frac{\eta + T}{W}, \qquad y = \frac{\eta + T}{R}.$$

Then

$$\boxed{T < \frac{1}{2}\min\{x, y\}.} \tag{65}$$

Proof. Equation (54) gives $T < \eta/E$. Moreover

$$E^2 - (2R-1)^2 = 3RW > 0,$$

so $|2R - 1| < E$ and therefore $|2R-1|T < \eta$. Hence

$$x - 2T = \frac{\eta - (1-2R)T}{W} > 0,$$

and

$$y - 2T = \frac{\eta - (2R-1)T}{R} > 0.$$

This is (65). $\square$

Theorem G.15 (Three short vertex roles)**.** *Call a vertex role short if it is supercritical or has positive-length intersection with an adjacent radial arm. If an exact-M_0 CE1/CE2 branch contains at least three short vertex roles, then the seven roles do not cover the full skeleton S.*

Proof. The center skeleton cap is

$$L_S(T_C) < \frac{3}{2}.$$

Every short role also has skeleton contribution strictly less than 3/2: this is the supercritical cap for a supercritical role and the translated center-role cap for an original positive-support role. Every remaining role is nonsupercritical and has no adjacent-ray support, so its skeleton contribution is at most 2.

If there are $q \geq 3$ short roles, then

$$\begin{aligned} L_S(T_C) + \sum_{i=0}^{5} L_S(T_i) &< \frac{3}{2} + q\frac{3}{2} + (6-q)2 \\ &= \frac{27-q}{2} \\ &\leq 12. \end{aligned}$$

The full skeleton has length 12, so subadditivity for a cover gives a contradiction. □

Lemma G.16 (Short-role count)**.** *Let m be the number of vertex roles of type Vd1, Vd2, or T3-like, and let q be the number of short vertex roles. Then*

$$\boxed{q = N_+ + m.} \tag{66}$$

Consequently, every exact-M_0 CE1/CE2 branch with $N_+ + m \geq 3$ is impossible.

Proof. The exhaustive vertex classification says that Vd0 is exactly the class with no positive-length adjacent-arm support, while Vd1, Vd2, and T3-like are exactly the three classes with such support. Proposition G.6 gives $A_i + B_i < 1/2$ for Vd1/Vd2, and Lemma C.8 gives $A_i + B_i \leq 1$ for T3-like. Thus none of these m roles is supercritical. Conversely, every supercritical role is Vd0, so it has no positive adjacent-arm support. The two disjoint sources of short roles are therefore the N_+ supercritical roles and the m non-Vd0 roles. This proves (66); Theorem G.15 gives the last assertion. □

Corollary G.17 (Length branches closed by the common budgets)**.** *The following branches are impossible.*

1. *CE0, $N_+ = 1$, and at least one Vd1/Vd2 row;*
2. *$N_+ = 0$ and at least one Vd1/Vd2 row, for either CE1 or CE2;*
3. *CE1, $N_+ = 1$, and at least one Vd1/Vd2 row;*
4. *CE2, $N_+ = 1$, and at least two Vd1/Vd2 rows;*
5. *CE2, $N_+ = 1$, with a Vd2 row containing a neighboring midpoint;*
6. *CE1 or CE2 with $N_+ + m \geq 3$, where m is the number of Vd1, Vd2, or T3-like roles;*
7. *CE1 or CE2 with $N_+ \geq 2$.*

Proof. For item 1, take $L_C = 0$, one supercritical cap $2/\sqrt{3}$, and one strict Vd1/Vd2 cap $1/2$ in Lemma G.12; the needed inequality is

$$\frac{2}{\sqrt{3}} - 1 < \frac{1}{2}.$$

For item 2, the center contribution is strictly less than $1/2$ and the chosen Vd1/Vd2 row has strict cap $1/2$; the other five rows contribute at most 1 each. This is Lemma G.12 with $n = 0$ and $q_1 = 1/2$.

For item 3, the unique supercritical row is Vd0, while the CE1 center cap, the Vd1/Vd2 cap, and the four remaining unit caps give

$$\left(\frac{\sqrt{3}}{2} - \frac{3}{4}\right) + \frac{1}{2} + \frac{2}{\sqrt{3}} + 4 < 6.$$

For item 4, the corresponding CE2 sum is strictly less than

$$\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{2}{\sqrt{3}} + 3 < 6.$$

For item 5, the neighboring-midpoint Vd2 cap is $1/3$, and the total is strictly less than

$$\frac{1}{2} + \frac{1}{3} + \frac{2}{\sqrt{3}} + 4 < 6.$$

Item 6 is Lemma G.16 and Theorem G.15.

Finally suppose $N_+ \geq 2$. Choose a supercritical index $s \neq 0$. Its own midpoint M_s is not covered by that row, and the exact-M_0 center does not cover it. Diameter locality leaves only rows $s-1, s, s+1$ as possible vertex roles containing M_s. Thus one of the adjacent rows contains M_s and, because it is the closure of an original open role, has positive-length support on the adjacent radial arm. By Lemma G.16, this rescuer is nonsupercritical and distinct from the two supercritical rows. Thus there are three short roles, and the theorem applies. □

Appendix H. Short Local Profiles and Exact Vd Placements

This appendix proves the local profile estimates used by the one-special-role branches. A T3-like rescuer and a Vd1 rescuer satisfy the same two scalar inequalities and therefore share one center-hiding theorem. The remaining Vd1/Vd2 placements use the interval residuals from Section D, one global radial envelope, and two immediate consequences of the corner normal form.

For $0 \leq c \leq 1/2$, put

$$B_{\mathrm{sc}}(c) = \frac{c + \sqrt{c^2 - 8c + 4}}{2}, \qquad A_{\mathrm{sc}}(c) = 1 - B_{\mathrm{sc}}(c). \tag{67}$$

The free strict-supercritical envelope says that a strict supercritical row with radial demand at least c has outgoing reach strictly below $B_{\mathrm{sc}}(c)$, and B_{sc} is strictly decreasing.

H.1. A global quarter radial envelope.

For $(p, h) \in [0,1]^2$, let $c_{\max}(p, h)$ be the maximal radial coordinate such that (p, h, c) belongs to the local admissible set.

Lemma H.1 (Quarter radial envelope)**.** *If*

$$0 \leq h \leq p, \qquad p + h < 1,$$

then

$$c_{\max}(p,h) \le 1 - \frac{h}{4}. \tag{68}$$

The inequality is strict on the selected $\mathcal{A}_T$ component.

Proof. Put $s = p+h$ and $q = s^4 - s^2 + ph$. Since $s < 1$, only the two nonsupercritical cells of Proposition J.2 occur.

On $\mathcal{A}_L$, the smaller boundary coordinate is h, and the selected radial frontier is the unique root in $[\sqrt{3}/2, 1]$ of

$$P_h(c) = c^4 - c^2 + hc - h^2 = 0.$$

At $c_0 = 1 - h/4$,

$$P_h(c_0) = \frac{h}{256}(h^3 - 16h^2 - 240h + 128).$$

The cubic decreases on $[0, 1/2]$ and has value $33/8$ at $1/2$, so this is positive for $h > 0$. Moreover

$$P_h\left(\frac{\sqrt{3}}{2}\right) = -\left(h - \frac{\sqrt{3}}{4}\right)^2 \le 0, \qquad P_h'(c) > 0 \quad (c \ge \sqrt{3}/2).$$

Hence the selected root is below c_0; for $h = 0$ equality gives $c_{\max} = 1$.

On $\mathcal{A}_T$, put

$$r = \sqrt{4s^2 - 3}, \qquad \gamma = \frac{p-h}{s}.$$

The exact identity

$$\frac{q}{s^2} = \frac{r^2 - \gamma^2}{4}$$

gives $0 \le \gamma < r$. Write $\gamma = wr$, $0 \le w < 1$. Then

$$p = \frac{s(1+wr)}{2}, \qquad h = \frac{s(1-wr)}{2}, \qquad c_{\max} = \frac{s(1+wr)}{1+r}.$$

For

$$\Psi(w) = c_{\max} + \frac{h}{4}$$

one has

$$\Psi'(w) = \frac{sr(7-r)}{8(1+r)} > 0,$$

so

$$\Psi(w) < \Psi(1) = \frac{s(9-r)}{8}.$$

Since $4s^2 = r^2 + 3$,

$$256 - (r^2+3)(9-r)^2 = (1-r)(r^3 - 17r^2 + 67r + 13) > 0$$

for $0 \le r < 1$; the cubic is increasing and positive on this interval. Thus $s(9-r) < 8$ and $c_{\max} + h/4 < 1$. □

H.2. **The rational T3-like rescuer profile.**

Lemma H.2 (T3-like rescuer profile)**.** *Let a translated T3-like role at V_0 contain M_1, have boundary trace $[0,a]$ on $e_{5,0}$, and have adjacent-radial trace $[c,u]$ on r_1 measured from V_1 toward O. Then*

$$a \le A_{\mathrm{sc}}(c), \qquad \frac{a}{a+1-u} \le A_{\mathrm{sc}}(c). \tag{69}$$

Proof. The translated T3-like normal form supplies

$$1 \le D \le \frac{2}{\sqrt{3}}, \qquad E = \sqrt{4-3D^2}, \qquad \varrho = \frac{D+E}{2},$$

and

$$\frac{E}{2D} \le a \le \frac{4-3D+E}{4D}.$$

Its radial interval is

$$c = \frac{D(1+a)-1}{\varrho}, \qquad u = 1 - \frac{\varrho}{D} + a.$$

Put

$$x = \frac{aD}{\varrho}, \qquad \theta = \frac{D-1}{\varrho}.$$

The relation $\varrho^2 - D\varrho + D^2 = 1$ gives

$$\varrho = \frac{1-2\theta}{1-\theta+\theta^2}, \qquad D = \frac{1-\theta^2}{1-\theta+\theta^2}, \qquad E = \frac{1-4\theta+\theta^2}{1-\theta+\theta^2},$$

with $0 \le \theta \le 2-\sqrt{3}$. Moreover

$$c = x+\theta, \qquad \frac{1-4\theta+\theta^2}{2(1-2\theta)} \le x \le \frac{1}{2} - \theta. \tag{70}$$

Let

$$\Phi(z) = \frac{z(2-z)}{1+z}.$$

It is increasing on $[0,1/2]$ and $\Phi(A_{\mathrm{sc}}(c)) = c$. Hence it is enough to prove $\Phi(x) \le x+\theta$, or equivalently

$$Q_\theta(x) = 2x^2 + (\theta-1)x + \theta \ge 0.$$

The vertex is $x_* = (1-\theta)/4$. If $0 \le \theta \le 1/5$, the lower endpoint x_L in (70) satisfies

$$x_L - x_* = \frac{1-5\theta}{4(1-2\theta)} \ge 0,$$

and

$$Q_\theta(x_L) = \frac{\theta(1-5\theta+11\theta^2-\theta^3)}{2(1-2\theta)^2} \ge 0.$$

If $1/5 \le \theta \le 2-\sqrt{3}$, the vertex lies in the feasible interval and

$$Q_\theta(x) \ge Q_\theta(x_*) = \frac{10\theta-1-\theta^2}{8} > 0.$$

Thus $x \le A_{\mathrm{sc}}(c)$. Since $D \ge \varrho$, one has $a \le x$, and

$$\frac{a}{a+1-u} = \frac{aD}{\varrho} = x.$$

This proves both inequalities. $\square$

H.3. **The Vd1 rescuer profile.**

Lemma H.3 (Vd1 rescuer profile)**.** *Let a Vd1 role at V_0 contain M_1, have boundary trace $[0,a]$ on $e_{5,0}$, and have adjacent-radial trace $[c,u]$ on r_1. Assume it has no positive support on r_5. Then*

$$a \le A_{\mathrm{sc}}(c), \qquad \frac{a}{a+1-u} \le A_{\mathrm{sc}}(c). \tag{71}$$

Proof. The Vd corner normal form gives $t>0$, $d=\sqrt{t^2+t+1}$, and

$$c=\frac{t(1-b)}{t+1}, \qquad u=\frac{d-a-tb-1}{t}.$$

The midpoint condition implies

$$tb=t-c(t+1), \qquad a+tb \le d-1-\frac{t}{2}.$$

One-sided support forces $t\ge 1$. Indeed, if $t<1$, then

$$d-a-tb-t \ge 1-\frac{t}{2} > \frac{1}{t+1} > \frac{1-a}{t+1},$$

so the raw r_5 interval has positive length, contrary to the Vd1 type.

Put

$$F(t,c)=\sqrt{t^2+t+1}-1-\frac{3t}{2}+c(t+1).$$

Then $a\le F(t,c)$. Since $c\le 1/2$,

$$\frac{\partial F}{\partial t}=\frac{2t+1}{2\sqrt{t^2+t+1}}-\frac{3}{2}+c \le \frac{2t+1}{2\sqrt{t^2+t+1}}-1<0,$$

where the last inequality follows from $(2t+1)^2<4(t^2+t+1)$. Thus F decreases for $t\ge 1$, and

$$a\le F(t,c)\le L(c):=\sqrt{3}-\frac{5}{2}+2c.$$

Every feasible row has $L(c)\ge 0$. The inequality $2L(c)\le A_{\mathrm{sc}}(c)$ is equivalent to

$$\sqrt{c^2-8c+4}\le 12-4\sqrt{3}-9c.$$

The right side is positive on $[0,1/2]$. Squaring is therefore legitimate, and the squared difference is $4Q(c)$, where

$$Q(c)=20c^2+(18\sqrt{3}-52)c+47-24\sqrt{3}.$$

The polynomial is decreasing on $[0,1/2]$ and $Q(1/2)=26-15\sqrt{3}>0$. Hence $2L(c)\le A_{\mathrm{sc}}(c)$, and therefore $a\le A_{\mathrm{sc}}(c)$.

Put $\varepsilon=1-u>0$. The endpoint formula gives

$$\varepsilon=\varepsilon_0+\frac{a}{t}, \qquad \varepsilon_0=\frac{2t+1-c(t+1)-d}{t}.$$

Since $c\le 1/2$ and $t\ge 1$,

$$2t+1-c(t+1)-d\ge\frac{3t+1}{2}-d>0;$$

the last inequality follows after squaring from $(3t+1)^2/4-d^2=(5t^2+2t-3)/4>0$. Thus $\varepsilon_0>0$. Moreover,

$$\varepsilon_0+\frac{F(t,c)}{t}=\frac{1}{2}.$$

Since $z/(z+\varepsilon_0+z/t)$ increases in z,

$$\frac{a}{a+\varepsilon} \leq \frac{F(t,c)}{F(t,c)+1/2} \leq \frac{L(c)}{L(c)+1/2} \leq 2L(c) \leq A_{\mathrm{sc}}(c).$$

□

H.4. **A common adjacent-rescuer theorem.**

Lemma H.4 (Common adjacent-rescuer obstruction)**.** *Let a local rescuer T_0 have trace $[0,a]$ on $e_{5,0}$ and adjacent-radial trace $[c,u]$ on r_1, measured from V_1 toward O. Put*

$$\varepsilon = 1-u > 0, \qquad \Theta = \frac{a}{a+\varepsilon}.$$

Assume

$$a \leq A_{\mathrm{sc}}(c), \qquad \Theta \leq A_{\mathrm{sc}}(c), \tag{72}$$

and assume radial isolation forces the center to cover the O-side gap of length ε. If T_1 is the unique strict supercritical row and T_2, T_3, T_4, T_5 are nonsupercritical, then the perimeter and r_1 are not covered.

Proof. The center exit on r_1 is δ/R, so radial isolation gives

$$\delta \geq R\varepsilon.$$

Let h be the reach forced on T_5 along $e_{5,0}$. If the companion center trace does not hide the gap after coordinate a, then

$$h \geq 1-a \geq B_{\mathrm{sc}}(c).$$

In the only hiding order,

$$\frac{k}{W} \leq a < R+\alpha.$$

The left endpoint inequality and radial isolation give

$$k \leq Wa, \qquad R\varepsilon \leq Wa,$$

so $R(a+\varepsilon) \leq a$ and $R \leq \Theta$. Also

$$\alpha \leq Wa - R\varepsilon - \eta,$$

whence

$$R+\alpha \leq a + R(u-a).$$

The hiding order forces $u > a$, and therefore

$$R+\alpha \leq a + \Theta(u-a) = \Theta \leq A_{\mathrm{sc}}(c).$$

Thus in every ordering

$$h \geq B_{\mathrm{sc}}(c).$$

Coverage of r_1 gives the supercritical row radial demand at least c, so its outgoing reach satisfies

$$b_1 < B_{\mathrm{sc}}(c) \leq h.$$

Lemma D.6, applied to T_2, T_3, T_4, T_5, gives

$$\sum_{i=2}^{5}(a_i+b_i) \geq 4+h-b_1 > 4,$$

contrary to their four nonsupercritical caps. □

Corollary H.5 (T3-like and Vd1 adjacent rescue)**.** *The common adjacent-rescuer lemma applies both to the exactly-one-T3-like branch and to the normalized Vd1 row containing the neighboring midpoint.*

Proof. Lemmas H.2 and H.3 give the two local inequalities. In both branches, midpoint and support isolation force the center to cover the O-side radial gap before the rescuer begins. ☐

H.5. **Adjacent Vd1/Vd2 placement.**

Lemma H.6 (Adjacent Vd1/Vd2 placement)**.** *Assume a reduced CE2 exact-M_0 candidate has $N_+ = 1$, row T_0 uniquely supercritical, row T_1 the unique Vd1/Vd2 row, and rows T_2, T_3, T_4, T_5 nonsupercritical Vd0. Then the perimeter together with r_2 is not covered. The reflected T_5 placement is also impossible.*

Proof. Let

$$I_L = \left[\frac{k}{W}, R+\alpha\right], \qquad I_R = \left[\frac{k}{R}, W+\delta\right].$$

For the supercritical reaches (a_0, b_0), put

$$A = \mathcal{R}_{I_R}(b_0), \qquad H = \mathcal{R}_{I_L}(a_0).$$

The endpoint-distance inequality is

$$a_0^2 + a_0 b_0 + b_0^2 \le 1.$$

It implies $a_0, b_0 < 1$. The signed center bounds give $R+\alpha < 1$, so $H > 0$. Boundary propagation gives

$$a_1 \ge A, \qquad b_i \ge H \quad (1 \le i \le 5).$$

The Vd cap gives

$$A + H < \frac{1}{2}. \tag{73}$$

The common perimeter budget gives

$$T := \alpha + \delta < \frac{1}{24}, \qquad T < \frac{\min\{R, W\}}{6}.$$

We claim

$$\delta < \frac{H}{4}. \tag{74}$$

If $H = W - \alpha$, then

$$4\delta + \alpha \le 4T < \frac{2W}{3} < W.$$

Suppose $H = 1 - a_0$. The other residual cannot be $1 - b_0$, since otherwise (73) would give $a_0 + b_0 > 3/2$, whereas the endpoint-distance inequality gives $a_0 + b_0 \le 2/\sqrt{3} < 3/2$. Hence

$$A = 1 - (W + \delta), \qquad b_0 \ge y := \frac{k}{R},$$

and $W + \delta > 1/2 + H$. Diameter transfer gives

$$H > \frac{y}{2} > \frac{\eta}{2R}.$$

Thus

$$\delta > J(R) := R - \frac{1}{2} + \frac{\eta}{2R}.$$

Writing $\eta/R = W/(1+E)$, exact differentiation gives

$$4E(1+E)^2 J'(R) = 2E(1+E)(1+2E) - W(2R-1).$$

If $R \le 1/2$, the second term is nonnegative after subtraction. If $R \ge 1/2$, then $W(2R-1) \le 1/8$, while the first term is larger than $1/8$. Thus $J'(R) > 0$. Since $J(3/8) = 1/24$, the bound $\delta < 1/24$ forces $R < 3/8$. Then

$$(2-3R)^2 - E^2 = (3-8R)(1-R) > 0,$$

so $\eta/R > 1/3$ and $H > 1/6$. Hence $4\delta < 1/6 < H$, proving (74).

The center interval on r_2 begins from the vertex side at $1-\delta$. If T_1 supports r_2, its exact endpoint satisfies

$$u_2 < 1 - b_1 - \frac{a_1}{t} \le 1 - H < 1 - \delta.$$

Thus it does not bridge to the center. For T_2, boundary handoff gives

$$a_2 > \frac{1}{2} + A, \qquad b_2 \ge H.$$

Put $p = 1/2 + A$. Equation (73) gives $0 \le H \le p$ and $p + H < 1$. Lemma H.1 gives

$$c_2 \le c_{\max}(p, H) \le 1 - \frac{H}{4} < 1 - \delta.$$

Thus neither possible radial bridge reaches the center interval, and r_2 is uncovered. $\square$

H.6. **Nonadjacent Vd1/Vd2 placements.**

Lemma H.7 (Nonadjacent Vd1/Vd2 placements)**.** *Under the same reduced CE2 branch, let the unique Vd1/Vd2 row be T_τ with $\tau \in \{2,3,4\}$. Then the perimeter together with r_τ is not covered.*

Proof. Put

$$T = \alpha + \delta, \qquad x = \frac{\eta + T}{W}, \qquad y = \frac{\eta + T}{R}.$$

Lemma G.14 gives

$$T < \frac{1}{2}\min\{x, y\}.$$

Let (a_0, b_0) be the reaches of the unique supercritical row. Then

$$a_0 + b_0 > 1, \qquad a_0^2 + a_0 b_0 + b_0^2 \le 1.$$

Let

$$B = e_{[y, W+\delta]}(b_0), \qquad U = e_{[x, R+\alpha]}(a_0), \qquad A = 1 - B, \qquad H = 1 - U.$$

Boundary propagation and the Vd cap give

$$a_\tau \ge A, \qquad b_\tau \ge H, \qquad A + H < \frac{1}{2}.$$

Let $\beta(q) = (-q + \sqrt{4 - 3q^2})/2$. The component endpoint B is either b_0 or $W + \delta$. In the latter case the center diameter gives $B \le \beta(x)$. In the former case, if $a_0 < x$, then $U = a_0$ and $A + H < 1/2$ would force $a_0 + b_0 > 3/2$, contrary to $a_0 + b_0 \le 2/\sqrt{3}$. Hence $a_0 \ge x$, and $B = b_0 \le \beta(a_0) \le \beta(x)$. Reflection gives $U \le \beta(y)$. Consequently

$$A > \frac{x}{2} > T, \qquad H > \frac{y}{2} > T.$$

Thus

$$d_\tau^C < \min\{A, H\} \qquad (\tau = 2, 3, 4),$$

because the three exits are δ, α, and $\min\{\alpha/R, \delta/W\} \le T$.

The Vd corner normal form gives the own-radial margin

$$c_\tau < 1 - \min\{A, H\}.$$

But the center interval begins from the vertex side at

$$1 - d_\tau^C > 1 - \min\{A, H\}.$$

The two radial traces are disjoint, and all other roles are excluded by Vd0 locality and diameter locality. □

H.7. **Placement assembly.**

Proposition H.8 (Short one-Vd1/Vd2 placement assembly)**.** *In the CE2, $N_+ = 1$, exactly-one-Vd1/Vd2 branch with no additional positive-support row, every placement is impossible directly or reduces to the all-Vd0 boundary-loss theorem.*

Proof. If the supercritical row is T_0, an adjacent Vd1/Vd2 row is eliminated by Lemma H.6, while a nonadjacent one is eliminated by Lemma H.7. If the unique Vd1/Vd2 row is T_0, midpoint forcing makes the supercritical row adjacent. The Vd2 alternative is excluded by the neighboring-midpoint boundary cap; the Vd1 alternative is Corollary H.5. If neither special row is T_0, midpoint forcing again makes them adjacent. A Vd2 neighbor rescue is impossible, while a Vd1–supercritical pair is replaced by two nonsupercritical Vd0 rows as in Lemma L.10; the resulting configuration is contradicted by Proposition L.2. The positive-support complement is already excluded by Theorem G.15. □

Appendix I. Strategy 2: Relaxations of g-Composition Chains

The signed center section supplies the exact CE1/CE2 traces, exits, and actual-row interfaces. Section D supplies the canonical maps

$$g_c, \qquad \widehat{g}_c, \qquad g_c^\vee, \qquad \widehat{g}_c^\vee,$$

together with center-assisted subscripts and certified lower relaxations. Every proof below begins with an exact actual-row chain. A path-budget or identity relaxation is used only after all center and nonincident contributions on its internal edges have been excluded. The complete T3-like endpoint audit and Vd1 replacement are in Appendix M.

I.1. **The common chain table.** For maps listed in geometric row order,

$$[\Phi_1 \mid \cdots \mid \Phi_r](x) = (\Phi_r \circ \cdots \circ \Phi_1)(x).$$

branch	exact skeleton	relaxations used	terminal certificate
$N_+ = 0$, all Vd0, $\mathrm{gr} = 0$	cyclic six-row $[\widehat{g}^{\vee}_{C_i}]$ chain	I^6	strict cyclic ascent
$N_+ = 0$, all Vd0, $\mathrm{gr} = 1$	two exact hatted endpoint caps and three ordinary rows	center-free middle I^3	endpoint sum < 1
CE2, all Vd0, $\mathrm{gr} = 2$, $N_+ = 0$ or 1	paired exact endpoint caps and three ordinary rows	center-free middle I^3	paired sum < 1
$N_+ = 0$, one or more T3-like, no Vd1/Vd2	two residual endpoint caps and three ordinary rows	center-free middle I^3	exact four-label sum < 1
CE1, $N_+ = 1$, all Vd0, one gap	$[\widehat{g}^{\vee}_{c_1} \mid \cdots \mid \widehat{g}^{\vee}_{c_5}](X)$	two affine slots and one threshold slot after duality	$Z > 1 - H$; equivalently reversed output $> 1 - X$
CE2, $N_+ = 1$, all Vd0, one gap	the same five-row chain	one available threshold slot and four identity slots	$Z > 1 - H$
one T3-like or adjacent Vd1 rescuer	free supercritical source and an ordinary four-row path	endpoint envelope plus path budget	$b_1 < g_c^{\mathrm{sc}} \le h$
adjacent Vd1/Vd2 placement	center residuals A, H and two possible radial bridges	quarter radial envelope	both bridges end before the center trace
nonadjacent Vd1/Vd2 placement	center residuals and Vd corner profile	identity propagation on ordinary edges	Vd-specific radial separation
Vd1–supercritical replacement	one exceptional adjacent pair	explicit open Vd0 axis replacements	all-Vd0 contradiction

Table 4. All Strategy 2 relaxations in the canonical g-family.

I.2. **The all-**Vd0 **gap-rank kernel.** Let gr be the number of positive center traces that contain a V-gap. Then

$$\mathrm{gr} \in \{0, 1, 2\}, \qquad \mathrm{gr} \le 1 \quad \text{for CE1.}$$

The two all-Vd0 row counts are governed by

	$\mathrm{gr} = 0$	$\mathrm{gr} = 1$	$\mathrm{gr} = 2$
$N_+ = 0$	strict identity cycle	one-side endpoint chain	paired endpoint chain
$N_+ = 1$	nine-point obstruction	five-row relaxed chain	paired endpoint chain.

The last column is CE2-only.

For $N_+ = 0$ and $\mathrm{gr} = 0$, no center trace contains a missed handoff; hence the six incident vertex traces alone cover every boundary edge. Thus

$$A_{i+1} > 1 - B_i \ge 1 - \widehat{g}_{C_i}(1 - A_i) = \widehat{g}^{\vee}_{C_i}(A_i) \ge A_i.$$

Iteration gives the impossible cycle

$$A_0 < A_1 < \cdots < A_5 < A_0.$$

For $N_+ = 0$ and $\mathrm{gr} = 1$, Proposition 4.1 retains the two endpoint caps exactly. The center trace lies on the endpoint edge, while the three intervening edges are center-free; Lemma D.6 therefore applies to the three middle nonsupercritical rows. The T3-like $N_+ = 0$ branch has the same ordinary path after the endpoint components are replaced by the exact residuals of Definition 4.3.

Proposition I.1 (Common CE2 two-gap application)**.** *Assume both CE2 center traces contain V-gaps and rows $T_1, \ldots, T_5$ are nonsupercritical Vd0 rows. Then the perimeter is not covered. No hypothesis on the criticality of T_0 is required.*

Proof. Put

$$p = W - \alpha, \qquad q = R - \delta.$$

The two gaps force endpoint inputs p, q and radial demands $p/W, q/R$. Proposition 4.1 gives

$$B_5 + B_1 \le \widehat{g}_{p/W}(1-p) + \widehat{g}_{q/R}(1-q) < 1.$$

The center traces lie only on the two endpoint edges. The three middle rows T_2, T_3, T_4 therefore form a center-free path through the four intervening edges. Lemma D.6 forces their total boundary contribution above three, contrary to their three unit caps. □

For $N_+ = 1$ and $\mathrm{gr} = 0$, the active proof is the direct nine-point obstruction of Strategy 4.

I.3. **One signed five-row chain.** Use the signed center data of Proposition F.1. Put

$$X = R - \delta, \qquad H = \frac{\eta + \alpha + \delta}{2R}, \qquad m = \min\left\{\frac{\alpha}{R}, \frac{\delta}{W}\right\},$$

and

$$c_1 = 1 - \frac{\delta}{R}, \quad c_2 = 1 - \delta, \qquad c_3 = 1 - m,$$
$$c_4 = 1 - \alpha, \quad c_5 = 1 - \frac{\alpha}{W}.$$

A gap in the normalized right trace starts the same exact chain for CE1 and CE2:

$$Z = [\widehat{g}^{\vee}_{c_1} \mid \widehat{g}^{\vee}_{c_2} \mid \widehat{g}^{\vee}_{c_3} \mid \widehat{g}^{\vee}_{c_4} \mid \widehat{g}^{\vee}_{c_5}](X). \tag{75}$$

Proposition F.5 connects the formal chain to actual reaches and gives

$$A_0 \ge Z, \qquad A_0 < 1 - H.$$

Thus the exact contradiction is $Z > 1 - H$. The first and fifth slots are extensive, and three capped-map duality steps reduce this target to

$$[\widehat{g}^{\vee}_{1-\alpha} \mid \widehat{g}^{\vee}_{1-m} \mid \widehat{g}^{\vee}_{1-\delta}](H) > 1 - X = W + \delta. \tag{76}$$

Proposition I.2 (CE1 one-gap obstruction)**.** *The CE1, $N_+ = 1$, all-*Vd0*, one-gap state is impossible.*

Proof. Here $\Delta_L \le 0$. The exact label analysis first disposes of the Full, L, and T_- ranges. On the sole surviving selected-T_+ ranges, the universal chord estimates give

$$p_1 > \widehat{g}^{\vee,\,1-4\alpha}_{1-\alpha}(H) =: L_1,$$
$$p_2 > \widehat{g}^{\vee,\,1-5m}_{1-m}(L_1) =: L_2.$$

The exact scalar calculation in Appendix K proves

$$\delta < \frac{1}{10}, \qquad p_2 > L_2 > 2\delta + 3\delta^2 > e(\delta).$$

The final threshold map therefore returns a value above $1 - X$, proving (76). The exact-row interface then gives $Z > 1 - H$, contradicting $A_0 < 1 - H$. □

Corollary I.3 (CE2 one-gap obstruction)**.** *The CE2, $N_+ = 1$, all-*Vd0*, exactly-one-gap state is impossible in either orientation.*

Proof. Proposition F.7 proves

$$X > e(\alpha), \qquad \min\{e(\alpha), e(\delta)\} < H.$$

If $e(\alpha) < H$, use the row-4 threshold slot and replace the other four slots by I. Otherwise $e(\delta) < H \leq e(\alpha) < X$, so use the row-2 threshold slot. At least one slot fires; both may do so. In either case $Z > 1 - H$. Reflection exchanges R, W, exchanges α, δ, and reverses the row order. □

I.4. **The remaining transfer branches.** For $N_+ = 0$, the all-Vd0 proof is the strict identity cycle, the one-side endpoint theorem, and Proposition I.1. The T3-like branch is Theorem M.1 followed by the center-free three-row path budget.

For the exactly-one-T3-like $N_+ = 1$ branch, the translated local profile gives

$$a \leq 1 - g_c^{\mathrm{sc}}, \qquad \frac{a}{a + 1 - u} \leq 1 - g_c^{\mathrm{sc}}.$$

The common center-hiding theorem forces an opposite endpoint reach at least g_c^{sc}, while the unique supercritical row has outgoing reach strictly below g_c^{sc}. Its four intervening internal edges form the ordinary center-free path specified in Corollary H.5.

For the one-Vd1/Vd2 branch, the preliminary perimeter budget removes CE1 and gives

$$\alpha + \delta < \frac{1}{24}, \qquad \alpha + \delta < \frac{\min\{R, W\}}{6}.$$

The adjacent and nonadjacent placements are the exact radial-separation lemmas in Appendix H. The remaining Vd1 adjacent pair is replaced by Proposition M.2; the resulting all-Vd0 configuration is impossible.

Proposition I.4 (Branches closed by relaxed g-chains)**.** *Strategy 2 closes:*

1. *the $N_+ = 0$ all-*Vd0 *and T3-like-without-Vd1/Vd2 branches for* CE1 *and* CE2*;*
2. *every positive-gap $N_+ = 1$ all-*Vd0 *state;*
3. *the exactly-one-T3-like $N_+ = 1$ state with no Vd1/Vd2 role;*
4. *the complementary transfer placements in the* CE2*, $N_+ = 1$, exactly-one-Vd1/Vd2 branch.*

Singleton gaps are included.

Proof. The first item is Propositions L.2 and L.5, with the endpoint proof completed by Theorem M.1. The second is Propositions I.2, I.3, and I.1. The third is Proposition L.1. The fourth is Proposition H.8, with its replacement step supplied by Proposition M.2. Every transfer is connected to actual rows by Lemmas D.2 and D.4. □

Appendix J. Exact Local Demand Calculus

This section develops the local calculus for configurations in which a coarse length sum does not by itself record enough information. The mechanism is always the same. A center trace fixes boundary inputs and complementary radial demands; an exact local map propagates these data through nonsupercritical rows; and the returning demand exceeds the remaining boundary or radial capacity.

Actual maximal reaches continue to be denoted by (A_i, B_i, C_i), while lowercase letters denote prescribed demands. Every estimate in this section is proved analytically. In particular, no empirical branch catalogue, machine-assisted certificate, or formal root from a discarded algebraic component is used.

J.1. **The exact local feasible set.** Put

$$u = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \qquad v = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right).$$

For $0 \le a, b, c \le 1$ define

$$K(a,b,c) = \operatorname{conv}\{0, au, bv, c(u+v)\}.$$

Thus a, b are lower demands on the two incident boundary rays and c is a lower demand on the inward radial ray.

Definition J.1. The local admissible set is

$$\begin{aligned}\mathcal{A} = \{(a,b,c) \in [0,1]^3 : K(a,b,c) &\text{ is contained}\\ &\text{in a closed unit equilateral triangle}\}.\end{aligned}$$

For $a, c \in [0,1]$ put

$$\begin{aligned}B_c(a) &= \max\{b : (a,b,c) \in \mathcal{A}\},\\ F_c(a) &= \min\{B_c(a), 1-a\},\\ G_c(a) &= 1 - F_c(a).\end{aligned}$$

The distinction between demands and actual reaches is important. If a role has actual reaches (A, B, C) and $A \ge a$, $C \ge c$, then (a, B, c) is a demand triple realized by the same role. If in addition $A + B \le 1$, then

$$B \le F_c(a). \tag{77}$$

No assertion about the type of a maximizing triangle is required.

Proposition J.2 (Exact admissible set and radial envelope)**.** *Put*

$$s = a+b, \qquad d = a^2+ab+b^2, \qquad m = \min\{a,b\}, \qquad M = \max\{a,b\},$$

and

$$q = s^4 - s^2 + ab.$$

Define

$$\begin{aligned}F_L(a,b,c) &= c^4 - c^2 + mc - m^2,\\ F_T(a,b,c) &= (s^2-1)c^2 + Mc - M^2,\\ F_S(a,b,c) &= (m^2-1)c^2 + (2mM^2+M)c + M^4 - M^2.\end{aligned}$$

Then $\mathcal{A} = \mathcal{A}_L \cup \mathcal{A}_T \cup \mathcal{A}_S$*, where*

$$\begin{aligned}\mathcal{A}_L &= \{d \le 1, s \le 1, q \le 0, F_L \le 0\},\\ \mathcal{A}_T &= \{d \le 1, s \le 1, q \ge 0, F_T \le 0, c \le 2M\},\\ \mathcal{A}_S &= \{d \le 1, s > 1, F_S \le 0, c \le \tfrac{1}{2}\}.\end{aligned}$$

All variables in these three sets are also constrained to $[0,1]$.

For $d \le 1$, the maximal radial demand is

$$c_{\max}(a,b) = \begin{cases} 1, & a=b=0, \\ C_L(m), & s\le 1, q\le 0, (a,b)\ne(0,0), \\ C_T(m,M), & s\le 1, q>0, \\ C_S(m,M), & s>1, \end{cases} \tag{78}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0,$$

and

$$C_T(m,M) = \frac{2M}{1+\sqrt{4s^2-3}}, \qquad C_S(m,M) = \frac{M(2mM+1) - M\sqrt{4d-3}}{2(1-m^2)}.$$

If $d > 1$, the radial envelope is undefined. At $q = 0$ the two subcritical formulas agree and equal s.

Proof. If $s > 0$ and $cs \le ab$, then $c/a + c/b \le 1$ (with the zero-coordinate cases obtained by continuity), so $c(u+v)$ lies in $\operatorname{conv}\{0, au, bv\}$. Its minimum enclosing equilateral side is the diameter

$$\sqrt{a^2+ab+b^2}.$$

Assume now that $cs > ab$. The four points occur cyclically as $0, bv, c(u+v), au$. For three outward unit normals separated by $120°$, the required side equals $2/\sqrt{3}$ times the sum of the three support values. On an angular cell with fixed support points this sum has second derivative equal to its negative. Hence an angular minimum occurs when a normal is perpendicular to one of the four hull edges. The four resulting side lengths are

$$L_{0A} = b + \max\{a,c\}, \qquad L_{B0} = a + \max\{b,c\},$$

$$L_{AC} = \begin{cases} (a^2+bc)/\sqrt{a^2-ac+c^2}, & a\ge c, \\ c(a+b)/\sqrt{a^2-ac+c^2}, & a<c\le a+b, \\ c^2/\sqrt{a^2-ac+c^2}, & c>a+b, \end{cases}$$

and L_{CB} is obtained by interchanging a, b. Thus the minimum side is the minimum of these four expressions.

Ordering a, b as $m \le M$, squaring the positive support equations gives $F_L = 0, F_T = 0, F_S = 0$ in the three respective contact regimes; the inactive support inequalities give $q \le 0, q \ge 0, s > 1$. Squaring creates one spurious component in each of the last two cells. In the T cell the two roots are

$$\frac{2M}{1+\sqrt{4s^2-3}}, \qquad \frac{2M}{1-\sqrt{4s^2-3}},$$

and $c \le 2M$ selects the first. In the S cell the smaller root lies below $1/2$ and the other above $1/2$; hence $c \le 1/2$ is exactly the geometric selector. The L fiber is the interval from zero to its unique selected root. Solving the selected equations gives (78). This also proves that $\mathcal{A}$ is coordinatewise down-closed. $\square$

J.2. **The exact capped map.** For $1/2 < c < 1$ put

$$\ell(c) = \frac{c}{2}\left(1 - \sqrt{4c^2 - 3}\right), \qquad r(c) = c - \ell(c)$$

when $c \geq \sqrt{3}/2$, and use the order-transition function $h(c)$ from (32). Also define

$$T_-(a,c) = \frac{\sqrt{a^2 - ac + c^2}}{c} - a,$$

$$T_+(a,c) = \frac{2(1-a^2)c^2}{2ac^2 + c + \sqrt{(2ac^2+c)^2 - 4(1-c^2)(1-a^2)c^2}}.$$

Proposition J.3 (Four-label capped map)**.** *For $1/2 < c \leq \sqrt{3}/2$,*

$$F_c(a) = \begin{cases} 1-a, & 0 \leq a \leq 1-c, \\ T_+(a,c), & 1-c < a < h(c), \\ h(c), & a = h(c), \\ T_-(a,c), & h(c) < a < c, \\ 1-a, & c \leq a \leq 1. \end{cases} \tag{79}$$

For $\sqrt{3}/2 < c < 1$,

$$F_c(a) = \begin{cases} 1-a, & 0 \leq a \leq 1-c, \\ T_+(a,c), & 1-c < a \leq \ell(c), \\ \ell(c), & \ell(c) < a < r(c), \\ T_-(a,c), & r(c) \leq a < c, \\ 1-a, & c \leq a \leq 1. \end{cases} \tag{80}$$

Thus there are exactly four labels

$$\text{Full}, \qquad L, \qquad T_-, \qquad T_+^{\text{hi}}.$$

At $a = \ell(c)$ in (80), $F_c(a) = r(c)$; immediately to its right the value is $\ell(c)$. This downward jump is genuine. The formal other root of the T_+ quadratic is not feasible because it violates $c \leq 2\max\{a, F_c(a)\}$.

The maps F_c are nonincreasing, the maps $G_c = 1 - F_c$ are nondecreasing and extensive, and

$$G_c(a) \leq z \iff G_c(1-z) \leq 1-a. \tag{81}$$

Proof. The cap $a + b \leq 1$ eliminates the S cell of Proposition J.2. On the cap $b = 1-a$, the T inequality reduces to $M(c-M) \leq 0$; hence the Full face is feasible exactly for $a \leq 1-c$ or $a \geq c$. Off that face a maximal point lies on $F_L = 0$ or $F_T = 0$. The L equation has roots $\ell(c), r(c)$ and the selector $q \leq 0$ keeps only the smaller coordinate. The ordered T halves give respectively the displayed lower quadratic root T_+ and the root T_-. Their order boundary is $a = b = h(c)$, while their $q = 0$ boundaries are $(\ell(c), r(c))$ and $(r(c), \ell(c))$. These observations give (79)–(80), including all equality assignments.

Down-closedness gives monotonicity, while $F_c(a) \leq 1-a$ gives $G_c(a) \geq a$. Finally,

$$G_c(a) \leq z \iff F_c(a) \geq 1-z \iff (a, 1-z, c) \in \mathcal{A}, \quad a + 1 - z \leq 1.$$

Reflection $a \leftrightarrow b$ gives (81). □

Put

$$\mu = \frac{2\sqrt{3}-3}{4},$$

and, for $0 < X < 1 - \sqrt{3}/2$, write

$$e(X) = \ell(1-X).$$

The root equation gives

$$\begin{aligned} &X < e(X) && \left(0 < X < 1 - \frac{\sqrt{3}}{2}\right), \\ &e(X) < \frac{2X}{1-2X} && (0 < X \le \mu), \\ &e(X) \le 2X + 5X^2 && (0 < X \le 1/8). \end{aligned} \tag{82}$$

Moreover,

$$a > e(X) \implies G_{1-X}(a) \ge 1 - e(X) \qquad \left(0 < X < 1 - \frac{\sqrt{3}}{2}\right). \tag{83}$$

Indeed, for

$$p_X(z) = z^2 - (1-X)z + (1-X)^2 - (1-X)^4,$$

one has

$$p_X(X) = X(1 - 3X + 4X^2 - X^3) > 0.$$

Since $X < (1-X)/2$, this puts X strictly below the smaller root $e(X)$. For $U = 2X/(1-2X)$, direct substitution gives

$$p_X(U) = -\frac{X^2}{(1-2X)^2}(4X^4 - 20X^3 + 37X^2 - 28X + 3).$$

The polynomial in parentheses is decreasing on $[0, \mu]$ and at the right endpoint equals $8217/64 - 74\sqrt{3} > 0$. Indeed, its derivative is at most $16/512 + 74/8 - 28 < 0$, and $(8217/64)^2 - 3 \cdot 74^2 = 230001/4096 > 0$. Hence $p_X(U) < 0$, which puts U strictly between the roots. Finally,

$$p_X(2X + 5X^2) = X^2(3X+4)(8X-1) \le 0,$$

and $X < 2X + 5X^2 < (1-X)/2$ on the stated range. Thus the trial point lies between the two roots, proving the third comparison. The implication (83) is read directly from (80). We shall also use $e(X) < 2X + 3X^2$ for $0 < X \le 1/10$, obtained from $p_X(2X + 3X^2) = X^2(8X^2 + 19X - 2) < 0$; here too the trial point lies strictly between X and $(1-X)/2$.

Appendix K. Cyclic Propagation in the All-Vd0 Cases

K.1. **The exactly-one-supercritical all-**Vd0 **branch.** For a center interval, call the nonempty closed set missed by its two open endpoint roles a *V-gap*. It may be a singleton. If the maximal closed center trace is $[s, t]$, open coverage gives the weak bounds $B_i \ge s$ and $A_{i+1} \ge 1 - t$; thus none of the arguments below discards a singleton gap.

Proposition K.1 (CE1 one-gap five-map obstruction)**.** *Assume that every vertex role is* Vd0*,* $N_+ = 1$*, and the center role is CE1. If its boundary trace contains a V-gap, the roles do not cover* H*.*

Proof. Use the signed center variables of Proposition F.1, and put

$$w = W, \qquad A = \alpha, \qquad D = \delta, \qquad X = R - D, \qquad H = \frac{\eta + A + D}{2R}.$$

The CE1 sign gives $D + RA \geq P$ and $RD > wA$, so

$$m = \min\left\{\frac{A}{R}, \frac{D}{w}\right\} = \frac{A}{R}.$$

The exact strict domain needed below is

$$\begin{gathered} A > 0, \quad D > 0, \quad A + wD < P, \quad D + RA \geq P, \\ A < \frac{w}{2}, \quad D < \frac{R}{2}, \quad RD - wA > 0, \end{gathered} \tag{84}$$

Proposition F.5 contains the five actual-row handoffs, the three duality steps, and the terminal diameter comparison. It remains only to prove its scalar target

$$(G_{1-D} \circ G_{1-m} \circ G_{1-A})\,(H) > 1 - X. \tag{85}$$

Here $A < P \leq \mu$, because $P = E(1-E)$ and $E \geq \sqrt{3}/2$. We first prove the two input comparisons used at row 4. Since $D = R - X$, the inequality $A + wD < P$ and the identity $Rw = \eta + P$ give

$$A < wX - \eta.$$

Put

$$\varphi(q) = wq - \frac{q}{2(1+q)}.$$

The bound $0 < D < R/2$ gives $R/2 < X < R$, and φ is convex. At the endpoints,

$$\varphi(R/2) < \frac{Rw}{2} < \eta,$$

while $\varphi(R) < \eta$ is equivalent to

$$E(1 - 2R^2) < 1,$$

which follows from $E < 1$. Convexity therefore gives $\varphi(X) < \eta$, and hence

$$A < \frac{X}{2(1+X)}.$$

Using the rational low-root bound in its valid range,

$$e(A) < \frac{2A}{1-2A} < X. \tag{86}$$

We also have

$$2R(H - A) = \eta + D + (1 - 2R)A.$$

This is positive if $R \leq 1/2$. If $R > 1/2$, then $A < P = \eta E$ gives

$$2R(H - A) > \eta\big(1 - (2R - 1)E\big) > 0.$$

Thus $H > A$. The positive signed right trace gives

$$\frac{\eta + A + D}{R} < w + D < 1,$$

so $H < 1/2 < 1 - A$.

Now consult the exact high-radial catalog for $c_4 = 1 - A$. The lower and upper Full ranges would require $H \leq A$ and $H \geq 1 - A$, respectively, so both are impossible. On the L or T_- range,

$$F_{1-A}(H) \leq e(A),$$

and therefore

$$G_{1-A}(H) \geq 1 - e(A) > 1 - X$$

by (86). Thus only the T_+^{hi} range remains. Its selector gives

$$H \leq e(A) \leq 2A + 5A^2.$$

We claim that this range forces $X > 1/2$. Indeed, if $X \leq 1/2$, the selector and the lower center inequality give

$$2RH = \eta + A + D \geq \eta + P + wA = w(R + A).$$

Together with $H < 2A/(1 - 2A)$, this implies

$$f_R(A) < 0, \qquad f_R(z) := w(R + z)(1 - 2z) - 4Rz.$$

But $f_R''(z) = -4w < 0$ and $f_R(0) = Rw > 0$. We check the other endpoint of the possible interval for A.

If $R \leq 1/2$, then $0 < A < P$. Direct simplification gives

$$f_R(P) = (1 - E)\left(4E^3 - 2E^2 + 1 - R(2E^3 - 2E^2 + 5E)\right).$$

The coefficient of R is positive, so

$$f_R(P) \geq \frac{1 - E}{2}\left(6E^3 - 2E^2 - 5E + 2\right).$$

For $\phi(E) = 6E^3 - 2E^2 - 5E + 2$ on $[\sqrt{3}/2, 1)$,

$$\phi'(E) = 18E^2 - 4E - 5 \geq \frac{17}{2} - 2\sqrt{3} > 0,$$

and

$$\phi\left(\frac{\sqrt{3}}{2}\right) = \frac{2 - \sqrt{3}}{4} > 0.$$

Hence $f_R(P) > 0$. Strict concavity puts f_R above its endpoint chord, so $f_R(A) > 0$ for $0 < A < P$, a contradiction.

If $R > 1/2$, the assumption $X \leq 1/2$ gives $D \geq R - 1/2$, and therefore

$$0 < A < A_0, \qquad A_0 := P - w\left(R - \frac{1}{2}\right) = \frac{w}{2} - \eta.$$

Direct simplification gives

$$f_R(A_0) = \frac{1 - E}{2}(E + 11R - 3ER - 5).$$

Since $11 - 3E > 0$ and $R > 1/2$,

$$E + 11R - 3ER - 5 > \frac{11 - 3E}{2} + E - 5 = \frac{1 - E}{2} > 0.$$

Thus $f_R(A_0) > 0$. Concavity again gives $f_R(A) > 0$ on $(0, A_0)$, a contradiction. Both ranges are impossible, proving $X > 1/2$.

Put $p_1 = G_{1-A}(H)$. If row 3 is not T_+^{hi}, then $p_1 \leq A + e(A) < 3A + 5A^2 < 29/64 < 1/2$, and $2R(H - m) = \eta + D - A > \eta - P > 0$. Extensivity gives $p_1 \geq H > m$, while $p_1 < 1/2 < 1 - m$; thus both Full ranges are impossible. On L, T_-, or the low-c tie one has $F_{1-m}(p_1) \leq p_1$, so

$$G_{1-m}(p_1) \geq 1 - p_1 > 1 - X.$$

Extensivity of the final reverse map G_{1-D} preserves this strict bound and proves (85). It remains to treat the selected row-3 T_+ branch. Proposition D.7 proves once for all that every selected T_+ arc is increasing and strictly concave.

When $d < 1-\sqrt{3}/2$, the arc endpoints are (d,d) and $(e(d), d+e(d))$, so concavity gives

$$q \geq p + \frac{d}{e(d)-d}(p-d). \tag{87}$$

For $0 < d \leq \mu$, the valid rational low-root bound gives

$$\frac{d}{e(d)-d} > \frac{1-2d}{1+2d} > 1-4d > 1-5d.$$

This covers row 4, because its parameter is $d = A < P \leq \mu$. For the row-3 parameter $d = m$, which need not be at most μ, the remaining high-radial range is handled without the rational estimate. If $\mu < d \leq 1/8$, then

$$\frac{d}{e(d)-d} \geq \frac{1}{1+5d} > 1-5d$$

by $e(d) \leq 2d + 5d^2$. If $1/8 < d < 1-\sqrt{3}/2$, then $e(d) < (1-d)/2$ and hence

$$\frac{d}{e(d)-d} > \frac{2d}{1-3d} > 1-5d.$$

The last comparison is equivalent to $15d^2 - 10d + 1 < 0$; its smaller root $(5-\sqrt{10})/15$ is less than $1/8$. In the low-c row-3 range $1-\sqrt{3}/2 \leq d < 1/5$, the other endpoint is $(h(1-d), 1-h(1-d))$, and the chord coefficient is

$$\frac{1-2h(1-d)}{h(1-d)-d}.$$

On $3/4 \leq c \leq \sqrt{3}/2$, direct differentiation of the formula for h shows that $h(c)$ is nonincreasing. Moreover,

$$h\left(\frac{3}{4}\right) = \frac{3}{2(\sqrt{6}+1)} < \frac{7}{16}.$$

Here low c gives $d \geq 1-\sqrt{3}/2 > 2/15 > 1/16$, and hence

$$h(1-d) < \frac{7}{16} < \frac{3+d}{7}.$$

Thus the displayed chord coefficient is greater than $1/3$, while $1-5d < 1/3$. For $d \geq 1/5$, extensivity alone is stronger than the row-3 estimate.

Applying these chord bounds first with $d = A$ and then with $d = m$ proves, without an omitted selected branch,

$$p_1 > L_1 := (2-4A)H - (1-4A)A,$$
$$p_2 := G_{1-m}(p_1) > L_2 := (2-5m)L_1 - (1-5m)m.$$

Finally let

$$D_0 = P - RA, \qquad D_h = A(4R-1) + 10RA^2 - \eta, \qquad x = Rw.$$

Feasibility puts $D_0 \leq D \leq D_h$. We first prove that the actual feasible value satisfies $D < 1/10$. Since

$$P = \frac{xE}{1+E} < \frac{x}{2}, \qquad \eta = \frac{x}{1+E} > \frac{x}{2},$$

the contrary assumption $D \geq 1/10$ and $A + wD < P$ give

$$A < \overline{A} := \frac{w(5R-1)}{10}.$$

The function $D_h(A)$ is increasing, so

$$D \leq D_h(A) < D_h(\overline{A}) < U(R) := \overline{A}(4R-1) + 10R\overline{A}^2 - \frac{x}{2}.$$

Direct expansion yields

$$\frac{1}{10} - U(R) = -\frac{R}{10}P_4(R), \qquad P_4(R) = 25R^4 - 60R^3 + 26R^2 + 22R - 14.$$

For $y = 2R - 1 \in [0,1]$,

$$16P_4(R) = 25y^4 - 20y^3 - 106y^2 + 124y - 39 \leq -101y^2 + 124y - 39 < 0.$$

The last quadratic has discriminant -380. Hence $U(R) < 1/10$, a contradiction, and therefore $D < 1/10$.

It remains to compare L_2 with $2D + 3D^2$. Put

$$J(D) = L_2(D) - \frac{23}{10}D.$$

Since

$$L_2 - 2D - 3D^2 - J(D) = 3D\left(\frac{1}{10} - D\right) > 0,$$

it is enough to prove $J(D) > 0$. Exact differentiation gives

$$J'(D) = \frac{S(R,A)}{10R^2}, \qquad S(R,A) = 100A^2 - (40R + 50)A + R(20 - 23R).$$

The inequality $D_0 \leq D_h$ is equivalent to

$$x \leq A(5R - 1 + 10RA).$$

Because $A < P < x/2 \leq 1/8$,

$$A > L := \frac{4x}{25R - 4}.$$

Also $\partial_A S = 200A - 40R - 50 < -45$, and direct substitution gives

$$S(R,L) = -\frac{Rq_3(R)}{(25R-4)^2},$$

where

$$q_3(R) = 8775R^3 - 14260R^2 + 7928R - 1120.$$

For $y = 2R - 1$,

$$8q_3(R) = 8775y^3 - 2195y^2 + 997y + 3007 \geq 812 > 0.$$

Thus $J'(D) < 0$ and $J(D) \geq J(D_h)$.

At $D = D_h$ one has $H = 2A + 5A^2$. Write $L_{2,h} = L_2(D_h)$. Expansion gives

$$L_{2,h} - A(1 + 4R) = \frac{A}{R^2}\mathcal{Q}(R,A),$$

where

$$\begin{aligned}\mathcal{Q}(R,A) = {}& 100A^3R - 40A^2R^2 - 30A^2R + 12AR^2 - 15AR + 5A \\ & - 4R^3 + 5R^2 - R.\end{aligned}$$

On $0 \le A \le x/2$,

$$\partial_A^2 \mathcal{Q} = 20R(30A - 4R - 3) < 0.$$

Its two endpoint values are

$$\mathcal{Q}(R,0) = x(4R-1) > 0, \qquad \mathcal{Q}\left(R, \frac{x}{2}\right) = \frac{x}{2}p(R),$$

where

$$p(R) = 25R^5 - 30R^4 + 20R^3 - 3R^2 - 7R + 3.$$

For $y = 2R - 1$,

$$32p(R) = 25y^5 + 65y^4 + 90y^3 + 106y^2 - 35y + 5 > 0,$$

because the terms of degree at least three are nonnegative and $106y^2 - 35y + 5$ has discriminant -895. Concavity therefore gives $L_{2,h} > A(1+4R)$.

On the other hand, $A < P = \eta E$ and $A < P < x/2$ imply

$$\frac{D_h}{A} = 4R - 1 + 10RA - \frac{\eta}{A} < C_0 := 4R - 2 + 5Rx - \frac{x}{2}.$$

Here we used

$$\frac{1}{E} > 1 + \frac{x}{2}, \qquad (1-x)\left(1 + \frac{x}{2}\right)^2 = 1 - \frac{x^2(3+x)}{4} < 1.$$

Finally,

$$1 + 4R - \frac{23}{10}C_0 = \frac{h_3(R)}{20}, \qquad h_3(R) = 230R^3 - 253R^2 - 81R + 112.$$

Since $h_3''(R) = 46(30R - 11) \ge 184$, strong convexity at $R_0 = 20/23$ gives

$$h_3(R) \ge \frac{788}{529} - \frac{(17/23)^2}{368} = \frac{289695}{194672} > 0.$$

Consequently

$$J(D_h) > A\left(1 + 4R - \frac{23}{10}C_0\right) > 0,$$

and hence

$$p_2 > L_2 > 2D + 3D^2.$$

The low root $e(D)$ is the smaller root of

$$H_D(z) = z^2 - (1-D)z + (1-D)^2 - (1-D)^4,$$

and direct substitution gives $H_D(2D + 3D^2) = D^2(8D^2 + 19D - 2) < 0$. Hence $e(D) < 2D + 3D^2 < p_2$. The one-hit part of Lemma D.8, applied to the row-2 map, gives

$$G_{1-D}(p_2) \ge 1 - e(D) > 1 - X,$$

because

$$e(D) < 2D + 3D^2 < \frac{23}{100} < \frac{1}{2} < X.$$

This is (85). The actual-row and terminal inequalities in Proposition F.5 now give the contradiction. $\square$

Proposition K.2 (Positive-gap all-Vd0, $N_+ = 1$)**.** *If the center role is CE1 or CE2, every vertex role is* Vd0*, and* $N_+ = 1$*, then no cover of* H *can have a boundary gap in the actual open vertex-role traces.*

Proof. The midpoint argument used above makes T_0 uniquely supercritical. In an assumed cover, every boundary gap in the vertex-role traces must lie in a center trace. CE1 therefore has the one-gap state excluded by Proposition K.1. For CE2, the exactly-one-gap state is Proposition F.7; the two-gap state is the common application in Proposition I.1. This exhausts the one- and two-gap states, including singleton gaps. The zero-gap state is intentionally owned by the center-independent obstruction of Strategy 4. □

Appendix L. Nonsupercritical, T3-like, and Vd1/Vd2 Demand Branches

L.1. **The exactly-one-$T3$ branch.**

Proposition L.1 (Exactly one T3-like row)**.** *Assume the center role is CE1 or CE2, $N_+ = 1$, exactly one vertex role is T3-like, and no role is Vd1 or Vd2. Then the roles do not cover H.*

Proof. Normalize the center midpoint to M_0. A T3-like row misses its own midpoint, a supercritical row misses its local midpoint, and a Vd0 row cannot rescue a neighboring midpoint. Consequently, after reflection,

$$T_0 \text{ is T3-like}, \qquad M_1 \in T_0, \qquad T_1 \text{ is uniquely supercritical},$$

and $T_2, \ldots, T_5$ are nonsupercritical Vd0 rows. The rational local calculation is Lemma H.2. It verifies the two local inequalities in (72), while midpoint and support isolation force the center to cover the O-side radial gap. The common adjacent-rescuer obstruction, Corollary H.5, therefore gives the contradiction. □

L.2. **The nonsupercritical CE1/CE2 boundary packages.** Figure 11 gives a common schematic for the package proved below.

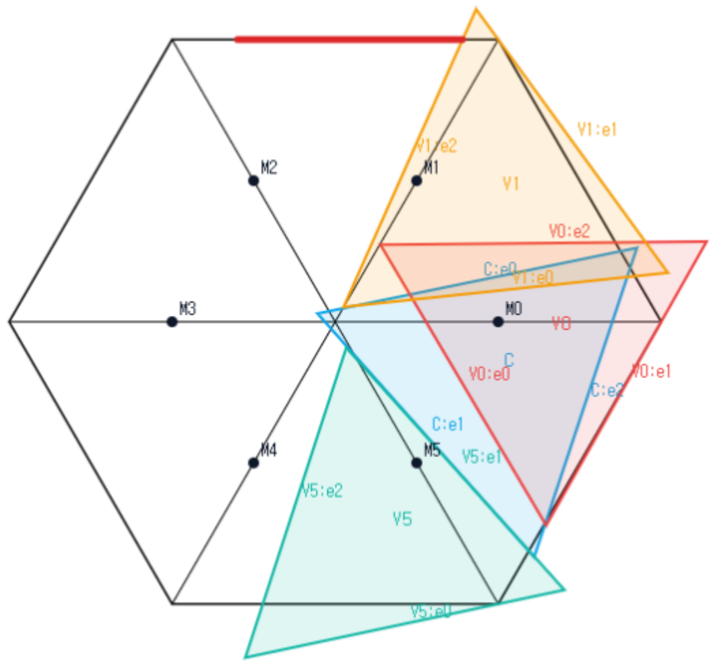


FIGURE 11. Schematic view of the common CE1/CE2, $N_+ = 0$, all-Vd0 boundary package, not to scale. The center role and three representative Vd0 vertex roles are shown; the remaining three roles are omitted for readability. The simultaneous placement is not a candidate cover, and no proof depends on visual inspection.

Proposition L.2 (All-Vd0 boundary loss)**.** *Assume that T_C is CE1 or CE2, every vertex role is Vd0, and*

$$A_i + B_i \le 1 \qquad (0 \le i \le 5).$$

Then the seven roles do not cover H.

Proof. Let g be the number of positive center traces containing an active V-gap. Then $g \in \{0,1,2\}$, with $g \le 1$ in CE1.

If $g = 0$, the six original open vertex traces cover the connected circle ∂H. Each has length at most one, and two nonempty members of a finite relatively open cover must overlap in positive length. Hence

$$6 = \mathcal{H}^1(\partial H) < \sum_{i=0}^{5} \mathcal{H}^1(U_i \cap \partial H) \le \sum_{i=0}^{5} (A_i + B_i) \le 6,$$

a contradiction.

Suppose $g = 1$ and reflect so that the active trace is

$$I_R = \left[\frac{k}{R}, W + \delta\right].$$

Put

$$s = \frac{k}{R}, \qquad q = R - \delta, \qquad w_0 = W + \delta - \frac{k}{R}.$$

Then $s + q + w_0 = 1$. Gap containment gives $B_0 \geq s$ and $A_1 \geq q$. The companion edge has no active gap, so $A_0+B_5 \geq 1$. Since row T_0 is nonsupercritical, $A_0+B_0 \leq 1$, and therefore $B_5 \geq B_0 \geq s$. The exact radial demands at the endpoint rows are

$$c_1 = 1 - \frac{\delta}{R}, \qquad c_5 = 1 - \frac{\alpha}{W}.$$

Lemma L.7 gives

$$F_{c_5}(s) + F_{c_1}(q) < 1.$$

Thus the two actual endpoint outputs satisfy $B_1 + B_5 < 1$. Lemma D.6, applied to T_2, T_3, T_4, forces their total boundary contribution above three, contrary to their three unit caps.

Finally, $g = 2$ is CE2-only and is exactly Proposition I.1. These three ranks are exhaustive. □

We next treat the additional T3-like rows in the $N_+ = 0$ branch. The following local facts also reappear later.

Lemma L.3 (T3-like midpoint and side tradeoff)**.** *Let T_i be T3-like. Then $M_i \notin T_i$. After applying the closed-trace domination of Proposition C.6, suppose the selected adjacent support is on r_{i+1} and M_{i+1} belongs to the dominated trace. If b is its boundary reach toward V_{i+1}, c its exit on its own arm, and ℓ its entry on r_{i+1}, all measured from the relevant vertex, then*

$$b + c < 1, \qquad b + \ell > 1. \tag{88}$$

Consequently two adjacent T3-like traces that cover each other's midpoint, whose selected supports point toward one another, cannot cover both outer half-arms if no other role has positive support there.

Proof. Apply Proposition C.6 and reflect if necessary. Its translated Type-II majorant has parameters

$$0 < t < 1, \qquad z = \sqrt{1 - t + t^2}, \qquad \alpha = 0, \qquad 0 < \beta < z.$$

Its wedge vertex has

$$Q_x = \frac{z - t\beta}{1 - t + t^2} > 1.$$

Hence

$$t\beta < z - z^2.$$

The majorant's own-radial exit is $c = \beta/(1 - t)$. Since

$$(1 - z)(1 + z) = t(1 - t),$$

we obtain

$$c < \frac{z - z^2}{t(1 - t)} = \frac{z}{1 + z} < \frac{1}{2}.$$

Here $z < 1$ because $0 < t < 1$. Thus the translated majorant misses M_0, and its closed-trace domination implies that the original T3-like role misses M_0 as well.

For the tradeoff, the translated T3-like trace has the exhaustive Type-II form

$$\begin{aligned} (1 - t)x + ty &\geq 0, \\ \beta + tx - y &\geq 0, \qquad 0 < t < 1, \quad z = \sqrt{1 - t + t^2}, \quad 0 < \beta < z. \\ \beta + x - (1 - t)y &\leq z, \end{aligned}$$

For the selected support on r_1, substitution in the Type-II half-planes gives

$$b = z - \beta, \qquad c = \frac{\beta}{1-t}, \qquad \ell = \frac{\beta + 1 - z}{1-t}.$$

The condition $M_1 \in T$ is $\beta \le z - (1+t)/2$. Therefore

$$b + c \le z + \frac{t}{1-t}\left(z - \frac{1+t}{2}\right) = 1 - \frac{(1-z)^2}{2(1-t)} < 1,$$

whereas

$$b + \ell - 1 = \frac{t(1 - z + \beta)}{1-t} > 0.$$

For a crossed adjacent pair, coverage forces $c_i \ge \ell_{i+1}$ and $c_{i+1} \ge \ell_i$. Combining these two inequalities with (88) gives both $b_i < b_{i+1}$ and $b_{i+1} < b_i$, a contradiction. $\square$

Lemma L.4 (The support-isolated four-label inequality)**.** *Assume T_C is CE1 or CE2, $N_+ = 0$, no role is Vd1 or Vd2, and, after normalization and reflection, T_0 is T3-like with selected adjacent support on r_1. Then the two endpoint rows in the isolated configuration satisfy*

$$F_{c_5^*}(a_5^*) + F_{c_1^*}(a_1^*) < 1, \tag{89}$$

where a_1^*, a_5^* are the exact boundary inputs and c_1^*, c_5^* the exact radial demands defined below.

Proof. Use the CE1 side variables $0 < \lambda < 1$, $\rho = \sqrt{1 - \lambda + \lambda^2}$ and center slacks

$$X = F_0(O), \qquad Y = F_2(O).$$

Then

$$a_C = \lambda - Y, \qquad \lambda s = X + Y + 1 - \rho,$$

and Proposition F.1 gives

$$\rho - X - Y > \frac{1}{2}, \qquad \frac{Y}{\lambda} < \frac{1}{2}, \qquad \frac{X}{1-\lambda} < \frac{1}{2}.$$

Consequently the apparent minima in the two radial exits collapse to

$$\gamma_1 = \frac{Y}{\lambda}, \qquad \gamma_5 = \frac{X}{1-\lambda}.$$

For CE2 the second boundary interval is $[S, T]$, where

$$S = \frac{X + Y + 1 - \rho}{1 - \lambda}, \qquad T = X + \lambda.$$

The T3-like normal form supplies $D > 1$, $0 < R < 1$, and $\alpha, p \ge 0$ with

$$R^2 - DR + D^2 = 1, \qquad \alpha + p = 1/D, \qquad q = 1 - p = \alpha + (D-1)/D < 1/2.$$

Its interval on r_1, measured from V_1, is

$$[c, u], \qquad c = Dq/R, \qquad u = q + (1 - R)/D.$$

For an initial trace $[0, p]$ and a center interval $J = [L, U]$, define

$$\Theta(p, J) = \begin{cases} 1 - p, & J = \varnothing \text{ or } p < L, \\ 1 - \max\{p, U\}, & p \ge L. \end{cases}$$

Then

$$a_1^* = \Theta(p, [s, t]), \qquad a_5^* = \Theta(\alpha, [S, T]),$$

with the second interval empty in CE1, and

$$c_5^* = 1 - \gamma_5, \qquad c_1^* = \begin{cases} c, & 1-u \le \gamma_1 \le 1-c, \\ 1-\gamma_1, & \text{otherwise.} \end{cases}$$

If $a_1^* + a_5^* > 1$, the two caps already imply $F_{c_5^*}(a_5^*) + F_{c_1^*}(a_1^*) < 1$. Assume the hard region $a_1^* + a_5^* \le 1$. The following table is an exhaustive audit of the four possible first labels; the middle column records the exact decisive inequality.

first label	right labels	conclusion
L	Full	$e(\gamma_5) < a_1^*$
L	L, T_-	$F_{c_5^*}(a_5^*) + F_{c_1^*}(a_1^*) < 1$
L	T_+^{hi}	$e(\gamma_5) < a_1^*$
T_-	L, T_-	$F_{c_5^*}(a_5^*) + F_{c_1^*}(a_1^*) < 1$
T_-	$\mathrm{Full}, T_+^{\mathrm{hi}}$	$F_{c_5^*}(a_5^*) < a_1^*$
Full	all	impossible
T_+^{hi}	$\mathrm{Full}, T_+^{\mathrm{hi}}$	impossible
T_+^{hi}	L, T_-	$F_{c_5^*}(a_5^*) + F_{c_1^*}(a_1^*) < 1.$

We verify the nonformal entries. Put $\kappa = (1-\rho)/(1-\lambda)$. From $X + (1-\lambda)Y < \rho(1-\rho)$ one obtains

$$\gamma_1 \ge a_C \Longrightarrow e(\gamma_5) < a_C, \tag{90}$$

$$S \le Y/\lambda \Longrightarrow e(\gamma_5) < S. \tag{91}$$

For (90), one has $\gamma_5 < a_C - \kappa$ and $a_C - \kappa < 1/8$; then (82) applies, and the remaining comparison is

$$(a_C - \kappa) + 5(a_C - \kappa)^2 \le \kappa.$$

After squaring its positive sides, the difference is $\lambda(23\lambda^3 + 58\lambda^2 + 7\lambda + 72) > 0$. For (91), combine the premise with the center inequality and put $x = \lambda$. This gives $0 < x < 3/8$ and

$$X < X_* = \frac{(1-\rho)(\rho(1-2x) - x(1-x))}{1 - x - x^2}.$$

Set

$$\delta = \frac{1-\rho}{1-x} = \frac{x}{1+\rho}, \qquad \eta = \frac{X}{1-x}, \qquad f = \rho(1-2x) - x(1-x), \qquad d = 1 - x - x^2.$$

Then $\gamma_5 \le \eta < \delta f/d$. Moreover

$$\frac{f}{d} < 1 - 2x,$$

because, after multiplication by positive denominators, this is just

$$\rho < d + \frac{x(1-x)}{1-2x} = 1 + \frac{2x^3}{1-2x}.$$

Hence $\eta < \delta(1-2x) < 1/8$. If $\eta = 0$, the assertion is immediate. Otherwise,

$$S \ge \frac{X + 1 - \rho}{1 - 2x} = \frac{1-x}{1-2x}(\eta + \delta) > 2\eta \left(\frac{1-x}{1-2x}\right)^2.$$

On the other hand,

$$5\eta < 5\delta(1-2x) < 2\left[\left(\frac{1-x}{1-2x}\right)^2 - 1\right].$$

For the middle sign use $\rho > 1-x$ and

$$2(2-3x)(2-x) - 5(1-2x)^3 = (4x-3)(10x^2-6x-1) > 0 \qquad (0 < x < 3/8).$$

Indeed both factors are negative; for the quadratic use $10x^2 \le 15x/4 < 6x+1$. The low-root estimate (82) now gives

$$e(\gamma_5) \le 2\gamma_5 + 5\gamma_5^2 \le 2\eta + 5\eta^2 < S,$$

proving (91) without any polynomial coefficient list. The L separation follows from its exact polynomial: if its output is $z = e(\eta)$ at input $1-\beta$, then

$$\beta \ge z + \eta.$$

Indeed the selector polynomial $P(x,z) = x^4 - 4x^3 + 5x^2 - (2+z)x + z - z^2$ is strictly decreasing on the low range and vanishes at $x = \eta$. Equations (90)–(L.2), with the three possibilities for Θ, prove every first-L entry. For a first T_- label the exact root test

$$T_-(a,c) < z \iff c^2(a+z)^2 - (a^2 - ac + c^2) > 0$$

is increasing in z. Substitution of a_1^*, followed by the center inequality, proves $F_{c_5^*}(a_5^*) < a_1^*$ except in the low–low entries, which are immediate from $T_-(a,c) \le \min\{a, 1-a\}$ and $L < 1/2$. First-coordinate Full would force either $X \ge (1-\lambda)\alpha$ and $Y \ge \lambda - \alpha$, contradicting $s \le p$, or, in the CE2 overlap case, $S \ge 1/2$, contradicting $S \le \alpha < 1/2$.

For a first T_+^{hi} label put $r = 1-\lambda$, $y = Y/\lambda$ and choose β so that

$$m = \sqrt{\beta^2 - \beta + 1}, \quad 1 - \gamma_5 = \frac{m}{r+\beta}, \quad F_{c_5^*}(a_5^*) = \frac{\beta m}{r+\beta}.$$

The exact selector is

$$\beta \ge \max\left\{r, \frac{1}{2}, \frac{1-r^2}{1+2r}\right\}, \qquad r \ge (1-\beta)(r+\beta)^2. \tag{92}$$

The center inequalities give $y < 1/10$ and, with

$$y_* = \frac{r}{1-r}\left(\frac{m}{r+\beta} - \frac{1}{2} - \frac{1-r}{1+\sqrt{r^2-r+1}}\right),$$

give $y < y_*$ and

$$3y_* \le 1 - \frac{\beta m}{r+\beta}.$$

This proves the right-L entry by $e(y) < 3y$. It excludes right Full and right high by the exact high-input filter. For right T_- with input a_C, direct substitution gives a sum below one. For input q, put $c = 1-y$, $q = tc$, and let $\widehat{b}_1 = F_{c_1^*}(q)$. The selected T_- equations give

$$\widehat{b}_1 \le \kappa := \frac{\sqrt{13}-1}{6}, \qquad q \le \tau < \frac{93}{200},$$

where $\tau \in (0, 1/2)$ is the unique root of $\tau^4 - 3\tau^3 + 3\tau^2 - 3\tau + 1 = 0$. In this CE2 overlap subcase the support ordering gives $q > S$. The common left-high identities are

$$S = S_0 + \frac{1-r}{r} y, \qquad S_0 = 1 - \frac{m}{r+\beta} + \frac{1-r}{1+\sqrt{r^2-r+1}}.$$

Since $y > 0$, we have the complete chain

$$S_0 < S < q \le \tau < \frac{93}{200}.$$

We now replace the former interval subdivision by a one-variable comparison. Write $s = r + \beta$ and $\rho_r = \sqrt{r^2 - r + 1}$. The selected Cell 2 inequality factors as

$$r - (1-\beta)(r+\beta)^2 = (s-1)(\beta^2 + r\beta - r).$$

The second factor is nonnegative because $r \le \beta$ and $\beta \ge 1/2$; if it vanishes, then $r = \beta = 1/2$. Hence $s \ge 1$. It follows that $\rho_r \le m$, since $\rho_r^2 - m^2 = (r-\beta)(s-1) \le 0$.

Define the one-variable lower bound

$$\overline{S}(\beta) = 1 - m + \frac{\beta}{1+m}.$$

Then

$$S_0 \ge 1 - \frac{m}{s} + \frac{1-r}{1+m} \ge \overline{S}(\beta).$$

Indeed, the second difference is

$$(s-1)\left(\frac{m}{s} - \frac{1}{1+m}\right) \ge 0.$$

For the last sign use $s \le 2\beta \le m(1+m)$: if $g = 3\beta - \beta^2 - 1$, then $g > 0$ and

$$m^2 - g^2 = -\beta(\beta-1)(\beta^2 - 5\beta + 5) \ge 0, \qquad m^2 + g = 2\beta.$$

Put $B_0 = 5657/10000$, $T = 93/200$, and $\beta_* = 161/250$. If $\beta m/s \ge B_0$, then $\beta m \ge B_0$. The function $\phi(\beta) = \beta m$ is strictly increasing, and the exact comparison

$$B_0^2 - \beta_*^2 m(\beta_*)^2 = \frac{22782769}{62500000000} > 0$$

therefore gives $\beta > \beta_*$. On $[\beta_*, 1]$ put

$$H(\beta) = \frac{107}{200} - Tm - (1-\beta)^2.$$

Since $H'' = -3T/(4m^3) - 2 < 0$, the function H is concave. Its endpoint values are positive: $H(1) = 7/100$, while, for $a_* = 107/200 - (1-\beta_*)^2 = 51033/125000$,

$$a_*^2 - T^2 m(\beta_*)^2 = \frac{1693881}{62500000000} > 0.$$

Thus $H > 0$ throughout the interval. The identity

$$(\overline{S}(\beta) - T)(1+m) = H(\beta)$$

shows that $\beta m/s \ge B_0$ would force $S_0 > T$, a contradiction. Consequently

$$F_{c_5^*}(a_5^*) < \frac{5657}{10000} < \frac{7-\sqrt{13}}{6}.$$

The final strict comparison follows by squaring $\sqrt{13} < 18029/5000$, whose squared difference is $44841/25000000 > 0$. Thus the final sum is also below one. The table is exhaustive, proving (89). $\square$

Proposition L.5 (The $N_+ = 0$ T3-like branch)**.** *Assume that T_C is CE1 or CE2, $N_+ = 0$, no vertex role is Vd1 or Vd2, and at least one vertex role is T3-like. Then the seven roles do not cover H.*

Proof. Normalize the unique center midpoint to M_0 and simultaneously apply the closed-trace domination of Proposition C.6 to every T3-like role. If T_0 were not T3-like, a maximal consecutive block of T3-like indices among $1, \dots, 5$ would have to match its roles bijectively to the same block of uncovered midpoints. The matching uses only adjacent indices, because a T3-like role misses its own midpoint and Vd0 roles cannot cover adjacent midpoints. Its first two roles therefore form a crossed pair forbidden by Lemma L.3. Hence T_0 is T3-like; reflect so that it supports r_1.

There are at most two T3-like roles. Indeed, with $N_+ = 0$ and no Vd1/Vd2 role, the short-role identity gives $q = k$, where k is the number of T3-like roles. If $k \geq 3$, Theorem G.15 gives an immediate full-skeleton contradiction. Midpoint locality then forces T_2, T_4, T_5 to be Vd0. Consequently r_1 has positive support only from T_C, T_0, T_1, and r_5 only from T_C, T_5.

The actual outgoing reaches of T_1, T_5 are bounded by the two safe maps in Lemma L.4; an extra adjacent-ray demand on a possible T3-like T_1 only shrinks its feasible set. Thus their sum is less than one. Rows T_2, T_3, T_4 must cover more than three units of the four intervening boundary edges, while their three nonsupercritical caps sum to at most three. This contradiction proves the proposition. $\square$

Lemma L.6 (Free strict-supercritical envelope)**.** *For $0 \leq c < 1/2$,*

$$\sup\{b : \exists a, (a,b,c) \in \mathcal{A}, a + b > 1\} = \mathcal{B}(c) := \frac{c + \sqrt{c^2 - 8c + 4}}{2}.$$

The supremum is not attained. For $c \geq 1/2$ the strict feasible set is empty. The function $\mathcal{B}$ is strictly decreasing on $[0, 1/2]$.

Proof. Lemma C.10 excludes $c \geq 1/2$. On the boundary $a = 1 - b$, a support rotation has strict slack precisely when

$$c < \frac{1 - b^2}{2 - b}.$$

The right side decreases from $1/2$ to zero. Solving equality gives the displayed larger root. Equality has no strict support slack, which explains nonattainment. Differentiation gives

$$\mathcal{B}'(c) = \frac{1}{2}\left(1 + \frac{c - 4}{\sqrt{c^2 - 8c + 4}}\right) < 0,$$

since $(4 - c)^2 - (c^2 - 8c + 4) = 12$. $\square$

L.3. **Two reusable capped-loss inequalities.**

Lemma L.7 (One-side capped loss)**.** *Let $0 < R < 1$, $S = \sqrt{1 - R + R^2}$, and let $s, u, w > 0$ satisfy $s + u + w = 1$ and $0 < u < R$. Put*

$$\delta = \frac{R}{1 + S}, \qquad \gamma = u - \delta - \frac{R}{1 - R} w,$$

and assume $\gamma > 0$. Set

$$c_1 = u/R, \qquad c_5 = 1 - \gamma.$$

Then

$$F_{c_5}(s) + F_{c_1}(u) < 1. \tag{93}$$

Proof. Put $\eta = (1-R)\delta = 1-S$ and

$$\widehat{b}_5 = F_{c_5}(s), \qquad \widehat{b}_1 = F_{c_1}(u).$$

The center identity may be written

$$c_5 = \frac{1-u+\eta-Rs}{1-R}. \tag{94}$$

The hypotheses imply $1/2 < c_1, c_5 < 1$: for example $u > \delta$ gives $c_1 > 1/(1+S) > 1/2$, and $\gamma < u-\delta < R-\delta = RS/(1+S) < 1/2$ gives $c_5 > 1/2$. Thus the four high-radial labels of Proposition J.3 are exhaustive.

Right low labels. If the right label is T_-, its exact frontier equation gives $(u+\widehat{b}_1)^2 = 1-R+R^2 = S^2$, so $\widehat{b}_1 = S-u$. For a right L label write $\widehat{b}_1 = kc_1$, $0 \le k \le 1/2$, and $C = 1-k+k^2 = c_1^2$. With $X_R = R+k$, the Cell-L selector factors as

$$q = c_1^2(X_R-1)G_k(X_R), \qquad G_k(X) = CX^3 + CX^2 - k(1-k)X + k^2.$$

For $X \ge k$, $G_k(X) > 0$: when $k > 0$, $G_k'(X) \ge k(1+2k-k^2+3k^3) > 0$ and $G_k(k) = k^2(1+k+k^3) > 0$. Hence $q \le 0$ gives $X_R \le 1$. If $h_R = 1-X_R$, then

$$S^2 - (u+\widehat{b}_1)^2 = h_R(1+2k^2+k(1-k)h_R) \ge 0.$$

Consequently either right low label satisfies

$$\widehat{b}_1 \le X := S-u. \tag{95}$$

A left low label against right high or Full. For left T_- put $z = s/c_5 = 1-p$ and $h = \sqrt{1-p+p^2}$. The component selector gives $0 \le p \le 1/2$, the frontier gives $s + \widehat{b}_5 = h$, and its selector gives $s \le (1-p)h$. Equation (94) becomes $c_5(1-Rp) = 1-u+\eta$, whence

$$u \ge \eta + (1-h) + Rph.$$

Put $q_0 = ph$. If $q_0 > R$, then $p(1-p) > R(1-R)$ and the preceding lower bound would give $u > R$. Thus $q_0 \le R$, and

$$\eta + 1 - h > \frac{R(1-R)+p(1-p)}{2} \ge (1-R)q_0.$$

Substitution in the exact formula

$$w = \frac{(1-R)(1-u)p - \eta(1-p)}{1-Rp}$$

now gives $w < 1-h$. A right high output has the form $H-u$, where $H = \sqrt{1-\beta+\beta^2} \le 1$; for Full take $H = 1$. Hence

$$\widehat{b}_5 + \widehat{b}_1 = h + H - (s+u) = 1 + w - (1-h) - (1-H) < 1.$$

For a left L label write

$$c_5 = h(x) := \sqrt{1-x+x^2}, \qquad \widehat{b}_5 = xh(x), \qquad 0 \le x \le \frac{1}{2},$$

and, along the center line,

$$s(x) = \frac{R-u+\eta+(1-R)(1-h(x))}{R}.$$

If $y = s(x)/h(x)$, the exact selector factorization is

$$\frac{q}{h(x)^2} = (y-(1-x))P_x(y),$$

where

$$P_x(y) = (1 - x + x^2)y^3 + (1 + 2x - 2x^2 + 3x^3)y^2 + (x + 2x^2 - x^3 + 3x^4)y + (x^2 + x^3 + x^5).$$

Every coefficient is nonnegative on $0 \le x \le 1/2$, and the polynomial is positive at every nondegenerate point. Thus a selected L point has $s(x) \le (1-x)h(x)$. The two curves meet once in $(0, 1/2)$: at $x = 0$ the left side is smaller, while at $x = 1/2$ it is enough to use

$$u \le \frac{R}{1+R} \quad \text{(Full)}, \qquad u \le \min\{RS, S/2\} \quad (T_+^{\mathrm{hi}}).$$

Writing $e_0 = 1 - \sqrt{3}/2$, the corresponding endpoint differences are

$$\frac{(1-R)(R^2 + 2Re_0 + 2e_0)}{2(1+R)} > 0$$

for Full, and respectively $[2e_0(1-R) - R^3]/2 \ge 0$ for $R \le 1/2$ and $(1-R)(3R + 4e_0 - 2)/4 \ge 0$ for $R \ge 1/2$ in the high case. At the unique transition x_0, the L frontier is the $q = 0$ endpoint of the preceding T_- branch. Since $xh(x)$ increases, every selected left L output is no larger than that transition output. The preceding strict loss therefore also covers (L, T_+^{hi}) and (L, Full).

Full and high exclusions. Left Full forces $s < 1/2$ and $\gamma \ge s$. Right Full or right high gives $u \le 1/2$, but

$$\gamma - s = 2u - 1 - \delta + \frac{1-2R}{1-R} w < 0 :$$

for $R \ge 1/2$ this is immediate, while for $R < 1/2$ use $w < 1 - u$ to bound it by $(u - R)/(1 - R) - \delta < 0$. Thus left Full cannot pair with either label. If the left label is high and the right one Full, then $s < 1/2$ and $u \le R/(1+R)$, whereas $\gamma > 0$ and $w > 1/2 - u$ force $u > R/2 + \eta > R/(1+R)$; the final inequality is equivalent to $2(1+R) > 1 + S$. This pair is also impossible.

A right low label. Use (95). If $s > X$, the cap gives $\widehat{b}_5 + \widehat{b}_1 < 1$. Suppose $s \le X$ and put

$$c_X = X + 2\delta, \qquad f(c, z) = c^4 - c^2 + zc - z^2.$$

At the limiting value $u = R$, with $\kappa = \delta$, one has

$$X_0 = \frac{1 - 2\kappa}{1 + \kappa}, \qquad c_0 = \frac{1 + 2\kappa^2}{1 + \kappa}.$$

Now $c_0 > \sqrt{3}/2$ because $16\kappa^4 + 13\kappa^2 - 6\kappa + 1 > 0$, and

$$f(c_0, X_0) = \frac{\kappa^2(1-2\kappa)^2(4\kappa^4 + 4\kappa^3 + 10\kappa^2 + 6\kappa + 5)}{(1+\kappa)^4} > 0.$$

Along the u-line, $\frac{d}{du} f(c_X, X) = -4c_X^3 + c_X + X < 0$, so the same two signs hold before the limiting endpoint. Therefore, for $\ell(c) = c(1 - \sqrt{4c^2 - 3})/2$,

$$\ell(c_X) < X < c_X - \ell(c_X), \qquad \ell(c_X) < 1 - X.$$

Write $c = h(z) = \sqrt{1 - z + z^2}$ and let z_X satisfy $h(z_X) = c_X$. The actual $c_5(s)$ corresponds to $0 < z \le z_X < 1/2$, and

$$s(z) = s_0 + \frac{1-R}{R}(1 - h(z)), \qquad s_0 = \frac{R - u + \eta}{R} > 0.$$

The transition input $(1-z)h(z)$ decreases, and the preceding root ordering gives $s(z_X) < (1-z_X)h(z_X)$, hence the same strict inequality for all $z \le z_X$. To check the order of the two boundary coordinates put

$$\phi(z) = s(z) - zh(z), \qquad k_R = \frac{1-R}{R}.$$

Its endpoint values are $s_0 > 0$ and $X - \ell(c_X) > 0$. Moreover

$$\phi'(z) = \frac{(k_R + z)(1-2z)}{2h(z)} - h(z).$$

For $k_R \le 2$ this is nonpositive; for $k_R > 2$ its zero is described by the strictly increasing function $K_0(z) = (2-3z+4z^2)/(1-2z)$, so the minimum is again at an endpoint. Thus $zh(z) < s(z)$. Applying the already displayed polynomial P_z gives the strict Cell-L selector, and $s + zh(z) \le X + \ell(c_X) < 1$. Since $\partial_b F_L = c(1-2z) > 0$ and each feasible fiber is an interval, the capped maximum is

$$\widehat{b}_5 \le zh(z) \le z_X h(z_X) = \ell(c_X) < 1 - X.$$

Together with $\widehat{b}_1 \le X$, this proves the loss for every right low label.

High–high. Let $c_L(z)$ be the unique root in $[\sqrt{3}/2, 1]$ of $f(c,z) = 0$. With $D_c = 4c^3 - 2c + z > 0$, implicit differentiation gives

$$c_L''(z) = \frac{2(c^2 - cz + z^2) + 16c^2 z(c-z)}{D_c^3} > 0.$$

A selected left high label requires $s < 1/2$ and $c_5(s) \le c_L(s)$. Define

$$s_0 = \frac{R - u + \eta}{R};$$

then $0 < s_0 < s$ and $c_5(s_0) = 1$. The explicitly named function $g(z) = c_5(z) - c_L(z)$ is concave, with $g(s_0) > 0$. The right high bound $u \le \min\{RS, S/2\}$ gives $c_5(1/2) > 7/8$. For $R \le 1/2$ this follows after squaring from $17 - 10R + 9R^2 - 64R^3 - 64R^4 > 0$ (a decreasing polynomial with value $9/4$ at $1/2$); for $R \ge 1/2$ it follows from $(3+R)^2 - 16S^2 = (1-R)(15R-7) > 0$. Finally

$$f(7/8, 1/2) = 33/4096 > 0$$

and f increases in c on this range, so $c_L(1/2) < 7/8$ and $g(1/2) > 0$. Concavity contradicts $g(s) \le 0$. Thus high–high is impossible.

If both labels are low, the cap already gives $\widehat{b}_5 + \widehat{b}_1 \le s + u = 1 - w < 1$. The preceding five paragraphs cover every other ordered pair of the four labels, proving (93). ☐

Lemma L.8 (CE2 two-endpoint capped loss)**.** *Let $[x, U]$ and $[y, v]$ be the two exact CE2 traces in the legacy coordinates of Remark F.2, where U denotes the endpoint called u there. Put*

$$p = 1 - U, \qquad q = 1 - v, \qquad r = \frac{x}{x+y}, \qquad w = \frac{y}{x+y},$$

and reflect if necessary so that $0 < w \le 1/2$ and $r = 1 - w$. Then

$$F_{p/w}(p) + F_{q/r}(q) < 1. \tag{96}$$

Proof. Put

$$S_0 = x + y, \quad K = \sqrt{1 - wr}, \quad z = \frac{w}{1+K}, \quad \zeta = \frac{r}{1+K}, \quad \alpha = \frac{p}{w}, \quad \beta = \frac{q}{r}.$$

Then $D = S_0 K$, $1 - K = wr/(1+K)$, and the exact midpoint inequalities give $1/2 < \alpha, \beta < 1$. Dividing the coupling equation $(U + v)S_0 - xy = D$ by S_0, and using successively $x < U$ and $y < v$, gives the strict endpoint inequalities

$$\beta + p > 1 + z, \qquad \alpha + q > 1 + \zeta. \tag{97}$$

Define

$$\Phi_w(\alpha) = F_\alpha(w\alpha), \qquad \Phi_r(\beta) = F_\beta(r\beta), \qquad h(t) = \sqrt{1 - t + t^2}.$$

Thus the desired sum is $\Phi_w(\alpha) + \Phi_r(\beta)$.

Selected branch parametrizations. For a side of weight $\sigma \in \{w, r\}$, every L label has

$$c = h(k), \qquad F_c(\sigma c) = kh(k), \qquad 0 \le k \le w, \tag{98}$$

on either side. On the large side the range follows from

$$Q_{\text{cell}}/h(k)^2 = (k - w)G_k(r + k),$$

where $G_k(X) = h(k)^2X^3 + h(k)^2X^2 - k(1-k)X + k^2 > 0$ for $X \ge k$. The remaining selected parametrizations are

$$\alpha = \frac{h(a)}{w+a}, \qquad \Phi_w(\alpha) = a\alpha, \qquad r \le a < 1, \tag{99}$$

$$\beta = \frac{h(b)}{r+b}, \qquad \Phi_r(\beta) = b\beta, \qquad r \le b < 1, \tag{100}$$

$$\beta = \frac{K}{r+t}, \qquad \Phi_r(\beta) = t\beta = K - r\beta, \qquad w \le t \le r. \tag{101}$$

For example, the small-high selector factors as

$$h(a)^2 a\left(\frac{w}{(w+a)^2} - (1-a)\right), \qquad w - (1-a)(w+a)^2 = (a-r)(a^2 + wa - w),$$

which gives $a \ge r$; the other two factorizations are $(b-w)(b^2+rb-r)$ and $(t-w)(r^2-wt)$, respectively. These formulas specify the selected geometric components, not discarded algebraic roots.

The small-side T_- label is absent for $w < 1/2$. Large-side Full is incompatible with the first inequality in (97). Small-side Full can pair only with L: either high or T_- on the large side has $\beta \le K$, which would force $p > (1+r)z > w/(1+w)$, contrary to Full. Thus, up to the symmetric $w = r = 1/2$ tie, the exact list is

$$(\text{Full}, L), (T_+^{\text{hi}}, L), (L, L), (L, T_-), (L, T_+^{\text{hi}}), (T_+^{\text{hi}}, T_-), (T_+^{\text{hi}}, T_+^{\text{hi}}). \tag{102}$$

Low–low sums are at most $K < 1$: an L output is at most wK, while (101) is at most rK. For Full–L, put

$$b_{\text{thr}} = 1 + z - \frac{w}{1+w}, \qquad \kappa = \frac{w}{1 + (1+w)z}.$$

Full and (97) give $\beta > b_{\text{thr}}$ and $(1+\kappa)b_{\text{thr}} = 1 + z$. Direct reduction yields

$$b_{\text{thr}}^2 - h(\kappa)^2 = \frac{w^2(A(w) + KC(w))}{D_0(w, K)} > 0,$$

where

$$A(w) = 8 - 5w + 22w^2 + 6w^3 + 5w^4 + 5w^5 - w^6,$$
$$C(w) = 8 - w + 14w^2 + 4w^3 - 2w^4 + w^5,$$
$$D_0(w, K) = ((1 + w)(1 + K))^2(1 + K + w + w^2)^2.$$

Both numerator polynomials are positive on $[0, 1/2]$. Since h decreases and $(1 + k)h(k)$ increases there, $\beta = h(k) > b_{\mathrm{thr}} > h(\kappa)$ implies $(1 + k)\beta < 1 + z$; combined with (97), this gives $k\beta < p$ and hence the strict Full–L loss.

It remains to verify the four pairs containing a high label on the small side.

(L, T_+^{hi}). Use parameters $0 \le k \le w$ and $r \le t < 1$:

$$\alpha = h(k), \quad \Phi_w = kh(k), \qquad \beta = \frac{h(t)}{r + t}, \quad \Phi_r = \frac{th(t)}{r + t}.$$

The first output and the last output increase in their parameters, while $h(t)/(r+t)$ decreases. If $w \le 2/5$, then

$$\Phi_w + \Phi_r < wK + \frac{1}{2 - w} < 1;$$

after using $K \le 1 - wr/2$, the positive cleared gap is $P(w) = w^4 - 3w^3 + 4w^2 - 6w + 2$, which decreases on $[0, 2/5]$ and has $P(2/5) = 46/625$. If $w \ge 2/5$, the first endpoint inequality gives $\beta > r + z \ge 3/4$. Since $h(3/4)/(r + 3/4) = \sqrt{13}/(7 - 4w) < 3/4$, monotonicity forces $t < 3/4$. Therefore

$$\Phi_w + \Phi_r < \frac{\sqrt{3}}{4} + \frac{3\sqrt{13}}{20} < 1.$$

(T_+^{hi}, T_-). Use a, t from (99) and (101). Both demands are at most K. The first endpoint inequality gives

$$\alpha > \frac{1 + z - K}{w} = \frac{1 + r}{1 + K} =: a_0.$$

Thus $K > 1 - w^2$, which forces $w > 1/3$, and comparison at $a = 2/3$ forces $a < 2/3$. The function

$$G(a) = \frac{(1 + a)h(a)}{w + a}$$

decreases on $[r, 2/3]$, because its derivative numerator $2a^3 + (4w - 1)a^2 + (1 - w)a + w - 2$ is at most its value $-7/54$ at $(2/3, 1/2)$. Hence $G(a) \le (1 + r)K$. Using the second endpoint inequality and $\Phi_r = K - r\beta$ gives

$$\Phi_w + \Phi_r < (2 + r)K - 1 - \zeta < 1.$$

After multiplication by $1 + K$, the last positive gap is $r(3w - w^2 - K)$; its squared comparison is

$$(3w - w^2)^2 - K^2 = w^4 - 6w^3 + 8w^2 + w - 1 > 0$$

for $w > 1/3$ (value $1/81$ at $1/3$, positive derivative thereafter).

$(T_+^{\mathrm{hi}}, T_+^{\mathrm{hi}})$. Use parameters a, b from (99)– (100). The endpoint inequalities first force $w > 2/5$: for $w \le 2/5$, the asserted contrary inequality $K(w + 1/(2r)) \le 1 + z$ reduces, after multiplication by $2r(1 + K)$, to

$$K(2w^2 - 4w + 1) + 2w^4 - 4w^3 + w^2 - w + 1 > 0,$$

which is immediate according as the coefficient of K is nonnegative or, using $K \le 1$, from the decreasing polynomial $2w^4 - 4w^3 + 3w^2 - 5w + 2 \ge 172/625$.

For $F_c(t) = (1+t)h(t)/(c+t)$, the derivative numerator $2t^3 + (4c-1)t^2 + (1-c)t + c - 2$ is increasing on the present rectangle, so F_c has at most one interior minimum. Endpoint comparison gives

$$\Phi_w + \alpha \le \frac{2}{1+w}, \qquad \Phi_r + \beta \le \frac{2}{1+r}.$$

The nontrivial endpoint signs are exactly

$$4 - (1+r)^2(1+w)^2K^2 = -w^2(w-1)^2(w^2 - w - 3) > 0$$

and

$$16r^2 - (1+r)^4 h(r)^2 = (1-r)(r^5 + 4r^4 + 7r^3 + 9r^2 - 4r - 1) > 0;$$

the bracket is increasing on $[1/2, 1]$ and equals $13/32$ at $1/2$. Weighting the two strict endpoint inequalities by w/K^2 and r/K^2 gives

$$\alpha + \beta > \frac{2K+1}{K(1+K)}.$$

Consequently

$$\Phi_w + \Phi_r < \frac{2}{1+w} + \frac{2}{1+r} - \frac{2K+1}{K(1+K)} < 1.$$

The numerator of the last gap is $3Kwr - Q(w)$, where $Q(w) = w^4 - 2w^3 + 7w^2 - 6w + 2 > 0$. Its squared difference is

$$D(w) = -w^8 + 4w^7 - 9w^6 + 13w^5 - 41w^4 + 65w^3 - 55w^2 + 24w - 4.$$

Here $D'(w) = (1-2w)H(w)$ with

$$H(w) = 4w^6 - 12w^5 + 21w^4 - 22w^3 + 71w^2 - 62w + 24 > 0$$

on $[2/5, 1/2]$; use $71w^2 - 62w + 24 \ge 743/71$ and the remaining cubic block at least $-11/4$. Thus D increases and $D(2/5) = 4904/390625 > 0$, proving the claim.

(T_+^{hi}, L). Use the fully specified parameters

$$\alpha = \frac{h(t)}{w+t}, \quad p = w\alpha, \quad b = t\alpha, \quad \frac{1}{2} \le t < 1,$$

and

$$\beta = H = h(k), \quad \widehat{b}_L = kH, \quad 0 \le k \le w.$$

Since $p + b = h(t)$, the first endpoint inequality gives

$$1 - (b + kH) > G := 2 + z - h(t) - (1+k)H. \tag{103}$$

Put

$$p_0 = 1 + z - K = \frac{w(2-w)}{1+K},$$

let $t_0 \in (1/2, 1)$ be the unique point with $p(t_0) = wh(t_0)/(w+t_0) = p_0$, and put $b_0 = t_0 p_0/w$. The exact endpoint lemma is

$$b_0 < 1 - wK. \tag{104}$$

To verify it, set $b_* = 1 - wK$ and $T_* = wb_*/p_0$. One has $0 < T_* < 1$, and

$$p_0^2(w + T_*)^2 - w^2 h(T_*)^2 = -\frac{w^3(KE(w) + J(w))}{(1+K)^2(2-w)^2} > 0,$$

where

$$E(w) = 2w^5 + 3w^3 - 20w^2 + 23w - 6,$$
$$J(w) = w^6 + 8w^5 - 22w^4 + 12w^3 + 6w^2 + 2w - 6.$$

Here $J \le -111/64$; if $E > 0$, then $KE + J < E + J < 0$, because $E + J$ is increasing and equals $-139/64$ at $w = 1/2$. Thus $p(T_*) < p(t_0)$. Since $p(t)$ decreases, $t_0 < T_*$ and (104) follows.

Finally put $P = 1 + z - p$. If $P < K$, then $t < t_0$ and $k \le w$, so

$$G > 2 + z - h(t_0) - (1 + w)K = 1 - b_0 - wK > 0.$$

If $P \ge K$, then $K \le P \le P_1 := 1 + z - w/(1 + w) \le 1$ and there is $a \in [0, w]$ with $h(a) = P$. Since $P < H$, one has $k < a$ and $(1+k)H < (1+a)P = P + \ell(P)$, where $\ell(P) = P(1 - \sqrt{4P^2 - 3})/2$. Hence

$$G > \mathcal{D}(P) := 1 - b(P) - \ell(P).$$

This function is concave. Indeed $\ell''(P) > 0$, and the high output b is convex in $p = 1+z-P$: with $A_t = 2+w-(1+2w)t > 0$ and $Q_t = 8wt^2+4t^2-8wt-13t-w+4 < 0$,

$$\frac{d^2 b}{dp^2} = -\frac{2h(t)(t + w)^3 Q_t}{w^2 A_t^3} > 0.$$

At $P = K$, (104) gives $\mathcal{D}(K) = 1-b_0-wK > 0$. At $P = P_1$, set $\kappa = w/(1+(1+w)z)$. The Full–L comparison already proved gives $P_1 > h(\kappa)$, hence $\ell(P_1) < \kappa P_1 = w/(1 + w)$, while $b(P_1) = 1/(1 + w)$. Thus $\mathcal{D}(P_1) > 0$, and concavity gives $G > 0$ throughout. Equation (103) proves the final hard pair.

The exact list (102) is now exhausted, proving (96). □

L.4. **The exactly-one-Vd1/Vd2 placements.** We now assume CE2, $N_+ = 1$, and exactly one row is Vd1 or Vd2. The corner normal form of Proposition G.6 will be used in the following form: for some $t > 0$ and $d = \sqrt{t^2 + t + 1}$,

$$x - (t + 1)y \le a, \qquad ty - (t + 1)x \le tb, \qquad tx + y \le d - a - tb. \tag{105}$$

In particular $a + b < 1/2$ for a positive-support row.

Lemma L.9 (Vd1 adjacent rescue)**.** *Assume an original open-role skeleton cover in the CE2 exact-M_0 branch, with T_0 the unique Vd1 row, $M_1 \in T_0$, T_1 uniquely supercritical, $T_2, \dots, T_5$ nonsupercritical Vd0, and no other positive-adjacent-support role. Then the perimeter together with r_1 is not covered.*

Proof. Lemma H.3 proves the two local inequalities in (72). Midpoint and support isolation force the center to cover the O-side radial gap before the Vd1 interval. The result is therefore Corollary H.5. □

Lemma L.10 (Vd1–supercritical replacement)**.**
Assume an original open-role perimeter-and-radial-skeleton cover in the CE2 exact-M_0 branch and the complementary no-additional-positive-support case. Let T_i, $i \ne 0$, be uniquely supercritical, and let the unique Vd1 row, distinct from T_0 and adjacent to T_i, contain M_i. Assume every other vertex row is nonsupercritical Vd0 and the center has no boundary trace on the shared edge of this pair. Then the pair can be replaced by two open nonsupercritical Vd0 roles preserving all boundary and radial demands used by Proposition L.2.

Proof. Normalize the Vd1 row at V_0 and the supercritical row at V_1. By the shared-edge hypothesis, no center trace intervenes on $e_{0,1}$; by the no-additional-positive-support hypothesis, no third vertex role has positive boundary or radial support in the handoffs used below. In (105), midpoint rescue and one-sided support imply $t \geq 1$ and, with

$$c = \frac{d-a-tb}{t+1}, \quad \lambda = \frac{t(1-b)}{t+1}, \quad \mu = \frac{d-a-tb-1}{t},$$

give the strict consequences

$$a + c < 1, \qquad a < \lambda \leq \frac{1}{2}, \qquad b < \frac{1}{2}, \qquad \mu < 1 - a. \tag{106}$$

The shared edge forces the supercritical input $a_i \geq 1 - b > 1/2$, and radial coverage forces $c_i \geq \lambda$. A direct consequence of Proposition J.2 is

$$(x, y, z) \in \mathcal{A}, \quad x \geq 1/2, \quad z \leq 1/2 \quad \Longrightarrow \quad y \leq 1 - z.$$

Indeed, substituting $x = 1/2, y = 1 - z$ in the selected supercritical polynomial gives $z^2(2z-5)(2z-1)/4 \geq 0$, and the polynomial is increasing in both relevant coordinates. Hence $b_i \leq 1 - c_i \leq 1 - \lambda$ and $a + b_i < 1$. Open radial handoff further gives

$$c_i^{\mathrm{req}} < \max\{c_i, \mu\} < \max\{1 - b_i, 1 - a\}.$$

Choose $a < p_1 < p_2 < 1 - b_i$ within these strict margins. In local coordinates use the unit triangles

$$\Delta_p^- = \operatorname{conv}\{(0, 1-p), (1, 1-p), (0, -p)\}, \quad p \leq 1/2,$$

and

$$\Delta_p^+ = \operatorname{conv}\{(p, 0), (p, 1), (p-1, 0)\}, \quad p \geq 1/2.$$

Their incident reaches are $p, 1-p$, their own-radial reach is respectively $1-p, p$, and they have zero adjacent support. Small translations, chosen below every strict margin above, make the two replacements open Vd0 roles with boundary sum < 1 while preserving the shared boundary and both radial handoffs. Thus the datum reduces to Proposition L.2. ☐

L.5. **Exact-demand terminal routing.**

Proposition L.11 (Exact-demand terminal branches)**.** *The exact-demand method proves the following terminal exclusions:*

1. *CE1 or CE2, $N_+ = 0$, all vertex roles Vd0;*
2. *CE1 or CE2, $N_+ = 0$, with at least one T3-like role and no Vd1/Vd2;*
3. *CE1 or CE2, $N_+ = 1$, all vertex roles Vd0, with at least one boundary gap;*
4. *CE1 or CE2, $N_+ = 1$, exactly one T3-like role and no Vd1/Vd2;*
5. *the CE2, $N_+ = 1$, exactly-one-Vd1/Vd2 branch.*

All gap alternatives include singleton gaps.

Proof. The five items are respectively Propositions L.2, L.5, K.2, L.1, and H.8. The strictness at open handoffs appears in the weak endpoint bounds, the nonattained supercritical envelope, and the strict replacement margins; no positive-length assumption is made about a V-gap itself. ☐

Appendix M. Strategy 2 Rigor Completion

This section supplies two pieces that must not be hidden by the concise reader presentation. First, it formally incorporates the complete status-bearing $407X$ four-label endpoint proof. Second, it gives the explicit Vd1–supercritical axis replacement in TeX. The incorporated proof-package objects are part of the proof at the exact Git blobs listed below.

M.1. **The incorporated $407X$ endpoint supplement.** All paths in the following table are relative to

`proof/4XXX_CE1CE2/40XX_Nplus0/`

`407X_T3_like_no_Vd1Vd2/`

proof-package object	Git blob SHA	role
`4073_boundary_loss_framework.md`	`f3f0395748f5649c62eff51f648e8052039a38c5`	residual reduction
`4074_L_Full_branch.md`	`880b59ec8d3c59b2d9ec036c188aebb2dfbea415`	(L, Full)
`4075_Tminus_low_lower_branch_obligations.md`	`ebe2b65f8840b8667c3dca672a8d0cdf2d4104ad`	first T_-
`4078_left_L_family_completion.md`	`f1323325b0690a1a9394d93911c28425ecfbca9b`	remaining first L
`4079_first_Full_branch.md`	`c66eea05277dbf2e1781f133f4f18c469f1b0909`	first Full
`407a_left_Thigh_branch_completion.md`	`aa5cb1a6dc635639972ba8f48cc92c9f519da296`	first T_+^{hi}
`407c_rigor_completion_details.md`	`c80243f67124020e94239745b23ebed2e50a8bcb`	analytic details
`407d_rigor_final_assembly.md`	`a6ef81f787c4c3d6933569ca89fcdf2eaa83dbc7`	final assembly

The exact capped-map theorem used by these files is Proposition J.3. Its four labels are

$$\mathrm{Full}, \qquad L, \qquad T_-, \qquad T_+^{\mathrm{hi}}.$$

The supplement retains the exact residual inputs a_1^*, a_5^* and radial demands c_1^*, c_5^* from Lemma L.4. It does not replace them by a symmetric or universal envelope.

first label	right labels	incorporated terminal source
L	Full	‘4074‘, low-root separation
L	L, T_-	‘4078‘, exact low-sheet inequalities
L	T_+^{hi}	‘4078‘ and ‘407c‘
T_-	L, T_-	‘4075‘, exact root ordering
T_-	Full, T_+^{hi}	‘4075‘, strict endpoint separation
Full	all labels	‘4079‘, first-Full exclusion
T_+^{hi}	Full, T_+^{hi}	‘407a‘, high-input exclusion
T_+^{hi}	L, T_-	‘407a‘ and ‘407c‘, analytic threshold and right-T_- bounds

Table 5. The exhaustive four-label endpoint audit. The listed exact source objects contain every polynomial selector, domain restriction, and strict inequality used in the corresponding entry.

Theorem M.1 (Complete support-isolated T3-like endpoint certificate)**.** *For every support-isolated state of Definition 4.3,*

$$\widehat{g}_{c_5^*}(1 - a_5^*) + \widehat{g}_{c_1^*}(1 - a_1^*) < 1.$$

Proof. If $a_1^* + a_5^* > 1$, the diagonal caps already give the strict inequality. Otherwise Proposition J.3 assigns each endpoint one of the four labels above. Table 5 is an exhaustive partition of the resulting ordered label pairs. The incorporated source in each row proves either that the pair is impossible or that its two exact capped outputs have sum strictly below one. File ‘407d‘ checks that the geometric residual reduction in ‘4073‘ reaches exactly these branches and no others. Therefore the displayed endpoint inequality holds throughout the complete support-isolated domain. □

M.2. **The explicit Vd1–supercritical replacement.**

Proposition M.2 (Vd1–supercritical axis replacement)**.** *Assume the CE2 exact-M_0 branch, with a unique supercritical row at V_1 and an adjacent Vd1 row at V_0 containing M_1. Assume every other vertex row is nonsupercritical Vd0, no additional role has positive support on the two relevant radial handoffs, and the center has no boundary trace on the shared edge $e_{0,1}$. Then the two special rows can be replaced by two open nonsupercritical Vd0 roles preserving all boundary and radial demands used by the all-Vd0 boundary-loss theorem.*

Proof. Use the Vd corner normal form with $t > 0$, $d = \sqrt{t^2 + t + 1}$ and exact Vd1 boundary reaches a, b:

$$x - (t+1)y \le a, \qquad ty - (t+1)x \le tb, \qquad tx + y \le d - a - tb.$$

Its own-radial reach and neighboring-arm endpoints are

$$c = \frac{d - a - tb}{t+1}, \qquad \lambda = \frac{t(1-b)}{t+1}, \qquad \mu = \frac{d - a - tb - 1}{t}.$$

Because the original open Vd1 role contains M_1,

$$b > \frac{t-1}{2t}, \qquad a + tb < d - 1 - \frac{t}{2}.$$

If $t < 1$, then

$$d - a - tb - t > 1 - \frac{t}{2} > \frac{1}{t+1} > \frac{1-a}{t+1},$$

which would create positive support on the other adjacent arm. Hence $t \ge 1$.

We need four strict margins:

$$a + c < 1, \qquad a < \lambda \le \frac{1}{2}, \qquad b < \frac{1}{2}, \qquad \mu < 1 - a. \tag{107}$$

For the first, maximize $a + c$ on the closed midpoint face $a = d - 1 - t/2 - tb$. Then

$$a + c - 1 = -\frac{2bt^2 + 2bt - 2dt - 2d + t^2 + 4t + 2}{2(t+1)}.$$

Using $b \ge (t-1)/(2t)$, the numerator is at least $2t^2 + 4t + 1 - 2(t+1)d$, which is positive because

$$(2t^2 + 4t + 1)^2 - 4(t+1)^2 d^2 = 4t^3 + 4t^2 - 4t - 3 > 0$$

for $t \ge 1$. The original midpoint inequalities are strict. The bound $\lambda \le 1/2$ follows from the lower bound for b. The closed upper bound for a, subtracted from λ, is minimized at $b = (t-1)/(2t)$ and equals $(t+1) - d > 0$, so $a < \lambda$. Furthermore

$$b < \frac{d - 1 - t/2}{t} < \frac{1}{2}$$

because $d < t + 1$. Finally

$$t(\mu + a - 1) = d - (t+1) - tb + (t-1)a \le t(d - t - 1) < 0.$$

This proves (107).

Let (a_1, b_1, c_1) be the actual reaches of the unique supercritical row. The center-free shared edge forces

$$a_1 \geq 1 - b > \frac{1}{2},$$

and radial coverage before the Vd1 interval forces $c_1 \geq \lambda$. We use the following admissibility consequence.

Half-square lemma. If $(x, y, z) \in \mathcal{A}$, $x \geq 1/2$, and $0 \leq z \leq 1/2$, then

$$y \leq 1 - z.$$

Indeed, suppose $y > 1 - z$. Then $x + y > 1$, so the selected supercritical cell applies. In the ordered half $x \leq y$, its necessary polynomial is

$$F(x, y, z) = (x^2 - 1)z^2 + (2xy^2 + y)z + y^4 - y^2 \leq 0.$$

It is nondecreasing in x. At $x = 1/2$ its y-derivative is $4y^3 - 2y + 2yz + z$, which is increasing for $y \geq 1 - z$ and positive at that endpoint. Hence

$$F(x, y, z) > F\left(\frac{1}{2}, 1 - z, z\right) = \frac{z^2(2z - 5)(2z - 1)}{4} \geq 0,$$

a contradiction. In the reflected half $y < x$, use $F(y, x, z)$. Both arguments exceed $r = 1 - z$, and on that quadrant the two partial derivatives are nonnegative; the second is bounded below by

$$4r^2z + z + 4r^3 - 2r = 2 - 5z + 4z^2 > 0.$$

Thus

$$F(y, x, z) > F(r, r, z) = z(1 - 2z) \geq 0,$$

again a contradiction.

A supercritical row has own-radial coordinate at most 1/2, so the lemma applied to (a_1, b_1, c_1) gives

$$b_1 \leq 1 - c_1 \leq 1 - \lambda.$$

Together with $a < \lambda$ this gives

$$a + b_1 < 1. \tag{108}$$

Let d_1^C be the center reach on the rescued arm, measured from O, and put $c_1^{\mathrm{req}} = 1 - d_1^C$. Only the center, the supercritical row, and the Vd1 row have positive intervals on this arm. Open coverage at the center handoff gives

$$c_1^{\mathrm{req}} < \max\{c_1, \mu\}.$$

By the half-square lemma and (107),

$$\max\{c_1, \mu\} \leq \max\{1 - b_1, 1 - a\}.$$

Therefore

$$c_1^{\mathrm{req}} < \max\{1 - b_1, 1 - a\}. \tag{109}$$

The strictness comes from the first inequality; no strictness is required in the second comparison.

Choose

$$a < p_1 < p_2 < 1 - b_1$$

within the strict margins so that

$$p_1 < \frac{1}{2}, \qquad 1 - p_1 > c, \qquad \max\{p_2, 1 - p_2\} > c_1^{\mathrm{req}}.$$

This is possible by choosing p_1 close to a and choosing p_2 close to a when the right side of (109) is supplied by $1-a$, or close to $1-b_1$ when it is supplied by $1-b_1$.

In the local metric define

$$\Delta_p^- = \operatorname{conv}\{(0,1-p),(1,1-p),(0,-p)\} \quad (p \le 1/2),$$
$$\Delta_p^+ = \operatorname{conv}\{(p,0),(p,1),(p-1,0)\} \quad (p \ge 1/2).$$

These are unit equilateral triangles. Their incident reaches are $p, 1-p$; their own-radial reaches are respectively $1-p, p$; and they have no positive adjacent-arm support.

Translate $\Delta_{p_1}^-$ by $(-\varepsilon, 0)$. Its relevant reaches become

$$(p_1-\varepsilon, 1-p_1, 1-p_1).$$

For the second row translate $\Delta_{p_2}^-$ by $(-\varepsilon, 0)$ if $p_2 < 1/2$, and translate $\Delta_{p_2}^+$ by $(0,-\varepsilon)$ if $p_2 \ge 1/2$. Its relevant reaches are respectively

$$(p_2-\varepsilon, 1-p_2, 1-p_2)$$

or

$$(p_2, 1-p_2-\varepsilon, p_2).$$

Choose $\varepsilon > 0$ smaller than every strict boundary, radial, and $p_2 - p_1$ margin. The first replacement still supplies at least a and c; the second supplies at least b_1 and c_1^{req}. The shared open edge is covered because

$$(1-p_1) + (p_2 - \varepsilon) > 1.$$

Both replacements contain their distinguished vertices in their interiors, have zero adjacent support, and have boundary sum strictly below one. Thus they are open nonsupercritical Vd0 roles, and every demand used by the all-Vd0 boundary-loss theorem is preserved. □

Appendix N. Authoritative CE2 One-Vd Placement Assembly

The earlier exact-demand chapter contains a compact placement summary. The following proposition is the authoritative assembly used by the final proof; it cites only the complete placement lemmas and the explicit replacement of Appendix M.

Proposition N.1 (Complete CE2 exactly-one-Vd1/Vd2 assembly)**.** *Assume the center has type CE2, $N_+ = 1$, exactly one vertex role has type Vd1 or Vd2, and no T3-like role is present. Then the seven roles do not cover H.*

Proof. Normalize the unique center midpoint to M_0. If an additional vertex role has positive adjacent-arm support, then the unique supercritical row, the Vd1/Vd2 row, and that additional role are three short roles; the skeleton contradiction of Theorem G.15 applies. Hence assume the no-additional-positive-support complement.

First suppose the unique supercritical row is T_0. If the Vd1/Vd2 row is adjacent, it is eliminated by Lemma H.6; the proof treats both possible supported arms and its reflection. If it is nonadjacent, its index lies in $\{2,3,4\}$ after reflection, and Lemma H.7 gives a strict radial separation.

Next suppose the unique Vd1/Vd2 row is T_0. It must rescue the midpoint of the unique supercritical row, so midpoint locality makes that row adjacent. If T_0 is Vd2, the neighboring-midpoint cap Lemma G.7, combined with the center and supercritical perimeter caps, is Proposition G.10. If T_0 is Vd1, the exact Vd1 local profile and the common center-hiding theorem give Corollary H.5.

Finally suppose neither special row is T_0. Since the exact-M_0 center covers no other radial midpoint, the Vd1/Vd2 row must rescue the midpoint missed by the unique supercritical row. Diameter locality makes the two rows adjacent. A Vd2 rescue is again eliminated by the neighboring-midpoint cap. For a Vd1 rescue, the shared edge has no center trace in this placement and the no-additional-support hypotheses are exactly those of Proposition M.2. Replace the pair by two open nonsupercritical Vd0 roles while preserving every boundary and radial demand used by Proposition L.2. All other vertex roles were already nonsupercritical Vd0, so that all-Vd0 theorem gives the contradiction.

These alternatives exhaust the relative positions of the center-midpoint row, the unique supercritical row, and the unique Vd1/Vd2 row. □

N.1. **Closure and correction of the legacy Strategy 2 references.**

Proposition N.2 (Authoritative closure of the compact CE2 citations)**.** *The CE2 exactly-one-Vd1/Vd2 item in Propositions 4.6 and I.4 is proved by Proposition N.1. The replacement conclusion stated in Lemma L.10 is proved in full by Proposition M.2.*

In the compact proof of Lemma L.10, the displayed chain

$$c_i^{\mathrm{req}} < \max\{c_i, \mu\} < \max\{1 - b_i, 1 - a\}$$

contains an overstrong second comparison. Its authoritative corrected form is

$$c_i^{\mathrm{req}} < \max\{c_i, \mu\} \leq \max\{1 - b_i, 1 - a\},$$

and therefore still gives

$$c_i^{\mathrm{req}} < \max\{1 - b_i, 1 - a\}.$$

No later argument uses the overstrong comparison.

Proof. The first assertion is the exhaustive placement proof of Proposition N.1. In its final placement, the shared edge is center-free and the complete two-ordered-half admissibility, strict-margin, and open-axis-replacement argument is exactly Proposition M.2. That proof establishes $c_i \leq 1 - b_i$ and $\mu < 1 - a$, which give the weak second comparison above. The first comparison is strict by open radial coverage, so the final bridge margin remains strict. Thus the earlier compact citations are summaries of these later complete proofs and are not independent proof dependencies. □

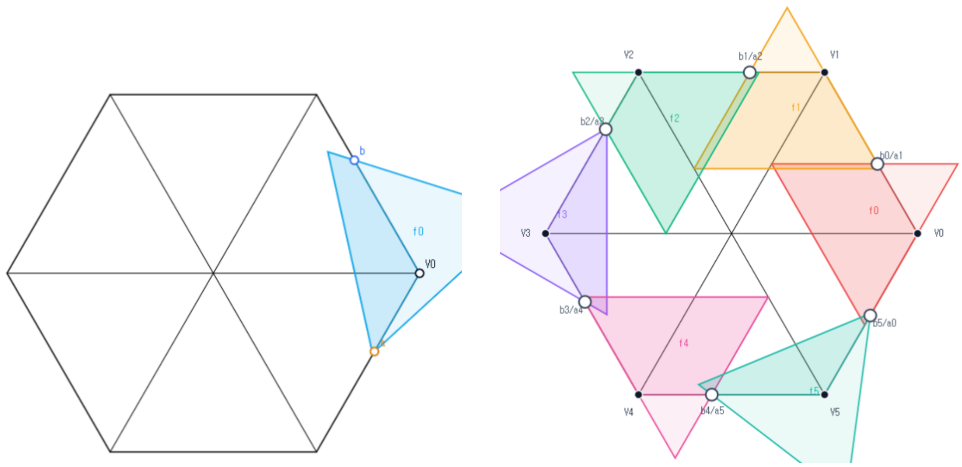
Appendix O. Area-Loss Obstructions

This section proves the local inequalities and cyclic estimates used in the area-loss argument.

Let

$$A_{\triangle} = \frac{\sqrt{3}}{4}$$

be the area of a unit equilateral triangle. Throughout this section areas are divided by $A_{\triangle}$. Thus a unit triangle has normalized area one and H has normalized area six.



(a) local retained area

(b) global retained areas

FIGURE 12. Schematic views of Strategy 3, not to scale. The local panel shows one vertex role with boundary reaches a, b, and the global panel shows the six cyclic roles. Only the darker portions represent area retained inside H; the proof uses the inequalities below, not the picture.

O.1. **The local square-loss inequalities.** Fix a vertex V_i. Let P_a and P_b be the points at distances a and b from V_i on its incoming and outgoing boundary edges. For a closed unit equilateral triangle T containing V_i, P_a, P_b, put

$$\mathcal{L}(T) = \frac{\operatorname{area}(T \setminus H)}{A_\triangle}.$$

Equivalently, if

$$f(a,b) = \max_T \frac{\operatorname{area}(T \cap H)}{A_\triangle}, \qquad \mathcal{L}(a,b) = 1 - f(a,b),$$

then $\mathcal{L}(a,b)$ is the least possible normalized loss. Reflection in the radial axis interchanges a and b, and increasing either demand can only decrease f.

Lemma O.1 (Local wedge reduction)**.** *After rotating indices to $i = 0$, write*

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0), \qquad W = \{(x,y) : x \geq 0,\ y \geq 0\}.$$

If a closed unit equilateral triangle T contains V_0, then

$$T \cap H = T \cap W.$$

Normalized Euclidean area is twice the corresponding (x,y)-coordinate area.

Proof. The two basis vectors have length one and angle $120°$, so

$$\|(x,y)\|^2 = x^2 + y^2 - xy, \qquad |\det(V_1 - V_0, V_5 - V_0)| = \frac{\sqrt{3}}{2}.$$

The hexagon is

$$H = \{0 \leq x, y \leq 2,\ |x - y| \leq 1\} \subset W.$$

Conversely, if $(x,y) \in T \cap W$, then the diameter-one condition gives $x^2+y^2-xy \le 1$. Hence

$$|x-y|^2 \le x^2+y^2-xy \le 1, \qquad \frac{3}{4}x^2 \le x^2+y^2-xy,$$

and the same estimate holds for y. Thus $|x-y| \le 1$ and $0 \le x, y \le 2/\sqrt{3} < 2$, proving $(x,y) \in H$. The determinant identity proves the area assertion. $\square$

Theorem O.2 (Unconditional local square loss)**.** *For every feasible pair* $0 \le a, b \le 1$ *and every realizing closed unit equilateral triangle* T*,*

$$\mathcal{L}(T) \ge \min(a,b)^2. \tag{110}$$

If $a+b>1$*, then*

$$\mathcal{L}(T) \ge \max(a,b)^2. \tag{111}$$

Consequently the same inequalities hold for $\mathcal{L}(a,b) = 1 - f(a,b)$*.*

Proof. In the coordinates of Lemma O.1, physical rotation through $60°$ is $R(x,y) = (x-y, x)$. Put

$$0 \le t \le 1, \qquad D = 1 - t + t^2, \qquad z = \sqrt{D}.$$

After choosing one of the six directed sides, every orientation has one of the following two forms:

$$\begin{array}{c|cc} & d & Rd \\ \hline \mathrm{I} & z^{-1}(1,t) & z^{-1}(1-t,1) \\ \mathrm{II} & z^{-1}(t,-(1-t)) & z^{-1}(1,t). \end{array} \tag{112}$$

Indeed these forms cover, respectively, the physical direction sectors $[120°, 180°]$ and $[60°, 120°]$; their endpoints cover the common boundary orientations.

For Type I introduce

$$U = \alpha + y - tx, \qquad V = \beta + x - (1-t)y.$$

Then T is the simplex $U, V \ge 0$, $U + V \le z$. Containment of $(0,0), (0,a), (b,0)$ gives

$$\alpha \ge tb, \quad \beta \ge (1-t)a, \quad \alpha + \beta + (1-t)b \le z, \quad \alpha + \beta + ta \le z. \tag{113}$$

The two inequalities defining W become

$$(1-t)U + V \ge (1-t)\alpha + \beta, \qquad U + tV \ge \alpha + t\beta.$$

For fixed t, lowering α or β therefore enlarges $T \cap W$. The outside loss is minimized at $\alpha = tb$, $\beta = (1-t)a$. At that placement the portion outside W has vertices, in the (U,V)-plane,

$$(0,0), \quad (a+tb, 0), \quad (tb, (1-t)a), \quad (0, b + (1-t)a).$$

Since the Jacobian of $(x,y) \mapsto (U,V)$ has absolute value D,

$$\mathcal{L}(T) \ge \mathcal{L}_{\mathrm{I}}(a,b;t) = \frac{(1-t)a^2 + tb^2 + 2t(1-t)ab}{D}. \tag{114}$$

If $a \le b$, then

$$D(\mathcal{L}_{\mathrm{I}} - a^2) = t\{b^2 - a^2 + (1-t)(a^2 + 2ab)\} \ge 0;$$

the reflected factorization proves $\mathcal{L}_{\mathrm{I}} \ge b^2$ when $b \le a$. This proves (110) in Type I.

Suppose now that $a+b>1$ and $a \le b$. Set

$$r = \frac{a}{b}, \qquad t_0 = \frac{1-r^2}{1+2r}.$$

Since $b > 1/(1+r)$, feasibility in (114) and (113) excludes $0 \le t < t_0$: for $0 < t < t_0$,

$$\left(\frac{1+r(1-t)}{1+r}\right)^2 - D = \frac{t(1+2r)(t_0-t)}{(1+r)^2} > 0,$$

so $b+(1-t)a > z$, while $t=0$ would give $a+b \le 1$. Hence $t \ge t_0$, and

$$D(\mathcal{L}_{\mathrm{I}} - b^2) = (1-t)b^2\{r^2 - 1 + t(2r+1)\} \ge 0.$$

If $b \le a$, put $r = b/a$ and $t_1 = (r^2+2r)/(1+2r)$. The identical reflected calculation excludes $t > t_1$ and gives

$$D(\mathcal{L}_{\mathrm{I}} - a^2) = ta^2\{r^2 + 2r - t(1+2r)\} \ge 0.$$

Thus (111) holds in Type I.

For Type II put

$$U = \alpha + (1-t)x + ty, \qquad V = \beta + tx - y.$$

Again T is the simplex $U, V \ge 0$, $U+V \le z$. Its exact reaches on the positive y- and x-axes are

$$A = \beta, \qquad B = z - \alpha - \beta, \qquad A + B \le z \le 1. \tag{115}$$

In particular Type II is impossible when $a+b>1$. Its wedge intersection is the quadrilateral with vertices

$$(0,0), \quad (B,0), \quad \left(\frac{B+(1-t)A}{D}, \frac{A+tB}{D}\right), \quad (0,A).$$

Hence

$$\mathcal{L}(T) = \mathcal{L}_{\mathrm{II}}(A,B;t) = 1 - \frac{(1-t)A^2 + tB^2 + 2AB}{D}. \tag{116}$$

Let $S = A+B$. If $A \le B$, write $A = rS$, $B = (1-r)S$ with $0 \le r \le 1/2$. Since $S^2 \le D$, and with

$$C = (1-t)r^2 + t(1-r)^2 + 2r(1-r),$$

we have $\mathcal{L}_{\mathrm{II}} \ge 1 - C$. The exact factorization

$$1 - C - r^2 D = (1-t)(1-2r+tr^2) \ge 0$$

gives $\mathcal{L}_{\mathrm{II}} \ge r^2 D \ge A^2$. Reflection gives $\mathcal{L}_{\mathrm{II}} \ge B^2$ when $B \le A$. Since $A \ge a$ and $B \ge b$, this proves (110). The two orientation types are exhaustive, so the theorem follows. □

O.2. **The direct** T3-like **loss.** The next theorem uses the classification data (o,n) from Section C: o counts triangle vertices outside H, and n counts adjacent radial arms met in positive length. Thus a T3-like role has $(o,n) = (2,1)$.

Theorem O.3 (Direct T3-like loss)**.** *Let T be a closed unit V_i-triangle of* T3-like *type, and let a, b be lower bounds for its adjacent boundary reaches. Then*

$$a + b \le 1. \tag{117}$$

Moreover, for $0 \le m \le 1/2$,

$$a, b \ge m \implies \mathcal{L}(T) \ge 2m - 4m^2. \tag{118}$$

Proof. The first assertion is Lemma C.8. For the loss bound apply the closed-trace domination in Proposition C.6. It gives a translated Type-II triangle $\widehat{T}$ of the same area with

$$T \cap H \subset \widehat{T} \cap H, \qquad \mathcal{L}(T) \geq \mathcal{L}(\widehat{T}).$$

In the notation of that proposition, $0 < t < 1$, $D = 1 - t + t^2$, $z = \sqrt{D}$, and after reflection the translated majorant has exact reaches

$$A = \beta, \qquad B = z - \beta$$

and unique wedge vertex

$$Q = \left(\frac{B + (1-t)A}{D}, \frac{A + tB}{D} \right), \qquad Q_x > 1, \quad Q_y \leq 1.$$

Thus $z - tA > D$, so

$$A < L := \frac{z - D}{t} = \frac{z(1-z)}{t}. \tag{119}$$

The exact normalized loss is

$$\mathcal{L}(\widehat{T}) = \mathcal{L}(z, A) = \frac{tA^2 + (1-t)(z-A)^2}{D}.$$

This decreases for $A < L < (1-t)z$, and therefore

$$\mathcal{L}(T) \geq \mathcal{L}(\widehat{T}) > \mathcal{L}(z, L). \tag{120}$$

Set $r = (1-z)/t$. Then $0 < r < 1/2$, and eliminating t, z gives

$$t = \frac{1-2r}{1-r^2}, \qquad z = \frac{r^2 - r + 1}{1 - r^2},$$

$$L = A_0(r) := \frac{r(r^2 - r + 1)}{1 - r^2}, \qquad z - L = B_0(r) := \frac{r^2 - r + 1}{1 + r}.$$

The limiting normalized inside area and loss are

$$F_0(r) = \frac{(r^2 - r + 1)^2}{1 - r^2}, \qquad \mathcal{L}_0(r) = 1 - F_0(r).$$

Since the exact reach A is at least the required demand $a \geq m$, (119) gives $A_0(r) > m$. Direct simplification yields

$$\mathcal{L}_0(r) - \{2A_0(r) - 4A_0(r)^2\} = \frac{r^2(5r^4 - 8r^3 + 13r^2 - 8r + 2)}{(1 - r^2)^2} \geq 0,$$

because the last polynomial is at least $9r^2 - 8r + 2 \geq 2/9$ on $[0, 1/2]$. Also $A_0(r) \leq r$, and

$$\mathcal{L}_0(r) - \frac{1}{4} = \frac{(1-2r)(2r^3 - 3r^2 + 6r - 1)}{4(1 - r^2)} \geq 0 \quad \text{when } A_0(r) \geq \frac{1}{4};$$

the cubic is increasing and has value $11/32$ at $r = 1/4$.

Finally $\psi(u) = 2u - 4u^2$ is increasing on $[0, 1/4]$ and never exceeds $1/4$ on $[0, 1/2]$. If $A_0(r) \leq 1/4$, then $\mathcal{L}_0(r) \geq \psi(A_0(r)) \geq \psi(m)$; if $A_0(r) \geq 1/4$, then $\mathcal{L}_0(r) \geq 1/4 \geq \psi(m)$. Together with (120), this proves (118). The reflected crossing is identical. $\square$

O.3. **The cyclic area certificates.** The selected handoffs of Proposition C.9 have the form

$$(a_i, b_i) = (1 - x_{i-1}, x_i), \qquad 0 < x_i < 1,$$

with cyclic indices. Row i is selected supercritical exactly when $x_i > x_{i-1}$.

Lemma O.4 (Cyclic area-loss certificate)**.** *Let the six actual vertex triangles realize selected handoffs*

$$(a_i, b_i) = (1 - x_{i-1}, x_i), \qquad 0 < x_i < 1.$$

Their total normalized loss is strictly greater than one if either

(1) *at least two selected rows are supercritical; or*
(2) *exactly one selected row is supercritical, every role is* Vd0 *or* T3-like*, and at least one role is* T3-like*.*

Proof. Put $m = \min_i x_i$ and $M = \max_i x_i$. The reflection $y_i = 1 - x_{-i-1}$ swaps each row's two coordinates and preserves both alternatives and the total loss. Hence assume $m \leq 1 - M$. Every selected row coordinate is then at least m, so (110) gives loss at least m^2 for every row.

In case (1), choose an ascent out of a minimum plateau, $x_{p-1} = m < x_p$. Its row contains the coordinate $1 - m$ and loses at least $(1 - m)^2$ by (111). A second ascent has one coordinate strictly larger than $1/2$ and hence loses strictly more than $1/4$. The other four rows lose at least m^2 each, so the total loss is strictly larger than

$$4m^2 + (1 - m)^2 + \frac{1}{4} = 5\left(m - \frac{1}{5}\right)^2 + \frac{21}{20} > 1.$$

In case (2), rotate so that the unique ascent leaves the minimum: $x_5 = m < x_0 = M$. The supercritical row loses at least $(1 - m)^2$. Lemma C.8 shows that a T3-like row is different from it, and Theorem O.3 gives that row loss at least $2m - 4m^2$. The remaining four rows lose at least m^2 each. Their total loss is at least

$$(1 - m)^2 + (2m - 4m^2) + 4m^2 = 1 + m^2 > 1.$$

□

Proposition O.5 (Branches closed by area)**.** *The following branches are impossible:*

(1) T_C *is* CE0*,* $N_+ \geq 2$*, with arbitrary vertex-role types;*
(2) T_C *is* CE0*,* $N_+ = 1$*, no role is* Vd1 *or* Vd2*, and at least one role is* T3-like*.*

Proof. In the CE0 class, the center role has no positive-length boundary trace. If the six vertex roles missed a boundary point, openness of the center role would give such a trace. Hence the vertex roles cover ∂H, so Proposition C.9 applies.

If $N_+ \geq 2$, its at-least-two clause selects handoffs with at least two supercritical rows. Lemma O.4 makes the six vertex triangles' total normalized inside area strictly less than five. The center triangle contributes at most one, so the total is strictly less than six.

If $N_+ = 1$, the exact-one clause preserves the unique actual supercritical index for every strict selection. The exhaustive vertex-type hypothesis gives case (2) of Lemma O.4; the six vertex triangles contribute less than five normalized area, and the center triangle contributes at most one. Again the sum is strictly below the normalized area six of H. □

Appendix P. The Direct Nine-Point Obstruction

This section supplies the exact frontier, witness, sign, and cap-overlap calculations for the direct nine-point obstruction.

This section proves a center-class-independent theorem. Its hypotheses are that all six vertex roles are Vd0, that they cover the full boundary, and that exactly one actual boundary row is supercritical. The theorem will therefore apply both to the CE0 branch and to the zero-gap subcases of CE1 and CE2. Six common radial points and three asymmetric points are excluded directly from all six actual vertex roles and are therefore forced into the center role. Only the unique supercritical row uses an AB-union; no nonsupercritical model core is introduced.

P.1. **The finite caliper criterion.** We first record the finite caliper certificate used below. It enters the proof through the exact statement and verification argument given here; neither a plot nor an unrecorded numerical convention is used.

Let R and J denote counterclockwise rotations through $2\pi/3$ and $\pi/2$, respectively. For a nonempty finite set $S \subset \mathbb{R}^2$ and a vector N, put

$$W_S(N) = \sum_{k=0}^{2} \max_{z\in S}\langle z, R^k N\rangle. \tag{121}$$

Theorem P.1 (Finite caliper certificate)**.** *If S has at least two distinct points, then S is contained in a closed unit equilateral triangle if and only if there are distinct $p, q \in S$ and $\varepsilon \in \{-1, 1\}$ for which*

$$W_S\big(\varepsilon J(q-p)\big) \le \frac{\sqrt{3}}{2}\|p-q\|. \tag{122}$$

If S consists of one point, it is contained in equilateral triangles of arbitrarily small positive side length.

For $S_x = \{O, A, B, x\}$, every clause of (122) is a finite polynomial clause of degree at most four in the coordinates of A, B, x; at most $12\cdot 4^3$ clauses are required, and repeated point labels merely delete zero-length pairs.

Proof. For a unit vector n, write

$$G_S(n) = \sum_{k=0}^{2} h_S(R^k n), \qquad h_S(n) = \max_{z\in S}\langle z, n\rangle.$$

The three supporting half-planes with normals n, Rn, R^2n form an equilateral triangle of side $2G_S(n)/\sqrt{3}$. Hence S fits in a closed unit equilateral triangle precisely when $\min_{\|n\|=1} G_S(n) \le \sqrt{3}/2$.

On an open normal-direction cell on which the three support points are fixed, $G_S(\theta) = c\cos\theta + d\sin\theta$ and $G_S'' = -G_S < 0$ unless all points coincide. Thus a minimum occurs at a cell boundary, where some support line contains two distinct points p, q. After a cyclic rotation of the three normals, its normal is $\varepsilon J(q-p)/\|p-q\|$. Homogeneity yields (122). A nondegenerate segment is checked directly with its two normals.

For the polynomial expansion fix (p, q, ε), put $N = \varepsilon J(q-p)$ and $N_k = R^k N$, and choose support labels $z_0, z_1, z_2 \in S_x$. The support cell is

$$\langle z_k, N_k\rangle \ge \langle z, N_k\rangle \quad (z \in S_x,\ k = 0, 1, 2), \tag{123}$$

with $\|p-q\|^2 > 0$. Since $W_S(N) \geq 0$, the remaining test is

$$4\left(\sum_{k=0}^{2}\langle z_k, N_k\rangle\right)^2 \leq 3\|p-q\|^2. \tag{124}$$

The points are affine-linear in the parameters, so (123)–(124) have degree at most four. Six unordered pairs, two signs, and 4^3 support-label triples give the asserted finite exhaustive list. □

P.2. **Corner-clipped AB-unions.** Let e_A, e_B be unit vectors with angle 120°, put

$$O = 0, \qquad A = ae_A, \qquad B = be_B, \qquad C = \{ue_A + ve_B : u, v \geq 0\},$$

and define

$$\mathcal{R}(a,b) = C \cap \bigcup\{T : T \text{ is a closed unit equilateral triangle and } O, A, B \in T\}. \tag{125}$$

The finite description in Theorem P.1 applies to the four-point set $\{O, A, B, x\}$ and makes membership in (125) an exact finite semialgebraic condition of degree at most four.

The strict supercritical branch admits a simpler closed form. In the next statement write $x = ue_A + ve_B$ and $\|x\|^2 = u^2 + v^2 - uv$.

Theorem P.2 (Strict supercritical AB-union)**.** *Assume*

$$a + b > 1, \qquad \rho = a^2 + ab + b^2 < 1,$$

put $h = \sqrt{3}/2$, $D = \sqrt{4\rho - 3}$, and define

$$\alpha = \frac{h(a + 2b - aD)}{2\rho}, \qquad \beta = \frac{h(a - b + (a+b)D)}{2\rho},$$
$$\gamma = \frac{h(-a + b + (a+b)D)}{2\rho}, \qquad \delta = \frac{h(2a + b - bD)}{2\rho}.$$

Then all four coefficients are positive and

$$\mathcal{R}(a,b) = (C \cap \mathcal{D}_A \cap \mathcal{H}_A) \cup (C \cap \mathcal{D}_B \cap \mathcal{H}_B), \tag{126}$$

where

$$\begin{aligned}
\mathcal{D}_A &= \{(u,v) : (u-a)^2 + v^2 - (u-a)v \leq 1\},\\
\mathcal{D}_B &= \{(u,v) : u^2 + (v-b)^2 - u(v-b) \leq 1\},\\
\mathcal{H}_A &= \{(u,v) : \alpha(u-a) + \beta v \leq 0\},\\
\mathcal{H}_B &= \{(u,v) : \gamma u + \delta(v-b) \leq 0\}.
\end{aligned}$$

Its non-axis frontier consists, in counterclockwise order, of the A-circle, the A-line, the B-line, and the B-circle.

Proof. The identities

$$4\rho - 3(a+b)^2 = (a-b)^2, \qquad (a+b)^2D^2 - (a-b)^2 = 4\rho((a+b)^2 - 1)$$

give $0 < D < 1$ and positivity of the four coefficients.

We apply the caliper criterion of Theorem P.1. For a point $x = (u,v)$ in the relative interior of C but outside $\triangle OAB$, the four hull vertices occur in the order

O, B, x, A. The two cone-axis hull edges cannot certify a unit triangle: their support sums, divided by h, are respectively

$$\max\{a, u\} + \max\{b, v - u\}, \qquad \max\{b, v\} + \max\{a, u - v\},$$

and are at least $a + b > 1$. It remains to test the Ax and Bx calipers.

For the Ax caliper put $p = u - a$ and $r^2 = p^2 + v^2 - pv$. Exact evaluation of the three supports gives

(127)
$$\frac{G_A}{h} = \frac{\max\{r^2, -ap + (a+b)v\} + N_A}{r}, \quad N_A = \begin{cases} 0, & p \leq 0, \\ ap, & 0 \leq p \leq v, \\ (a+b)p - bv, & p \geq v. \end{cases}$$

No support label is omitted in these three sectors. If $p \leq 0$, $G_A \leq h$ is equivalent to

$$r \leq 1, \qquad -ap + (a+b)v \leq r.$$

The second inequality is the half-plane condition in (126), because

$$r^2 - (-ap + (a+b)v)^2 = \frac{(cp + dv)(c'p + d'v)}{4\rho},$$

where

$$c = a + 2b - aD, \quad d = a - b + (a+b)D, \quad c' = a + 2b + aD, \quad d' = a - b - (a+b)D,$$

and $c, c', d > 0 > d'$. Thus the low sector is exactly $\mathcal{D}_A \cap \mathcal{H}_A$. In the middle sector $0 < p \leq v$ one has $r \leq v$, whereas the numerator in (127) is at least $(a+b)v > r$, so no fit is possible.

In the high sector $p \geq v$, the inequalities $G_A \leq h$ imply

$$r^2 + (a+b)p - bv \leq r, \qquad bp + av \leq r. \tag{128}$$

Write $x - A = ry(\theta)$, $0 \leq \theta \leq 60°$, with cone coordinates

$$y(\theta) = te_A + we_B, \quad t = \frac{2}{\sqrt{3}}\cos(\theta - 30°), \quad w = \frac{2}{\sqrt{3}}\sin\theta.$$

Then (128) gives

$$r \leq f(\theta) = 1 - (a+b)t + bw, \qquad S(\theta) = bt + aw \leq 1.$$

Let n_B be the unit vector with dual coordinates (γ, δ). The identities

$$b\gamma + (a+b)\delta = h, \quad (a+b)\gamma + a\delta = \frac{h}{2}(1 + D), \quad b\delta - a\gamma = \frac{h}{2}(1 - D)$$

show that the unique angle θ_B of this normal satisfies $S(\theta_B) = 1$ and $f(\theta_B) = (1 - D)/2$. The function S has at most one critical point, a maximum, so $S(\theta) \leq 1$ implies $\theta \leq \theta_B$. On the feasible part $f, f' \geq 0$; hence $f(\theta)\cos(\theta_B - \theta)$ is increasing. Consequently

$$\langle n_B, x - B\rangle \leq a\gamma - b\delta + \frac{h}{2}(1 - D) = 0.$$

The diameter bound also gives $\|x - B\| \leq 1$. Thus every high sector fit belongs to $\mathcal{D}_B \cap \mathcal{H}_B$. Reflection exchanges a, u with b, v and supplies the corresponding result for the Bx caliper. This proves (126) in the cone interior; the axes follow by taking limits, since both sides are closed.

For completeness, the line–circle junctions are obtained by solving the two displayed line and circle equations. The line normals satisfy

$$\alpha^2 + \alpha\beta + \beta^2 = h^2, \qquad \gamma^2 + \gamma\delta + \delta^2 = h^2,$$

and the line determinant is

$$\alpha\delta - \gamma\beta = \frac{3(1-D)(D+3)}{16\rho} > 0.$$

The successive cone slopes are

$$+\infty, \quad \frac{2}{1-D}, \quad \frac{a(1-D)+2b}{2a+b(1-D)}, \quad \frac{1-D}{2}, \quad 0;$$

the two middle comparisons reduce to $a(4-(1-D)^2) > 0$ and $b(4-(1-D)^2) > 0$. This proves the asserted four-piece order. □

P.3. **Exact-one handoffs and common demands.**

Lemma P.3 (Exact-one handoff chain)**.** *Assume the six vertex roles cover ∂H and exactly one actual row is supercritical. Rotate its index to* 4. *The strict handoffs supplied by Proposition C.9 satisfy*

$$x_3 < x_2 < x_1 < x_0 < x_5 < x_4.$$

With

$$a = 1 - x_3, \qquad b = x_4, \qquad p = 1 - b, \qquad q = 1 - a,$$

one has

$$0 < a, b < 1, \qquad a + b > 1, \qquad a^2 + ab + b^2 < 1, \tag{129}$$

and every selected row obeys

$$a_i \geq p, \qquad b_i \geq q.$$

Proof. Every selected row other than 4 is strictly nonsupercritical, so $x_i < x_{i-1}$ for $i \neq 4$, which is precisely (P.3). In particular x_3 is the global minimum and x_4 the global maximum. The two anchors of row 4 lie in its open role; their mutual squared distance is $a^2 + ab + b^2$, strictly below the diameter squared of the closed unit triangle. This proves (129). Finally every $x_j \in [x_3, x_4]$, so $1 - x_{i-1} \geq 1 - x_4 = p$ and $x_i \geq x_3 = q$. □

P.4. **Six directly forced radial points.** Put $h = \sqrt{3}/2$ and

$$s = p + q, \qquad m = \min(p, q), \qquad M = \max(p, q), \qquad \chi = s^4 - s^2 + pq.$$

The strict domain (129) gives

$$pq > (1-s)(2-s), \qquad m > 1 - s, \qquad s > 1 - m. \tag{130}$$

Let $c_* = c_{\max}(p, q)$ be the exact radial envelope of Proposition J.2. On the present subcritical domain, equation (78) becomes

$$c_* = \begin{cases} C_L(m), & \chi \leq 0, \\ \dfrac{2M}{1 + \sqrt{4s^2 - 3}}, & \chi > 0, \end{cases} \tag{131}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

At $\chi = 0$ the two expressions agree and equal s.

Lemma P.4 (Direct radial forcing)**.** *One has*

$$0 < c_* < 1, \qquad c_* > 1 - m. \tag{132}$$

Assume that every U_i *is* Vd0 *and that* $U_C, U_0, \ldots, U_5$ *cover* H*. Then, for every* i*,*

$$D_i = (1 - c_*)V_i \in U_C.$$

Consequently U_C *contains the centered closed disk*

$$\mathcal{D}_\eta = \{x : \|x\| \le \eta\}, \qquad \eta = h(1 - c_*). \tag{133}$$

Proof. On the C_L branch, $C_L(m) < 1$ because the defining polynomial is positive at 1; if $s < \sqrt{3}/2$, then $c_* \ge \sqrt{3}/2 > s > 1 - m$, while if $s \ge \sqrt{3}/2$ its value at s is $\chi \le 0$, so $c_* \ge s > 1 - m$. On the other branch put $E = \sqrt{4s^2 - 3}$. The selector gives $sE > M - m$, whence $c_* < s < 1$. Moreover, with

$$A_0 = \frac{2M}{1 - m} - 1,$$

one has

$$A_0^2 - E^2 = \frac{4(1 - s)\{(1 - m)(1 - 2m) + m(2 - m)(1 - s)\}}{(1 - m)^2} > 0.$$

Thus $A_0 > E$ and $c_* > 1 - m$. This proves (132).

Fix i. If $D_i \in U_i$, openness supplies $\varepsilon > 0$ with $c_* + \varepsilon < 1$ such that $(1 - c_* - \varepsilon)V_i \in U_i$. The same role contains its two selected handoff anchors. Since their incoming and outgoing demands dominate p and q, convexity and passage to the closure give a closed unit triangle realizing $(p, q, c_* + \varepsilon)$. This contradicts the definition of $c_* = c_{\max}(p, q)$. Hence $D_i \notin U_i$.

If D_i belonged to U_{i-1} or U_{i+1}, a small plane ball inside that open role would meet the adjacent radial arm r_i in a positive-length interval. This contradicts the Vd0 definition. For the remaining roles put $r = 1 - c_* > 0$. Directly,

$$\|rV_i - V_{i\pm 2}\|^2 = 1 + r + r^2 > 1, \qquad \|rV_i - V_{i+3}\| = 1 + r > 1.$$

Since $V_j \in U_j$ and two points of an open unit triangle are at distance strictly below 1, these inequalities exclude D_i from the other three vertex roles. Thus no vertex role contains D_i. The point lies in $\operatorname{int}(H)$ by (132), so the assumed cover forces $D_i \in U_C$.

The six points D_i form a regular hexagon of circumradius $1 - c_*$. Their convex hull contains its centered incircle of radius $h(1 - c_*)$, and convexity of U_C gives (133). $\square$

We next define the three asymmetric witnesses. At V_4 use local coordinates

$$P = V_4 + u(V_5 - V_4) + v(V_3 - V_4), \qquad \mathsf{q}(u, v) = u^2 + v^2 - uv.$$

The outgoing demand is b and the incoming demand is a. Accordingly, set

$$\alpha = \frac{h(b + 2a - bD)}{2\rho}, \qquad \beta = \frac{h(b - a + (a + b)D)}{2\rho},$$

$$\gamma = \frac{h(-b + a + (a + b)D)}{2\rho}, \qquad \delta = \frac{h(2b + a - aD)}{2\rho},$$

where now $\rho = a^2 + ab + b^2$ and $D = \sqrt{4\rho - 3}$. The two line pieces of Theorem P.2 are

$$L_-(\lambda) = (b - \beta\lambda, \alpha\lambda), \qquad L_+(\mu) = (\delta\mu, a - \gamma\mu).$$

They meet at $\lambda_* = \mu_* = 8h\rho/(3(D+3))$ in

$$J = \left(\frac{2b + a - aD}{D+3}, \frac{b + 2a - bD}{D+3} \right). \tag{134}$$

Put

$$E_- = (1-b, 2-b), \qquad E_+ = (2-a, 1-a).$$

The actual adjacent handoffs have local centers

$$C_5(t) = (1+t, t), \qquad 1-a \le t \le b, \quad \text{and} \quad C_3(y) = (y, 1+y), \qquad 1-b \le y \le a.$$

The restrictions of the two unit-circle equations are

$$g_-(\lambda) = \frac{3}{4}\lambda^2 - 3(\alpha + b\beta)\lambda + 3b^2 - 3b + 2,$$

$$g_+(\mu) = \frac{3}{4}\mu^2 - 3(\delta + a\gamma)\mu + 3a^2 - 3a + 2. \tag{135}$$

Let λ_- and μ_+ be their first roots; explicitly,

$$\lambda_- = 2(\alpha + b\beta) - \frac{2}{3}\sqrt{9(\alpha + b\beta)^2 - 9b^2 + 9b - 6},$$

$$\mu_+ = 2(\delta + a\gamma) - \frac{2}{3}\sqrt{9(\delta + a\gamma)^2 - 9a^2 + 9a - 6}.$$

Lemma P.5 (Moving adjacent-role signs)**.** *Throughout* (129)*,*

$$\lambda_* < \lambda_- < h^{-1}, \qquad \mu_* < \mu_+ < h^{-1}. \tag{136}$$

Moreover $J_u, J_v < 1/2$. The point $L_+(\mu_+)$ satisfies

$$(L_+(\mu_+))_u + (L_+(\mu_+))_v < 3 - 2a,$$

and the reflected bound holds for $L_-(\lambda_-)$. Writing $F(P,t) = \mathsf{q}(P - C_5(t)) - 1$, one has, for every $t \in [1-a, b]$,

$$F(L_+(\mu_+), t) \ge 0, \qquad F(J,t) > 0, \qquad F(L_-(\lambda_-), t) > 0. \tag{137}$$

Under the reflection $(u,v,a,b) \leftrightarrow (v,u,b,a)$, the analogous signs hold for every $y \in [1-b, a]$: the first-root point $L_-(\lambda_-)$ has nonnegative sign against $C_3(y)$, and the other two witnesses have strictly positive sign.

Proof. Put

$$U = a + b - 1, \qquad z = a - b, \qquad D^2 = z^2 + 6U + 3U^2.$$

Then $0 < U < 1/6$ and $z^2 < \Omega := 1 - 6U - 3U^2$. We give the fixed-line calculation for the circle centered at $E_+ = (2-a, 1-a)$; reflection gives the calculation for E_-. Put

$$F(P,t) = \mathsf{q}(P - (1+t, t)) - 1, \qquad t_0 = 1 - a,$$

so this circle is $F(P, t_0) = 0$. At the junction, exact expansion gives

$$(D+3)^2 F(J, t_0) = 3D(z^2 + Uz + 1 - 3U) + P_J,$$

where

$$P_J = z^4 + 5z^2 + 6Uz^2 + 3U^2z^2 + 9Uz + 3U + 15U^2 > 0;$$

indeed $z^2 + Uz + 1 - 3U \ge 1 - 3U - U^2/4 > 0$, and $5z^2 + 9Uz + 15U^2$ is positive definite. Thus $F(J, t_0) > 0$. Also

$$J_u + J_v = \frac{(1+U)(3-D)}{3+D} < 1.$$

Indeed this inequality is $3U < D(2+U)$, which follows from $D^2 \geq 3U(2+U)$ and $(2+U)^3 > 3U$. Since

$$\frac{\partial F}{\partial t}(P,t) = 1 + 2t - P_u - P_v,$$

$F(J,t)$ is strictly increasing for $t \geq 0$, and therefore is positive throughout $[1-a, b]$.

On the opposite line L_-, the derivative at the junction, after multiplication by a positive factor, has at $t = 1-a$ the numerator

$$N_0 = D(9 + 3z^2 + 6Uz - 9U^2) + P_0,$$

where

$$\begin{aligned} P_0 = {} & 18U^3 + 6U^2z^2 + 63U^2 + 18Uz^2 + 18Uz + 54U \\ & + 2z^4 + 9z^2 + 9. \end{aligned}$$

The coefficient of D is at least $9 - 6U - 9U^2 > 0$, while $P_0 \geq 9 + 54U - 18U > 0$. Hence $N_0 > 0$. At the other endpoint $t = b$, the corresponding numerator is

$$N_1 = D(9 + 3z^2 + 6U - 3U^2) + P_1,$$

where

$$\begin{aligned} P_1 = {} & 9U^4 - 3U^3z + 45U^3 + 9U^2z^2 - 6U^2z + 72U^2 \\ & - Uz^3 + 21Uz^2 + 9Uz + 45U + 2z^4 + 9z^2 + 9. \end{aligned}$$

Its D-coefficient is positive and, using $|z| < 1$,

$$P_1 \geq 9 + 45U - 9U - U - 6U^2 - 3U^3 > 0.$$

The junction derivative is affine in t, so it is positive throughout $[1-a, b]$. Since the restriction of F to L_- is a convex quadratic, is positive at the junction, and is increasing there, it has no intersection on the opposite line side for any actual adjacent handoff.

On the correct line L_+, the value at its outer endpoint h^{-1} has, after multiplication by a positive factor, numerator

$$A_0 = P_2 - 4DU(1 + U + z),$$

where

$$\begin{aligned} P_2 = {} & 3U^4 + 6U^3z + 6U^3 + 4U^2z^2 + 12U^2z + 12U^2 \\ & + 2Uz^3 + 6Uz^2 + 2Uz + 6U + z^4 + 2z^2 - 3. \end{aligned}$$

For fixed U, replace the odd powers of z by $x = |z|$. The resulting polynomial $P_2^+(U,x)$ satisfies

$$\partial_x P_2^+ = 2(3U^3 + 4U^2x + 6U^2 + 3Ux^2 + 6Ux + U + 2x^3 + 2x) > 0,$$

and therefore its supremum for $z^2 < \Omega$ occurs at $x = \sqrt{\Omega}$. There

$$P_2^+(U, \sqrt{\Omega}) = 4U(U + \sqrt{\Omega} - 3) < 0.$$

As $1 + U + z = 2a > 0$, this gives $A_0 < 0$. The positive junction value and negative endpoint value force the first root strictly between them, proving the L_+ half of (136); reflection proves the other half.

For the coordinate bounds, the exact identities

$$D^2 - (3U - z)^2 = 6U(1 + z - U), \qquad D^2 - (3U + z)^2 = 6U(1 - z - U)$$

give $J_u, J_v < 1/2$. It remains to control the coordinate sum along L_+. For $E = L_+(h^{-1})$, the sign of $3 - 2a - E_u - E_v$ is the sign of

$$D(6U + 2z + 6) + B(U,z),$$

where

$$B(U,z) = -9U^3 - 9U^2 z - 9U^2 - 3Uz^2 - 18Uz + 3U - 3z^3 + 3z^2 - 3z + 3.$$

Here

$$B_z = -3(3z^2 + 2Uz + 6U - 2z + 3U^2 + 1) < 0,$$

because the quadratic in parentheses has discriminant $-32U^2 - 80U - 8 < 0$. Hence its minimum on $z^2 < \Omega$ is at $z = \sqrt{\Omega}$, where $B(U, \sqrt{\Omega}) = 6(1 - 3U - \sqrt{\Omega}) > 0$ because $(1-3U)^2 - \Omega = 12U^2$. Thus $E_u + E_v < 3 - 2a$. Also $J_u + J_v < 1 < 3 - 2a$, and the coordinate sum is affine on the line segment, so the same strict bound holds at the first root. Now

$$\frac{\partial F}{\partial t}(P,t) = 1 + 2t - P_u - P_v.$$

Since $F(L_+(\mu_+), 1-a) = 0$, the coordinate-sum bound makes its t-derivative strictly positive for $t \geq 1 - a$. Together with the no-opposite-line result, this proves (137). Reflection gives both the bound on L_- and the complete $C_3(y)$ statement. □

Define the local and global witnesses by

$$Q_-^{\mathrm{loc}} = L_-(\lambda_-), \qquad Q_0^{\mathrm{loc}} = J, \qquad Q_+^{\mathrm{loc}} = L_+(\mu_+), \tag{138}$$

and by the displayed conversion from (u,v) to the plane.

Lemma P.6 (Direct asymmetric forcing)**.** *Assume that $U_C, U_0, \ldots, U_5$ cover H. Then*

$$Q_-, Q_0, Q_+ \in \mathrm{int}(H) \setminus \bigcup_{i=0}^{5} U_i,$$

and consequently all three witnesses belong to U_C.

Proof. The root order and positivity of the line coordinates give $0 < u, v < 1$ for all three points. Each belongs to $\mathcal{R}(b,a)$ and hence lies in a closed unit triangle containing V_4, so $\mathsf{q}(u,v) \leq 1$; hence $|u - v| < 1$ and all three points lie in $\mathrm{int}(H)$.

The closure of U_4 is one of the triangles represented by $\mathcal{R}(b,a)$, since it contains V_4, X_3, X_4. All three witnesses lie on its exact non-axis frontier. If one belonged to the open triangle U_4, a sufficiently small plane ball about it would lie both in U_4 and in the interior of the corner cone, making the witness an interior point of $\mathcal{R}(b,a)$. This contradiction excludes U_4.

For the roles U_0, U_1, U_2, their contained vertices have local coordinates

i	0	1	2
V_i	$(2,1)$	$(2,2)$	$(1,2)$.

For $P = (u,v)$,

$$\|P - (2,1)\|^2 - 1 = \mathsf{q}(u,v) + 2 - 3u, \quad \|P - (1,2)\|^2 - 1 = \mathsf{q}(u,v) + 2 - 3v.$$

The inequalities $J_u, J_v < 1/2$ and the line order give distance greater than one from V_0 for Q_-, Q_0 and from V_2 for Q_0, Q_+. For the remaining pair, the first-root circle and the sum bound in Lemma P.5 give

$$\|Q_+ - V_0\|^2 - 1 = a(3 - a - (Q_+)_u - (Q_+)_v) > a^2,$$

and, by reflection, $\|Q_- - V_2\|^2 - 1 > b^2$. All three points are farther than one from $V_1 = (2,2)$, because writing $u = 1 - \xi$, $v = 1 - \eta$ gives

$$\|P - V_1\|^2 = 1 + \xi + \eta + \xi^2 + \eta^2 - \xi\eta > 1.$$

Since two points in one open unit equilateral triangle have distance strictly less than one, these inequalities exclude U_0, U_1, U_2.

The actual handoff $X_5 = C_5(x_5)$ belongs to U_5, and $1 - a < x_5 < b$ by (P.3). The three signs in (137) therefore put every witness at distance at least one from X_5, excluding U_5. Similarly, if $y = 1 - x_2$, then $1 - b < y < a$, $X_2 = C_3(y) \in U_3$, and the reflected signs exclude U_3. Thus no vertex role contains any witness. The assumed cover now forces all three interior witnesses into U_C. □

The nine directly forced points are $D_0, \ldots, D_5, Q_-, Q_0, Q_+$. Under a putative cover, Lemmas P.4 and P.6 therefore force the compact set

$$K_{\mathrm{wit}}(a, b) = \mathcal{D}_\eta \cup \{Q_-, Q_0, Q_+\} \subset U_C. \tag{139}$$

P.5. **Tangent reduction and exact mixed overlaps.** For the nontrivial regime $c_* > 2/3$, we first remove the square radicals in the first-root witnesses. Normalize $\bar{\alpha} = \alpha/h$, and similarly for the other three coefficients, and put

$$\xi = \lambda/h, \quad \zeta = \mu/h, \quad \xi_* = \zeta_* = \frac{8\rho}{3(D+3)}.$$

Let g_-, g_+ be the rescaled forms of (135), and take one Newton step at the junction:

$$\widehat{\xi}_- = \xi_* - \frac{g_-(\xi_*)}{g_-'(\xi_*)}, \qquad \widehat{\zeta}_+ = \zeta_* - \frac{g_+(\zeta_*)}{g_+'(\zeta_*)}.$$

Define, in local coordinates,

$$\begin{aligned} A &= \left(b - \frac{3}{4}\bar{\beta}\widehat{\xi}_-, \frac{3}{4}\bar{\alpha}\widehat{\xi}_-\right), \\ B &= J, \\ C &= \left(\frac{3}{4}\bar{\delta}\widehat{\zeta}_+, a - \frac{3}{4}\bar{\gamma}\widehat{\zeta}_+\right), \end{aligned}$$

and convert them to global vectors.

Lemma P.7 (Newton inner reduction)**.** *The points above have coordinates rational in a, b, D and satisfy*

$$A \in (Q_0, Q_-), \qquad C \in (Q_0, Q_+).$$

Consequently

$$\widehat{K} = \mathrm{conv}(\mathcal{D}_\eta \cup \{A, B, C\}),$$

obeys $\widehat{K} \subseteq \mathrm{conv}(K_{\mathrm{wit}})$.

Proof. Each quadratic g_-, g_+ is strictly convex, positive, and decreasing at the junction. Its tangent zero therefore lies strictly between the junction and its first root. This gives the two open-segment inclusions. All displayed coefficients and Newton quotients are rational in a, b, D, and convexity gives the asserted containment. □

Put $W = C + RA$. For a disk $\mathbb{D}(X, r)$ let

$$I(X, r) = \{n : \|n\| = 1,\ \langle X, n\rangle + r \geq h\}.$$

P.6. **The exact global certificate for the mixed overlaps.** This subsection proves the two mixed-overlap signs used in Proposition 6.2. It first replaces the exact radial value by rational upper envelopes, then reduces the mixed residuals to eight integer-polynomial signs. Three fixed analytic charts and twenty global Bernstein expansions prove those signs. No interval arithmetic, box subdivision, pruning, or branch-and-bound is used.

Retain the notation of Section P. Thus

$$p = 1 - b, \qquad q = 1 - a, \qquad s = p + q, \qquad m = \min\{p, q\}, \qquad M = \max\{p, q\},$$

$$\chi = s^4 - s^2 + mM, \qquad h = \frac{\sqrt{3}}{2}, \qquad c_* = c_{\max}(p, q).$$

The strict domain gives

$$mM > (1 - s)(2 - s), \qquad m > 1 - s, \qquad s > \frac{2}{3}. \tag{140}$$

Put

$$F_m(x) = x^4 - x^2 + mx - m^2, \qquad \kappa = F'_m(s) = 4s^3 - 2s + m.$$

Since $m > 1 - s$,

$$\kappa > 4s^3 - 3s + 1 = (s + 1)(2s - 1)^2 > 0. \tag{141}$$

P.6.1. *Rational radial upper envelopes.* Define

$$\bar{c}_L = s - \frac{\chi}{\kappa} \quad (\chi \le 0), \qquad \bar{c}_T = s - \frac{\chi}{1 - m} \quad (\chi \ge 0). \tag{142}$$

Lemma P.8 (Branchwise radial upper bounds)**.** *On the corresponding radial cells,*

$$c_* \le \bar{c}_L < 1, \qquad c_* \le \bar{c}_T < 1.$$

At $\chi = 0$ both upper bounds equal $s = c_*$.

Proof. On the $C_L(m)$ cell, $F_m(s) = \chi \le 0$ and $F_m(c_*) = 0$. For $x \ge s > 2/3$ one has $F''_m(x) = 12x^2 - 2 > 0$, so convexity and (141) give

$$0 = F_m(c_*) \ge F_m(s) + F'_m(s)(c_* - s),$$

whence $c_* \le s - \chi/\kappa = \bar{c}_L$. Let $U = 1 - s$. From (140), $mM > U(1 + U)$, and therefore

$$\begin{aligned}\chi + U\kappa &> U\{m + 3s^3 - s^2 - 3s + 2\}\\ &= U\{m + 1 - 4U + 8U^2 - 3U^3\} > 0.\end{aligned}$$

Indeed $0 < U < 1/3$ and $1 - 4U + 8U^2 - 3U^3 > 1 - 4U + 7U^2 > 0$. Thus $\bar{c}_L < 1$.

On the other cell put

$$E = \sqrt{4s^2 - 3}, \qquad m_0 = \frac{s(1 - E)}{2}, \qquad M_0 = \frac{s(1 + E)}{2}.$$

Then

$$\chi = (m - m_0)(M_0 - m), \qquad s - c_* = \frac{2(m - m_0)}{1 + E}.$$

Hence

$$\frac{\chi}{s - c_*} = \frac{1 + E}{2}(M_0 - m) \le 1 - m,$$

because $0 \le E \le 1$ and $M_0 \le 1$. This proves $c_* \le \bar{c}_T$, while $\chi > 0$ gives $\bar{c}_T < s < 1$. □

Reflect so that $z = a - b \geq 0$ and put $U = a + b - 1$. Then

$$s = 1 - U, \qquad m = \frac{1 - U - z}{2}, \qquad D^2 = z^2 + 6U + 3U^2 =: K(U, z),$$

$$\Xi(U, z) := 4\chi = 1 - 10U + 21U^2 - 16U^3 + 4U^4 - z^2,$$

$$G(U, z) := 2\kappa = 5 - 21U + 24U^2 - 8U^3 - z.$$

Consequently

$$\bar{c}_L = 1 - U - \frac{\Xi(U, z)}{2G(U, z)}, \qquad \bar{c}_T = 1 - U - \frac{\Xi(U, z)}{2(1 + U + z)}. \tag{143}$$

Neither expression contains the quartic root or the radial radical $\sqrt{4s^2 - 3}$.

P.6.2. *The Gram residuals and exact core reduction.* On the active cell write $\bar{c}$ for the corresponding expression in (143), and put

$$\bar{\eta} = h(1 - \bar{c}), \qquad e = 1 - \bar{c}, \qquad t = 2\bar{c} - 1.$$

Lemma P.8 gives $0 < \bar{\eta} \leq \eta = h(1 - c_*)$ when $c_* > 2/3$. Thus the three disks obtained from $\mathbb{D}(C, 2\eta)$, $\mathbb{D}(W, \eta)$, and $\mathbb{D}(RA, 2\eta)$ by replacing η with $\bar{\eta}$ are contained in the disks already present in the Minkowski sum.

Represent a global vector as (x, hy) and put $C' = R^{-1}C$,

$$N_A = \|A\|^2, \qquad N_C = \|C\|^2, \qquad \nu = \langle A, C' \rangle, \qquad \Delta_0 = x_A y_{C'} - y_A x_{C'}.$$

The Gram identity is

$$h^2 \Delta_0^2 = N_A N_C - \nu^2.$$

The two mixed square residuals for the smaller disks factor as

$$\bar{P}_A = h^2 N_A Q_A, \qquad Q_A = \Delta_0^2 - \|tA - eC'\|^2, \tag{144}$$

$$\bar{P}_C = h^2 N_C Q_C, \qquad Q_C = \Delta_0^2 - \|tC' - eA\|^2. \tag{145}$$

All witness coordinates are rational in U, z, D. Substituting (143) into (144)–(145) and reducing modulo $D^2 - K(U, z)$ gives, for $B \in \{L, T\}$ and $X \in \{A, C\}$,

$$\operatorname{num}(Q_{B,X}) = 32(K + 3)^2 \{A_{B,X}(U, z) + D B_{B,X}(U, z)\}, \tag{146}$$

where $A_{B,X}, B_{B,X} \in \mathbb{Z}[U, z]$. Every omitted denominator is strictly positive: it is a positive constant times $(D + 3)^4$, the square of the relevant upper-envelope denominator, and squares of the two nonzero Newton-derivative numerators. The term counts are

(B, X)	$\#A_{B,X}$	$\#B_{B,X}$
(L, A)	347	298
(L, C)	347	298
(T, A)	342	294
(T, C)	341	294.

Since $D \geq 0$, it remains to prove all eight integer polynomials nonnegative.

P.6.3. *Three fixed charts and global positive bases.* Set

$$\mathcal{G}(U) = 1 - 6U - 3U^2, \qquad H(U) = 1 - 10U + 21U^2 - 16U^3 + 4U^4,$$

$$U_H = 1 - \frac{\sqrt{3}}{2}, \qquad U_{\max} = \frac{2}{\sqrt{3}} - 1.$$

The identities

$$4\chi = H(U) - z^2, \qquad H(U) = (U-1)^2(4U^2 - 8U + 1)$$

and

$$\mathcal{G}(U) - H(U) = 4U(1 - 6U + 4U^2 - U^3) \geq 0 \quad (0 \leq U \leq U_H)$$

show that the radial cells are exactly covered by

$$\begin{aligned} T: &\quad 0 \leq U \leq U_H, \quad 0 \leq z^2 \leq H(U), \\ L_0: &\quad 0 \leq U \leq U_H, \quad H(U) \leq z^2 \leq \mathcal{G}(U), \\ L_1: &\quad U_H \leq U \leq U_{\max}, \quad 0 \leq z^2 \leq \mathcal{G}(U). \end{aligned} \tag{147}$$

We use the elementary positive-basis fact that a polynomial whose coefficients in the tensor Bernstein basis on $[0,1]^2$ are all nonnegative is nonnegative on that square. This is one global polynomial identity, not a subdivision procedure. All coefficients below lie in $\mathbb{Q}(\sqrt{3})$, and their signs are decided exactly by rational comparisons.

For the T cell use

$$U = \frac{(x-1)(3-x)}{4x}, \qquad z = y\frac{9-x^4}{8x^2}, \qquad 1 \leq x \leq \sqrt{3}, \quad 0 \leq y \leq 1. \tag{148}$$

Here

$$H\left(\frac{(x-1)(3-x)}{4x}\right) = \left(\frac{9-x^4}{8x^2}\right)^2,$$

so the chart covers the whole T cell. Clearing the positive powers of 2 and x, then putting $x = 1 + (\sqrt{3}-1)r$, gives four global Bernstein expansions on $[0,1]^2$:

	bidegree	coefficients	zero coefficients
$A_{T,A}$	$(88,22)$	2047	2
$B_{T,A}$	$(80,20)$	1701	0
$A_{T,C}$	$(88,22)$	2047	2
$B_{T,C}$	$(80,20)$	1701	0.

Every nonzero coefficient is strictly positive.

For an L-cell polynomial $F(U,z)$, write

$$F(U,z) = E_F(U,w) + zO_F(U,w), \qquad w = z^2,$$

and put

$$R_F(U,w) = E_F(U,w)^2 - wO_F(U,w)^2.$$

The implications $E_F \geq 0$ and $R_F \geq 0$ give $E_F \geq \sqrt{w}\,|O_F|$, hence $F \geq 0$. Use the two fixed charts

$$U = U_H r, \qquad w = H(U) + s\{\mathcal{G}(U) - H(U)\} \tag{149}$$

for L_0, and

$$U = U_H + (U_{\max} - U_H)r, \qquad w = s\mathcal{G}(U) \tag{150}$$

for L_1, with $0 \le r, s \le 1$. The global Bernstein expansions of E_F and R_F have the following bidegrees and coefficient counts:

F	$L_0 : E_F$	$L_0 : R_F$	$L_1 : E_F$	$L_1 : R_F$
$A_{L,A}$	$(46, 11); 564$	$(92, 22); 2139$	$(26, 11); 324$	$(52, 22); 1219$
$B_{L,A}$	$(42, 10); 473$	$(84, 20); 1785$	$(24, 10); 275$	$(48, 20); 1029$
$A_{L,C}$	$(46, 11); 564$	$(92, 22); 2139$	$(26, 11); 324$	$(52, 22); 1219$
$B_{L,C}$	$(42, 10); 473$	$(84, 20); 1785$	$(24, 10); 275$	$(48, 20); 1029.$

Every coefficient in these sixteen expansions is strictly positive. Therefore all eight polynomials in (146) are nonnegative on the corresponding radial cells.

Lemma P.9 (Exact mixed-cap overlaps)**.** *In the regime $c_* > 2/3$, the tangent witnesses defined above satisfy*

$$I(C, 2\eta) \cap I(W, \eta) \neq \varnothing, \qquad I(W, \eta) \cap I(RA, 2\eta) \neq \varnothing.$$

Proof. Since $D \ge 0$, the eight signs prove $Q_A, Q_C \ge 0$ for $z \ge 0$ by (146); reflection supplies $z \le 0$. Equations (144)–(145) give the two mixed cap overlaps for the smaller disks, hence for the original disks. $\square$

Exact replay record. The canonical data are in

`proof/3XXX_CE0/31XX_Nplus1/310X_all_Vd0/`

`3105X_self_contained_direct_Vd0_nine_point/3105X_computation/`

The module `mixed_overlap_core_polynomials.py` assembles the six numbered data shards. The exact checks

`verify_mixed_overlap_core_derivation.py`

`verify_global_core_positivity.py`

derive the eight core polynomials and verify the twenty global Bernstein identities over $\mathbb{Q}(\sqrt{3})$. They authenticate SHA-256 digest

`dc46aaf263655d5159ecd3a81db72ee82477951d06172f4743b248df37209485`

and return

```
adaptive_subdivision: false
branch_and_bound: false
result: PASS
```

using exact integer, rational, and $\mathbb{Q}(\sqrt{3})$ arithmetic.

P.7. **Ray order and four overlaps.**

Proposition P.10 (Technical four-overlap verification)**.** *Assume* (129) *and $c_* > 2/3$. Then*

$$\|A\|, \|C\| > \eta,$$

the rays A, B, C, W, RA occur in that cyclic order, and the four consecutive cap pairs

$$I(A, 2\eta) \leftrightarrow I(B, 2\eta) \leftrightarrow I(C, 2\eta) \leftrightarrow I(W, \eta) \leftrightarrow I(RA, 2\eta)$$

overlap across their intervening short sectors.

Proof. Put $\tau = h - 2\eta = h(2c_* - 1)$. The rays of

$$A, \ B, \ C, \ W, \ RA \tag{151}$$

occur in this cyclic order from A to RA. Indeed the two line distances give

$$\frac{\operatorname{cross}(A,B)}{\|B-A\|}=\alpha(1-b)+\beta>0,\qquad \frac{\operatorname{cross}(B,C)}{\|C-B\|}=\gamma+\delta(1-a)>0.$$

If $A=(-X,-Y)$ and $C=(-P,-Q)$ in the centered cone basis, then $0<X,Y,P,Q<1$, $X,Q>1/2$, and

$$\frac{\operatorname{cross}(C,RA)}{h}=PX+(Q-P)Y>0.$$

The last sign is immediate unless $Q<P$ and $X<Y$, in which case it is larger than $Q-P/2>0$. The ray of $W=C+RA$ lies between those of C and RA. The same coordinates give

$$\|A\|,\|C\|>h/2>\eta. \tag{152}$$

For the two equal-radius overlaps it is enough to prove

$$\frac{\operatorname{cross}(A,B)}{\|B-A\|}\geq\tau,\qquad \frac{\operatorname{cross}(B,C)}{\|C-B\|}\geq\tau. \tag{153}$$

We record the exact analytic verification. By reflection take $a\leq b$ and set $m=1-b$, $M=1-a$, $U=a+b-1$, $s=m+M$, and $w=M-m$. The smaller of the two normalized left sides in (153) is

$$d=\frac{\mathcal{A}+U(2m-1)+D(\mathcal{A}+U)}{2\rho},\qquad \mathcal{A}=1-m+m^2. \tag{154}$$

On the C_L sheet, write $F_L(z)=z^4-z^2+mz-m^2$. This defining polynomial evaluated at $r=1-m/2+m^2/2$ is

$$F_L(r)=\frac{m^2(m-1)}{16}(m^5-3m^4+11m^3-17m^2+28m-12)>0.$$

Indeed the polynomial in parentheses is increasing on $(0,1/2)$ and has value $-33/32$ at $1/2$; its derivative is $28-34m+m^2(5m^2-12m+33)>0$. Since $F_L(h)=-(m-h/2)^2\leq 0$ and the selected root is unique, $2c_*-1\leq\mathcal{A}$. Put

$$X_0=\mathcal{A}+U,\qquad Y_0=2\rho\mathcal{A}-\mathcal{A}-U(2m-1).$$

Then

$$d-\mathcal{A}=\frac{D(\mathcal{A}+U)-Y_0}{2\rho},$$

and exact expansion gives

$$\begin{aligned}Y_0&=2(m^2-m+1)U^2+(2m^3-2m+3)U\\&\quad+(m^2-m+1)(2m^2-2m+1)>0.\end{aligned}$$

Furthermore

$$D^2X_0^2-Y_0^2=4m\rho\Phi(U,m),$$

where

$$\begin{aligned}\Phi(U,m)&=(1-m)(m^2-m+2)U^2\\&\quad+(2-m^2)(m^2-m+1)U\\&\quad-m(1-m)^2(m^2-m+1).\end{aligned}$$

In these variables

$$\chi=U^4-4U^3+5U^2-(m+2)U+m(1-m).$$

Since $U^2(U^2-4U+5)>0$, the selector $\chi\le 0$ implies $U>m(1-m)/(m+2)$; Φ is increasing in U and at that lower endpoint is

$$\frac{m^2(1-m)(m^4-2m^3+6m^2-6m+4)}{(m+2)^2}>0.$$

The last quartic equals $(m^2-m)^2+5m^2-6m+4>0$. Thus $d>\mathcal{A}\ge 2c_*-1$.

On the other radial sheet let $E=\sqrt{4s^2-3}$. Then $0\le w<sE$ and $c_*=(s+w)/(1+E)$. At fixed U, differentiation of (154) can be checked without an implicit estimate. Namely,

$$d=\frac{1+D}{2}-UB(D,w),\qquad B(D,w)=\frac{(1+U)(3+D)+w(1-D)}{D^2+3}.$$

Here $D^2=w^2+6U+3U^2$, so $0\le D'=w/D\le 1$, and

$$B_w=\frac{1-D}{D^2+3}\ge 0.$$

Writing $N=(1+U)(3+D)+w(1-D)$, direct differentiation gives

$$B_D=\frac{(1+U-w)(D^2+3)-2DN}{(D^2+3)^2}>-\frac{34}{27}.$$

Indeed the first term in the numerator is positive, while $1+U-w=2a>0$, $N<(7/6)4+1=17/3$, $D<1$, and $D^2+3\ge 3$. If $B_D<0$, then $B_DD'\ge B_D$; otherwise $B_DD'\ge 0$. Hence $B'=B_w+B_DD'>-34/27$, and $U<1/6$ yields

$$d'<\frac{1}{2}+\frac{1}{6}\frac{34}{27}=\frac{115}{162}<1,$$

whereas $(2c_*-1)'=2/(1+E)\ge 1$. Hence their difference decreases with w. At the boundary $w=sE$, where $\chi=0$ and $c_*=s$, the preceding sheet applies: this is an admissible point because $h<s<1$ and $D^2=4s^4-12s+9<1$. Indeed $s^4-3s+2=(s-1)(s^3+s^2+s-2)<0$ on this interval; the second factor is increasing and already equals $(7h-5)/4>0$ at $s=h$. The preceding sheet gives $d\ge 2s-1$. Therefore (153) holds on both sheets, and reflection supplies the other line.

It remains to check the two mixed overlaps. On the two radial cells use the exact rational envelopes and global Bernstein identities established in Subsection P.6. Lemma P.9 gives precisely the overlaps for $\mathbb{D}(C,2\eta)$ with $\mathbb{D}(W,\eta)$ and for $\mathbb{D}(W,\eta)$ with $\mathbb{D}(RA,2\eta)$.

Thus the four consecutive pairs of caps centered on the rays (151) overlap. $\square$

Appendix Q. Exact Computer-Assisted Certificate for the Mixed Cap Overlaps

The two mixed cap overlaps in Strategy 4 are certified by exact symbolic arithmetic. This section states the complete mathematical reduction, explains the verification algorithms, and formally incorporates the code and sparse integer data listed below as an electronic proof supplement. No floating-point arithmetic, interval arithmetic, adaptive subdivision, branch-and-bound, or unrecorded numerical tolerance is used.

Q.1. **Certificate manifest.** All paths below are relative to

`proof/3XXX_CE0/31XX_Nplus1/310X_all_Vd0/`
`3105X_self_contained_direct_Vd0_nine_point/3105X_computation/`

The exact files incorporated in the certificate are

file	Git blob SHA
`mixed_overlap_core_data_00.py`	`359b219f9deda8586217dcbf9ccc10de0ad48578`
`mixed_overlap_core_data_01.py`	`bfc807f906359636035634f799d74d552cf847fc`
`mixed_overlap_core_data_02.py`	`92887696ef94aeb93f3d95ab9b16a5388406fff4`
`mixed_overlap_core_data_03.py`	`c2791a8a6add132ad7a17655cbb13987124cb24e`
`mixed_overlap_core_data_04.py`	`7eeabecbbd17543b728130a1f2bc84f79e11a8d2`
`mixed_overlap_core_data_05.py`	`d676114292e722b60d1b792f615ee88ad58b6982`
`mixed_overlap_core_polynomials.py`	`07f57b178cf74980f8d956541842d0c4c4b2faa7`
`verify_mixed_overlap_core_derivation.py`	`0a55211ae37f56e1fc461560b0668a01456c7417`
`verify_global_core_positivity.py`	`821f9a6dc750fbdb80e02e40282b32cc845c3383`

The six data shards decode to one canonical sparse-polynomial transcript with SHA–256 digest

`dc46aaf263655d5159ecd3a81db72ee82477951d06172f4743b248df37209485`

The first verifier independently derives the transcript from the witness formulas and compares every coefficient with the authenticated object. The second verifier derives all Bernstein coefficients from that same object and checks their signs exactly.

Q.2. **Rational radial upper envelopes.** Retain the notation of Section P. Thus

$$p = 1 - b, \qquad q = 1 - a, \qquad s = p + q, \qquad m = \min\{p, q\}, \qquad M = \max\{p, q\},$$

$$\chi = s^4 - s^2 + mM, \qquad h = \frac{\sqrt{3}}{2}, \qquad c_* = c_{\max}(p, q).$$

The strict witness domain gives

$$mM > (1 - s)(2 - s), \qquad m > 1 - s, \qquad s > \frac{2}{3}. \tag{155}$$

Put

$$F_m(x) = x^4 - x^2 + mx - m^2, \qquad \kappa = F_m'(s) = 4s^3 - 2s + m.$$

Then

$$\kappa > 4s^3 - 3s + 1 = (s + 1)(2s - 1)^2 > 0.$$

Define

$$\bar{c}_L = s - \frac{\chi}{\kappa} \quad (\chi \le 0), \qquad \bar{c}_T = s - \frac{\chi}{1 - m} \quad (\chi \ge 0). \tag{156}$$

Lemma Q.1 (Branchwise radial upper bounds)**.** *On the corresponding radial cells,*

$$c_* \le \bar{c}_L < 1, \qquad c_* \le \bar{c}_T < 1.$$

At $\chi = 0$ both upper bounds equal $s = c_*$.

Proof. On the L cell, $F_m(s) = \chi \le 0$ and $F_m(c_*) = 0$. For $x \ge s > 2/3$, $F_m''(x) > 0$, so convexity gives

$$0 \ge F_m(s) + F_m'(s)(c_* - s),$$

and hence $c_* \le \bar{c}_L$. If $U = 1 - s$, then

$$\chi + U\kappa > U\{m + 1 - 4U + 8U^2 - 3U^3\} > 0$$

by (155); therefore $\bar{c}_L < 1$.

On the T cell put

$$E=\sqrt{4s^2-3},\qquad m_0=\frac{s(1-E)}{2},\qquad M_0=\frac{s(1+E)}{2}.$$

Then

$$\chi=(m-m_0)(M_0-m),\qquad s-c_*=\frac{2(m-m_0)}{1+E}.$$

Consequently

$$\frac{\chi}{s-c_*}=\frac{1+E}{2}(M_0-m)\le 1-m,$$

which gives $c_*\le \bar{c}_T$. The strict inequality $\chi>0$ gives $\bar{c}_T<s<1$. □

Reflect so that $z=a-b\ge 0$ and put $U=a+b-1$. Then

$$s=1-U,\qquad m=\frac{1-U-z}{2},\qquad D^2=z^2+6U+3U^2=:K(U,z),$$

and

$$\Xi(U,z)=4\chi=1-10U+21U^2-16U^3+4U^4-z^2,$$
$$G(U,z)=2\kappa=5-21U+24U^2-8U^3-z.$$

Thus

$$\bar{c}_L=1-U-\frac{\Xi(U,z)}{2G(U,z)},\qquad \bar{c}_T=1-U-\frac{\Xi(U,z)}{2(1+U+z)}. \tag{157}$$

Q.3. **Gram reduction and the geometric implication.** On the active cell let $\bar{c}$ denote the corresponding expression in (157), and put

$$\bar{\eta}=h(1-\bar{c}),\qquad e=1-\bar{c},\qquad t=2\bar{c}-1.$$

Lemma Q.1 gives $0<\bar{\eta}\le\eta=h(1-c_*)$. It therefore suffices to prove the mixed intersections for the smaller disks with radii $2\bar{\eta},\bar{\eta},2\bar{\eta}$.

Represent a global vector as (x,hy) and put $C'=\mathsf{R}^{-1}C$. Define

$$N_A=\|A\|^2,\qquad N_C=\|C\|^2,\qquad \nu=\langle A,C'\rangle,$$
$$\Delta_0=x_Ay_{C'}-y_Ax_{C'}>0.$$

Thus the Euclidean cross product is $h\Delta_0$, and the Gram identity is

$$h^2\Delta_0^2=N_AN_C-\nu^2. \tag{158}$$

Define

$$Q_A=\Delta_0^2-\|tA-eC'\|^2,\qquad Q_C=\Delta_0^2-\|tC'-eA\|^2. \tag{159}$$

Lemma Q.2 (Residual signs imply the mixed cap intersections)**.** *If $Q_A,Q_C\ge 0$, then*

$$I(C,2\bar{\eta})\cap I(C+\mathsf{R}A,\bar{\eta})\ne\varnothing,$$

and

$$I(C+\mathsf{R}A,\bar{\eta})\cap I(\mathsf{R}A,2\bar{\eta})\ne\varnothing.$$

Consequently the same intersections hold with η in place of $\bar{\eta}$.

Proof. Rotate the first pair by R^{-1}. Its centers become C' and $A + C'$, so their difference is A. An outer common tangent normal n_A is chosen with

$$\langle A, n_A\rangle = \bar\eta.$$

Since $\|A\| > \bar\eta$ and $\Delta_0 > 0$, choose the tangent side for which

$$\langle C', n_A\rangle = \frac{\bar\eta\nu + \sqrt{N_A - \bar\eta^2}\, h\Delta_0}{N_A}.$$

The common support of the two disks is therefore

$$2\bar\eta + \frac{\bar\eta\nu + \sqrt{N_A - \bar\eta^2}\, h\Delta_0}{N_A}.$$

Let $\bar\tau = h - 2\bar\eta = ht$. Expanding (159) and using (158) gives

$$\bar P_A := (N_A - \bar\eta^2)h^2\Delta_0^2 - (\bar\tau N_A - \bar\eta\nu)^2 = h^2 N_A Q_A. \tag{160}$$

Thus $Q_A \geq 0$ implies

$$\sqrt{N_A - \bar\eta^2}\, h\Delta_0 \geq \bar\tau N_A - \bar\eta\nu,$$

and the common support is at least $2\bar\eta+\bar\tau = h$. Hence the two level-h caps intersect. The same calculation with A and C' exchanged gives

$$\bar P_C := (N_C - \bar\eta^2)h^2\Delta_0^2 - (\bar\tau N_C - \bar\eta\nu)^2 = h^2 N_C Q_C, \tag{161}$$

and proves the second intersection. Enlarging the disk radii from $\bar\eta$ to η only enlarges the caps. ☐

Q.4. **Eight exact integer-polynomial signs.** The Newton witnesses A, C are rational functions of U, z, D. Substituting (157) into (159), clearing denominators, and reducing by

$$D^2 - K(U, z) = 0$$

gives, for $B \in \{L, T\}$ and $X \in \{A, C\}$,

$$\operatorname{num}(Q_{B,X}) = 32(K + 3)^2\{A_{B,X}(U, z) + DB_{B,X}(U, z)\}, \tag{162}$$

where all eight core polynomials lie in $\mathbb{Z}[U, z]$. Every omitted denominator is strictly positive on the strict domain: its factors are positive constants, $(D+3)^4$, $G(U, z)^2$ or $(1+U+z)^2$, and squares of the two Newton-derivative numerators. Here $G = 2\kappa > 0$, and the fixed-line sign theorem proves both Newton derivatives strictly negative.

The derivation verifier constructs A, C, both rational envelopes, and all four residuals directly in the exact field $\mathbb{Q}(U, z, D)$. It performs the reduction modulo $D^2 - K$, removes the common positive factor, primitive-normalizes the eight integer polynomials, and compares their sparse term lists coefficient by coefficient with the authenticated transcript. Thus the data are checked against the geometric formulas rather than trusted as an independent precomputation.

Q.5. **Three global positive-basis certificates.** Set

$$\mathcal{G}(U) = 1 - 6U - 3U^2, \qquad H(U) = 1 - 10U + 21U^2 - 16U^3 + 4U^4,$$

$$U_H = 1 - \frac{\sqrt{3}}{2}, \qquad U_{\max} = \frac{2}{\sqrt{3}} - 1.$$

The complete reflected domain is the union of the three closed cells

$$\begin{aligned} T:&\quad 0\le U\le U_H,\quad 0\le z^2\le H(U),\\ L_0:&\quad 0\le U\le U_H,\quad H(U)\le z^2\le \mathcal{G}(U),\\ L_1:&\quad U_H\le U\le U_{\max},\quad 0\le z^2\le \mathcal{G}(U).\end{aligned}$$

For the T cell use

$$U=\frac{(x-1)(3-x)}{4x},\qquad z=y\frac{9-x^4}{8x^2},\qquad 1\le x\le\sqrt{3},\quad 0\le y\le 1,$$

and then $x=1+(\sqrt{3}-1)r$. This maps $[0,1]^2$ onto the complete cell. After multiplication by positive powers of 2 and x, the four transformed core polynomials have global tensor Bernstein expansions with data

	bidegree	coefficient count	zero count
$A_{T,A}$	$(88,22)$	2047	2
$B_{T,A}$	$(80,20)$	1701	0
$A_{T,C}$	$(88,22)$	2047	2
$B_{T,C}$	$(80,20)$	1701	0.

Every nonzero coefficient is strictly positive.

For an L-cell polynomial write

$$F(U,z)=E_F(U,w)+zO_F(U,w),\qquad w=z^2,$$

and

$$R_F(U,w)=E_F(U,w)^2-wO_F(U,w)^2.$$

The implications $E_F\ge 0$ and $R_F\ge 0$ give $E_F\ge\sqrt{w}\,|O_F|$, hence $F\ge 0$ for $z\ge 0$. Use the exact charts

$$U=U_Hr,\qquad w=H(U)+s\{\mathcal{G}(U)-H(U)\}$$

for L_0, and

$$U=U_H+(U_{\max}-U_H)r,\qquad w=s\mathcal{G}(U)$$

for L_1. The positivity verifier forms one global Bernstein expansion for each of E_F,R_F on each square. The resulting sixteen expansions have the bidegrees and coefficient counts recorded in Appendix P.6; every coefficient is strictly positive.

The verifier implements polynomial addition, multiplication, substitution, and power-to-Bernstein conversion over exact integers. Coefficients in $\mathbb{Q}(\sqrt{3})$ are represented as rational pairs and signed by exact rational comparisons. If $P(r,s)=\sum a_{ij}r^is^j$ has bidegree (m,n), its exact tensor Bernstein coefficients are

$$b_{k\ell}=\sum_{i\le k}\sum_{j\le\ell}a_{ij}\frac{\binom{k}{i}}{\binom{m}{i}}\frac{\binom{\ell}{j}}{\binom{n}{j}}.$$

This is precisely the conversion implemented by the verifier.

Theorem Q.3 (Exact mixed-overlap certificate)**.** *On the complete strict witness domain with $c_*>2/3$, all eight polynomials in* (162) *are nonnegative. Hence $Q_A,Q_C\ge 0$, and the two mixed cap intersections required by Proposition P.10 hold.*

Proof. The T chart proves the four T-cell core signs by nonnegative Bernstein coefficients. On each L chart the positive Bernstein certificates for E_F and R_F prove the four L-cell signs. Since $D\ge 0$, (162) gives $Q_A,Q_C\ge 0$ for $z\ge 0$; reflection supplies $z\le 0$. Lemma Q.2 then gives the two cap intersections. □

Remark Q.4 (Reproduction status). The files above constitute the exact certificate used by the proof. A reproduction run should execute

```
verify_mixed_overlap_core_derivation.py
verify_global_core_positivity.py
```

from their package directory. This manuscript revision does not claim that those commands were rerun during the present source-only edit; the certificate is incorporated by its exact repository objects and authenticated transcript.

Appendix R. Nonstandard Symbols

Standard geometric notation is defined at first use. The following symbols recur across proof mechanisms.

symbol	meaning
$\mathrm{CE0}, \mathrm{CE1}, \mathrm{CE2}$	The three exhaustive center-role boundary-trace classes.
$\mathrm{Vd0}, \mathrm{Vd1}, \mathrm{Vd2}, \mathrm{T3\text{-}like}$	The four exhaustive vertex-role classes.
(A_i, B_i, C_i)	Actual maximal incoming boundary, outgoing boundary, and radial reaches.
(a_i, b_i, c_i)	Selected lower-bound demands; these do not define N_+.
N_+, m, q	The number of actual supercritical rows, non-Vd0 roles, and short roles; in CE1/CE2 one has $q = N_+ + m$.
gr	The active-gap rank; bare g is reserved for transfer maps.
$\Lambda(K)$	The least side length of a closed equilateral triangle containing K.
$\omega(x), \sigma(x)$	The radical $\sqrt{1-x+x^2}$ and its deficit $1-\sqrt{1-x+x^2}$.
$\mathcal{A}$	The exact local admissible set of boundary–radial demand triples.
$g_c(x)$	The historical raw defect map: $\max\{y : (1-x, y, c) \in \mathcal{A}\}$.
$\widehat{g}_c(x)$	The nonsupercritical cap $\min\{g_c(x), x\}$.
$f^\vee(a)$	The complement dual $1 - f(1-a)$; in particular $\widehat{g}_c^\vee$ is the extensive incoming-reach transfer.
$\mathcal{R}_J(p)$	The far-side residual demand after an initial trace and center interval J.
$g_{c,J}^\vee, \widehat{g}_{c,J}^\vee$	The raw and capped center-assisted reach transfers.
g_c^{sc}	The single free strict-supercritical outgoing envelope $\sup_{\{x: g_c(x)>x\}} g_c(x)$.
$\widehat{g}_{1-d}^{\vee,\lambda}$	A certified affine selected-T_+ lower relaxation.
$\widehat{g}_{1-d}^{\vee,\mathrm{th}}$	The low-root threshold lower relaxation.
$[\Phi_1 \mid \cdots \mid \Phi_r]$	The composition $\Phi_r \circ \cdots \circ \Phi_1$, with maps listed in geometric row order.
B_c, F_c, G_c	Technical appendix aliases: $B_c(a) = g_c(1-a)$, $F_c(a) = \widehat{g}_c(1-a)$, and $G_c(a) = \widehat{g}_c^\vee(a)$.
$e(d)$	The high-radial low-root threshold for deficit d.
$R, W, E, \alpha, \delta, \Delta_R, \Delta_L$	The common signed CE1/CE2 center variables and trace surpluses.
$f(a,b), \mathcal{L}(a,b)$	Normalized maximal retained area and the corresponding local area loss.
$c_*, \mathcal{D}_\eta, Q_-, Q_0, Q_+$	The radial envelope, forced disk, and exact asymmetric witnesses in Strategy 4.
$A, B, C, \widehat{K}, K_{\mathrm{wit}}$	The Newton inner points, inner witness hull, and exact forced witness set.
$I(X, r)$	The level-$\sqrt{3}/2$ support cap associated with the disk $\mathbb{D}(X, r)$.